SECJ2203: Software Engineering

System Documentation (SD)

System Cooperative Of KADA

Version 1.0

Date 30/1/2025

Faculty of Computing

Prepared by: <Data Alchemist>

Revision Page

a. **Overview**

This version of the SD includes the Software Requirements Specification (SRS) details and user types within the KADA Cooperative System.

b. **Target Audience**

The target audience includes stakeholders of the KADA Cooperative System, specifically Members, Clerks, and the Board of Directors.

c. **Project Team Members**

Member Name	**Role**	**Task**	**Status**
Muhammad Naim Bin Abdullah	team leader	1.5 overview, 2.3 Launch Phase, 2.2 System Features, 2.4.9 US009: User Story <Create an Account> User Story description of US009 Activity Diagram of US009 System Sequence Diagram of US009 2.4.10 US010: User Story <Manage List of Members> User Story description of US010 Activity Diagram of US010 System Sequence Diagram of US010 2.4.11 US011: User Story <Manage Loan Application> User Story description of US011 Activity Diagram of US011 System Sequence Diagram of US011 , 2.4.14 US014: User Story <Interest> User Story description of US014	complete

		Activity Diagram of US014 System Sequence Diagram of US014 , 2.5 Performance and Other Requirements , 2.6 Design Constraints , 4.1 Use Case View , 4.3 Logical View , 6 user interface design	
Muhammad Mukhritz Al Iman Bin Mohd Raffi	group member	1.1 purpose , 1.2 scope, 2.2 System Features , 2.4.7 US007: User Story <Verify Loan Application> User Story description of US007 Activity Diagram of US007 System Sequence Diagram of US007 2.4.8 US008: User Story <View Annual Report> User Story description of US008 Activity Diagram of US08 System Sequence Diagram of US008 , 2.4.13 US013: User Story <Payment> User Story description of US013 Activity Diagram of US013 System Sequence Diagram of US013 , 2.5 Performance and Other Requirements , 2.6 Design Constraints , 4.4 Process View , 4.5 Physical/ Deployment	complete

		View, 5 data design , 6 user interface design	
Syarifah Dania Binti Syed Abu Bakar	group member	1.3 Definitions, Acronyms and Abbreviations, 2.1 Persona , 2.1.1 Persona 1 (member) 2.1.2 Persona 2 (clerk) 2.1.3 Persona 3 (ALK), 2.4.1 US001: User Story <Manage Profile> User Story description of US001 Activity Diagram of US001 System Sequence Diagram of US001 , 2.4.4 US004: User Story <Register> User Story description of US004 Activity Diagram of US004 System Sequence Diagram of US004 2.4.5 US005: User Story <Login> User Story description of US005 Activity Diagram of US005 System Sequence Diagram of US005 2.4.6 US006: User Story <Logout> User Story description of US006 Activity Diagram of US006 System Sequence Diagram of US006 , 2.4.15 US015: User Story <Add Staff>	complete

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Yasmin Batrisya Binti Zahiruddin	group member	1.4 reference , 2.4.2 US002: User Story <Applying Loan> User Story description of US002 Activity Diagram of US002 System Sequence Diagram of US002 2.4.3 US002: User Story <Check Loan Status> User Story description of US003 Activity Diagram of US003 System Sequence Diagram of US003 , 2.4.12 US012: User Story <Calculate loan Repayment> User Story description of US012 Activity Diagram of US012 System Sequence Diagram of US012 , 4.2 Implementation View, 6 user interface design	complete

d. **Version Control History**

Version	**Primary Author(s)**	**Sprint/Iteration**	**Description of Version**	**Date Completed**

1.0	Team leader (Muhammad Naim Bin Abdullah)	Sprint 1	Completed Chapter 9.	30/1/2024

Note:
This System Documentation (SD) template is adapted from IEEE Recommended Practice for Software Requirements Specification (SRS) (IEEE Std. 830-1998), Software Design Descriptions (SDD) (IEEE Std. 10161998 1), and Software Test Documentation (IEEE Std. 829-2008) that are simplified and customized to meet the need of SECJ2203 course at School of Computing, UTM. Examples of models are from Arlow and Neustadt (2002) and other sources stated accordingly.

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Introduction

1.1 Purpose

A Software Requirements Specification (SRS) serves as a guide for the development team and clients to ensure an agreement of the system's objectives, features, and limitations by precisely defining and documenting the functional and non-functional requirements of a software system.

In the case of the proposed KADA Registration and Loan Application System, this SRS document defines the requirements for digitizing KADA's existing manual processes for membership registration and loan applications.The document will serves as a reference and guidance for stakeholders, including developers, project managers and testers ,outlining the project's scope, functional and non-functional requirements, and design considerations, providing a roadmap for the development and implementation phases.

1.2 Scope

The proposed system at KADA means to digitize and ease up the present manual procedures about membership registration and loan applications, respectively. This enhances efficiency in terms of minimizing paperwork, reducing error cases, and providing a practical platform for members, applicants, and KADA personnel. There are three roles that the system plays at the front-applicants, clerks, and ALK.

The system provides the facility for online membership and loan application forms that can be filled out anytime and sent without necessarily going to submit them physically, especially to those who are far away from KADA's headquarters. Immediately after successful submission, applicants get confirmation to reduce uncertainty about the status of their forms.

It facilitates clerks' work through one common platform that will enable them to handle membership and loan applications efficiently. One can access applicant details, verify information submitted, and change the status of an application with ease. The system enables clerks to generate various reports and track applications, enhancing productivity.

The application will be reviewed and approved by ALK members remotely via the system, which is a very significant improvement compared to the present system where their presence in meetings is absolutely necessary. The efficiency at which decision-making has been improved by providing access to ALK members from any location greatly reduces approval times.

In this respect, automated notifications, document storage, and real-time data access will make operations within KADA smoother and more effective. While the system focuses on registration and loan management, it is also designed with heavy security for the safety of sensitive personal and financial data. This is a modernization project that seeks to increase productivity among the staff in KADA, as well as improve the experience for applicants and members.

1.3 **Definitions, Acronyms and Abbreviation**

- SD – System Documentation
- KADA - Lembaga Kemajuan Pertanian Kemubu
- ALK - Ahli Lembaga Koperasi
- SRS - System Requirement Specification
- Functional requirement - technical details on specification, data analysis and processing, and explain what a system must do
- Non-functional Requirement - criteria that will be used to judge the system operation

1.4 **References**

Here are the references in APA format:

1. Sommerville, I. (2011). *Software engineering* (9th ed.). Addison-Wesley.
2. IEEE Software Engineering Standards Committee. (2018). *IEEE Std 830-1998, IEEE recommended practice for software requirements specifications*.

3. Bennett, S., McRobb, S., & Farmer, R. (2010). *Object-oriented systems analysis and design* (4th ed.). McGraw-Hill.

1.5 **Overview**

This SRS document is divided into sections that provide a comprehensive overview of the KADA system. It involves an overview of the system requirements, functional and non-functional specifications, user interfaces, data management rules, system constraints, and potential improvements. The system's goals and objectives are thoroughly examined in each section, with detailed descriptions, specifications, and requirements provided to assure understanding.

This document provides an overview for project stakeholders, including project managers and development teams, to better understand and visualize the KADA system's requirements and specifications. This description of the planned KADA system promotes effective data storage, data accessibility and decision making.

2. Specific Requirements

2.1 Persona

The system is designed to serve multiple user groups, each with unique roles, responsibilities and levels of interaction. Identifying these users and understanding their needs is essential for ensuring an intuitive and efficient system design. This section outlines the primary personas, detailing their specific requirements, technical expertise, and expected system interactions. By addressing these user needs, the system aims to streamline processes, improve accessibility, and enhance overall user experience.

2.1.1 Persona 1 (User type 1/Actor 1)

User Need

- Apply membership or loan from KADA without having to visit KADA's headquarters.
- Track loan status on the website

User Stories

- An applicant wants to track the status of their loan application so that they can know if it has been approved or needs further action. A status indicator is displayed in the dashboard and emails are sent for status updates

2.1.2 Persona 2 (Clerk)

User Need

- Manage and verify applications efficiently
- Create user accounts for the ones whose membership applications are approved

User Stories

- A clerk wants to review membership applications to review and verify applicants' details

2.1.3 Persona n (Ahli Lembaga Pengarah (ALK))

User Need

- Review and approve loan applications.
- Access financial reports for further evaluation

User Stories

- ALK who wants to review and approve loan applications remotely to participate in decision making even when they are not in the office.

2.2 System Features

The KADA Cooperative System is a flexible software solution that works with all devices, including Android and iOS platforms. It improves the management of member registrations and loan applications by addressing issues about data storage and accessibility. The system's real-time cloud database ensures simple access and an improved user experience.

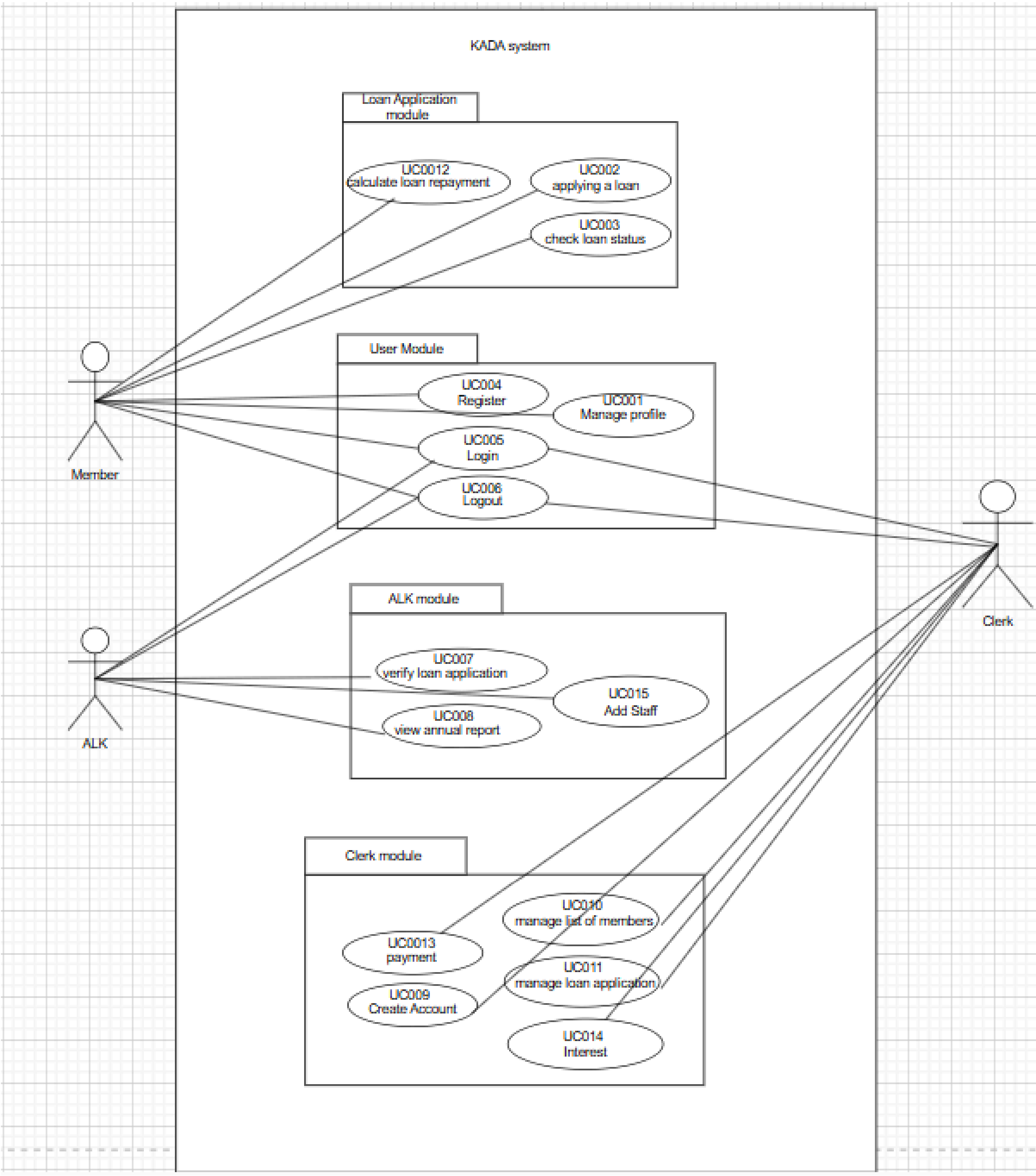


Figure 1.1: Use Case Diagram for KADA cooperative system

Table 1.1: Description of Module and Functions for KADA cooperative system

Use Case ID	Name	Description
UC001	Manage Profile	Allows an applicant to edit personal information.
UC002	Applying Loan	Allows an applicant to make a loan application.
UC003	Check Loan Status	Allows an applicant to check the status of their loan (approved or declined).
UC004	Register	Allows members to register as members of the KADA Cooperative. it also cover on adding the Clerk and ALK Account
UC005	Login	Allows users (applicants, clerks, and ALK) to access their respective interfaces.
UC006	Logout	Allows all users to log out of the system.
UC007	Verify Loan Application	Enables ALK to view loan application details and decide whether to approve or decline.
UC008	View Annual Report	Enables ALK to view the annual report of the KADA Cooperative, including profit or loss data.
UC009	Create an Account	Enables the clerk to create an account for an applicant after verifying their staff status.
UC010	Manage List of Members	Enables the clerk to delete, update, or view personal information of members.
UC011	Manage Loan Application	Enables the clerk to filter and review loan applications but not make approval decisions.
UC012	Calculate loan repayment	Allow member to calculate their own loan repayment
UC013	Payment	Enables the clerk to oversee the payment of loans and savings.
UC014	Interest	Allow the clerk to update their interest based on the company requirement
UC015	Add Staff	Allow ALK to add more staff accounts so they can access the website.

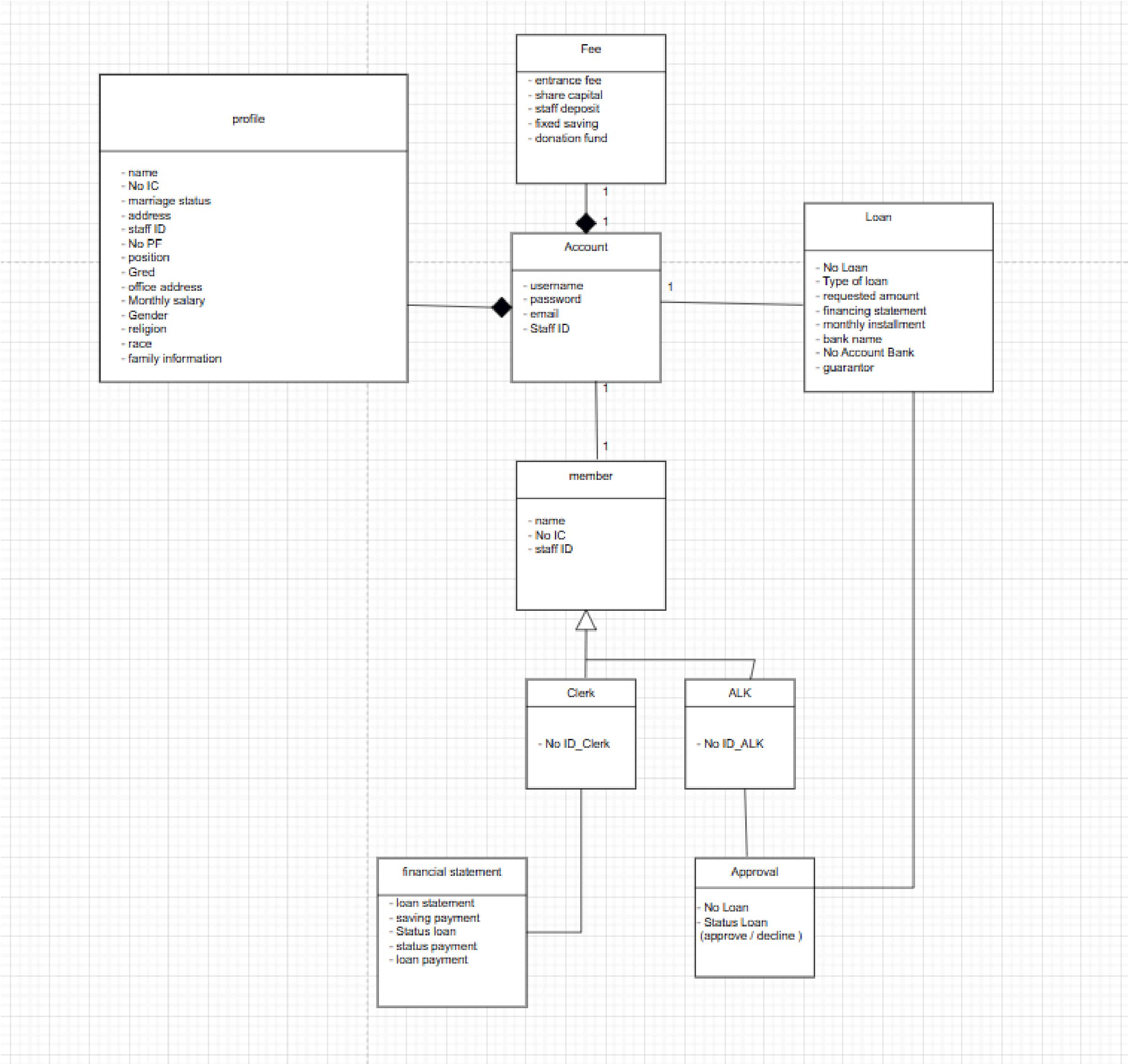


Figure 1.2: Domain Model for KADA cooperative system

Figure 2.3 shows the state machine diagram for the KADA Cooperative System.

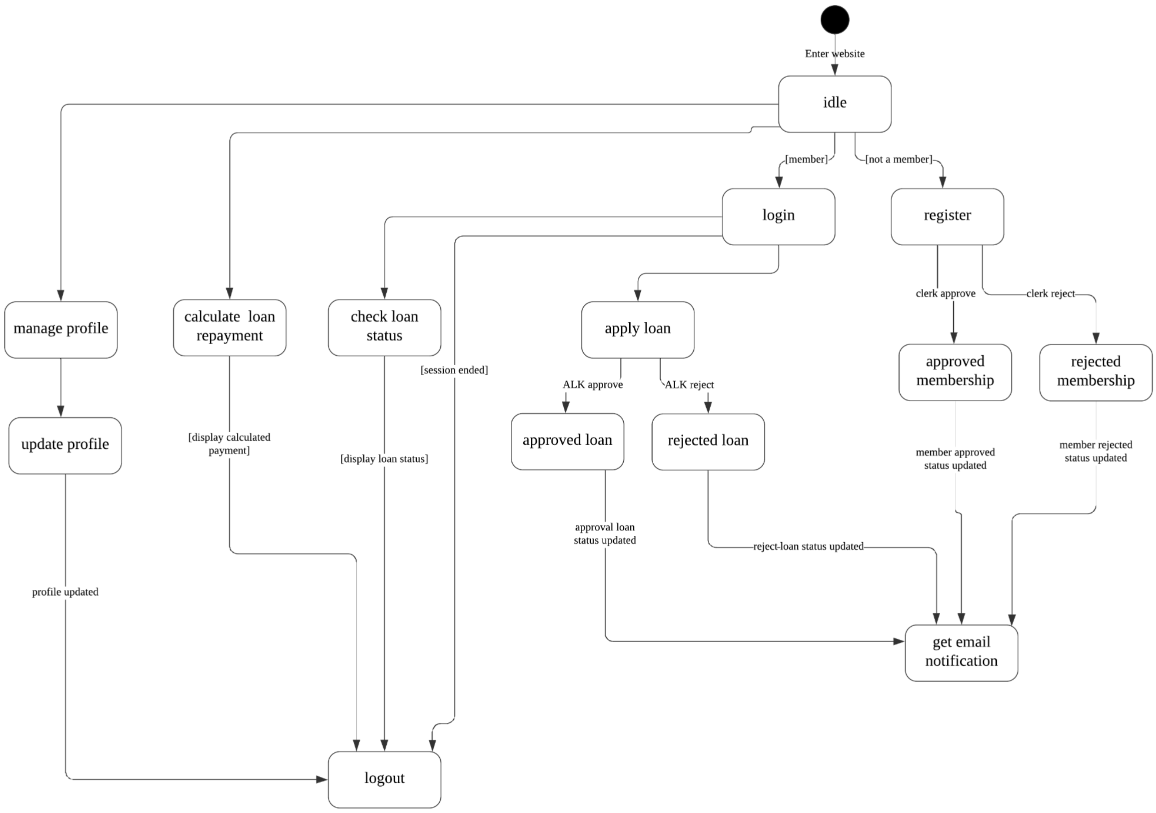


Figure 2.3 : State Machine Diagram for <KADA system>

2.3 **Launch Phase**

Product Backlog [describe more on product backlog];

Sprint	**Team members assigned**
UC001 - Manage Profile *UC004 - login* *UC005 - register* *UC006 - Logout* *UC015 - Add Staff*	Syarifah Dania Binti Syed Abu Bakar
UC002 - applying loan *UC003 - check loan status* *UC012 - calculate loan payment*	Yasmin Batrisyia Binti Zahiruddin
*UC009 - create an account ** *UC010 - manage list of member* *UC011 - manage loan application* *UC014 - Interest*	Muhammad Naim Bin Abdullah
UC007 - verify loan application *UC008 - view annual report* *UC013 - Payment*	Muhammad Mukhritz Al Iman bin Mohd Raffi

2.4 User Story Details

User stories describe how different users will interact with the system to accomplish their tasks. Each story focuses on a specific feature or functionality, outlining what the user wants to do and why it matters. These stories help ensure the system is designed to be user-friendly and meets real-world needs. They include step-by-step flows, alternative scenarios, and conditions to handle different situations. The following sections break down each user story in detail.

2.4.1 US001: User Story <Manage Profile>

Table 2.1: User Story Description for <Manage Profile>

User story: <Manage Profile>
ID: UC001
User Story Description **As a** member of the cooperative **I want** to be able to manage my personal information in the system **So that** I can ensure my details are accurate and up to date
Flow of events: 1. The member logs into the system using their credentials (delegated username and password) 2. The member navigates to the "Profile" section in the dashboard 3. The member updates the desired fields (name, address, contact info) 4. The system validates the input and saves changes 5. A confirmation message is displayed to the customer …
Alternative flow *n*: **If a member attempts to update their email address with an invalid format** **1. The system displays an error message indicating the email format is incorrect.** **2. The member re-enters the email address in the correct format and submits again.**
Acceptance Criteria Postcondition - The member's profile is updated in the system database Precondition - The member must be logged into the system with valid credentials Other conditions - The system must ensure that only the logged-in member can update their profile

Exception flow: If the system encounters a server error during the update: 1. The changes are not saved. 2. An error message is displayed to the customer

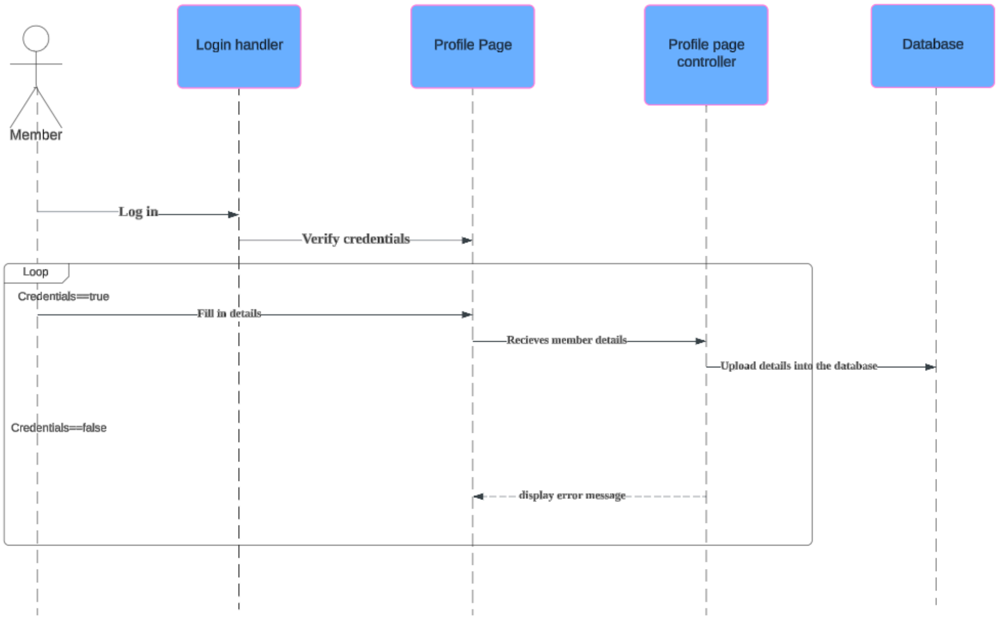


Figure 2.5: Sequence Diagram for <Manage Profile>

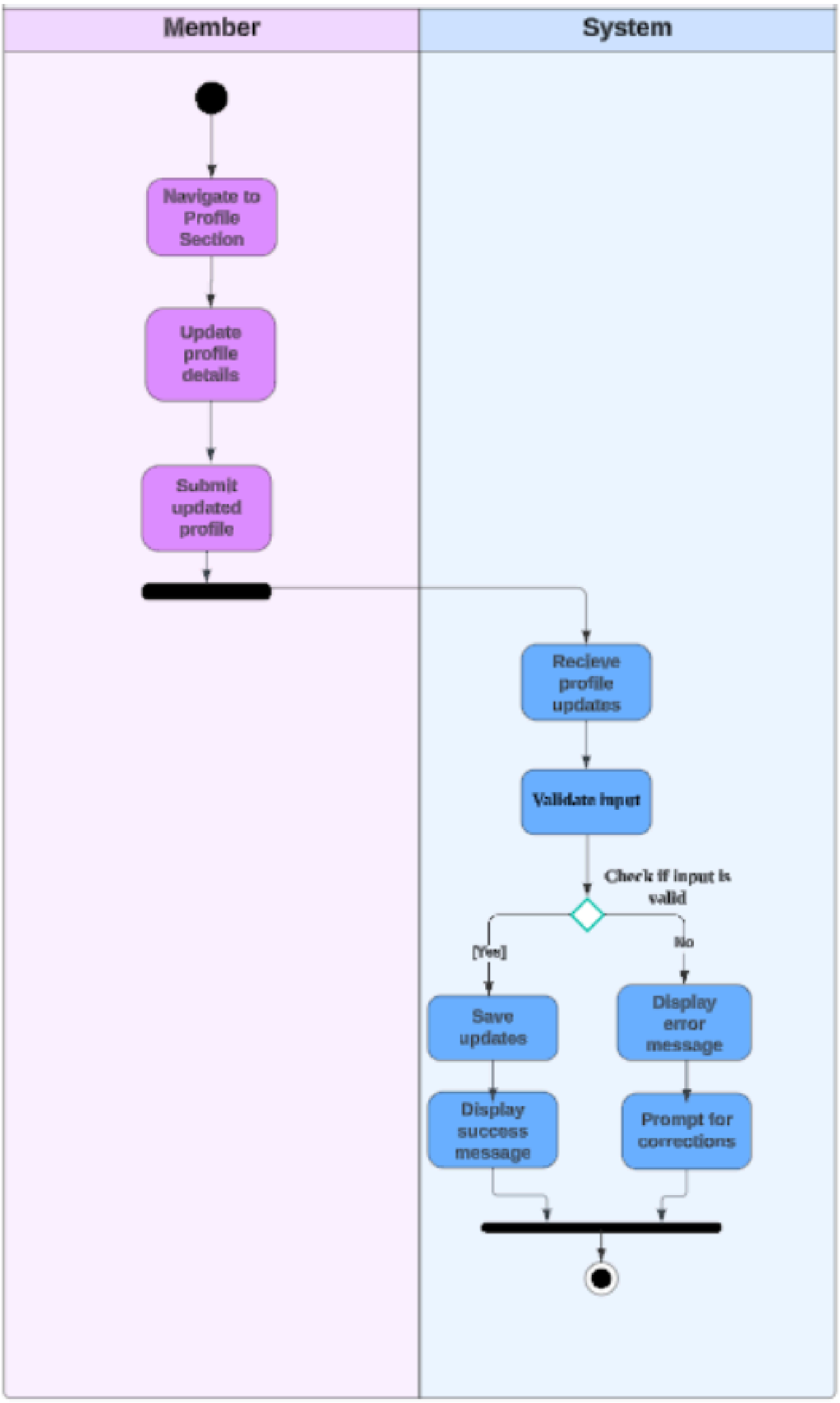


Figure 2.2: Activity Diagram for <<Manage Profile>>

2.4.2 US002: User Story <Applying Loan>

Table 2.2: User Story Description for <Applying Loan>

User story: <Applying Loan>
ID: US002
User Story Description **As a** cooperative member of KADA **I want** to be able to apply for a loan **So that** I can use the money for my needs without using up all my savings
Flow of events: 1. The member navigates to the loan application page 2. The member filled in required details 3. The member submit registration form 4. The system will receive the details and validate 5. If the details are filled in correctly, details is saved to the database 6. User will receive the successful application message …
Alternative flow *n*: If the member provide an invalid information: 1. The system highlights the invalid or incomplete fields with error messages. 2. The customer corrects the information and resubmits.
Acceptance Criteria Postcondition - The member will get the notification says that the form has been submitted Precondition - User must be a membership of KADA
Exception flow: If the system fails during loan application submission - An error message is displayed: “Loan application failed, please try again later.”

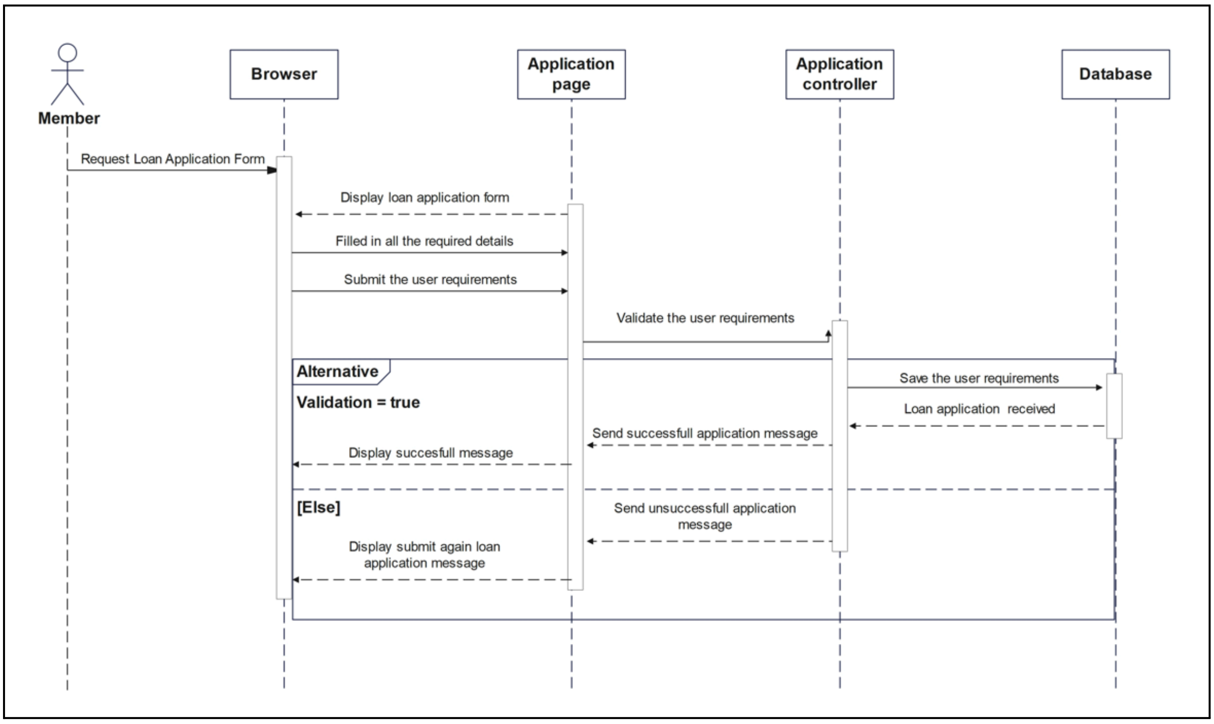


Figure 2.4.1: Sequence diagram for <Applying Loan>

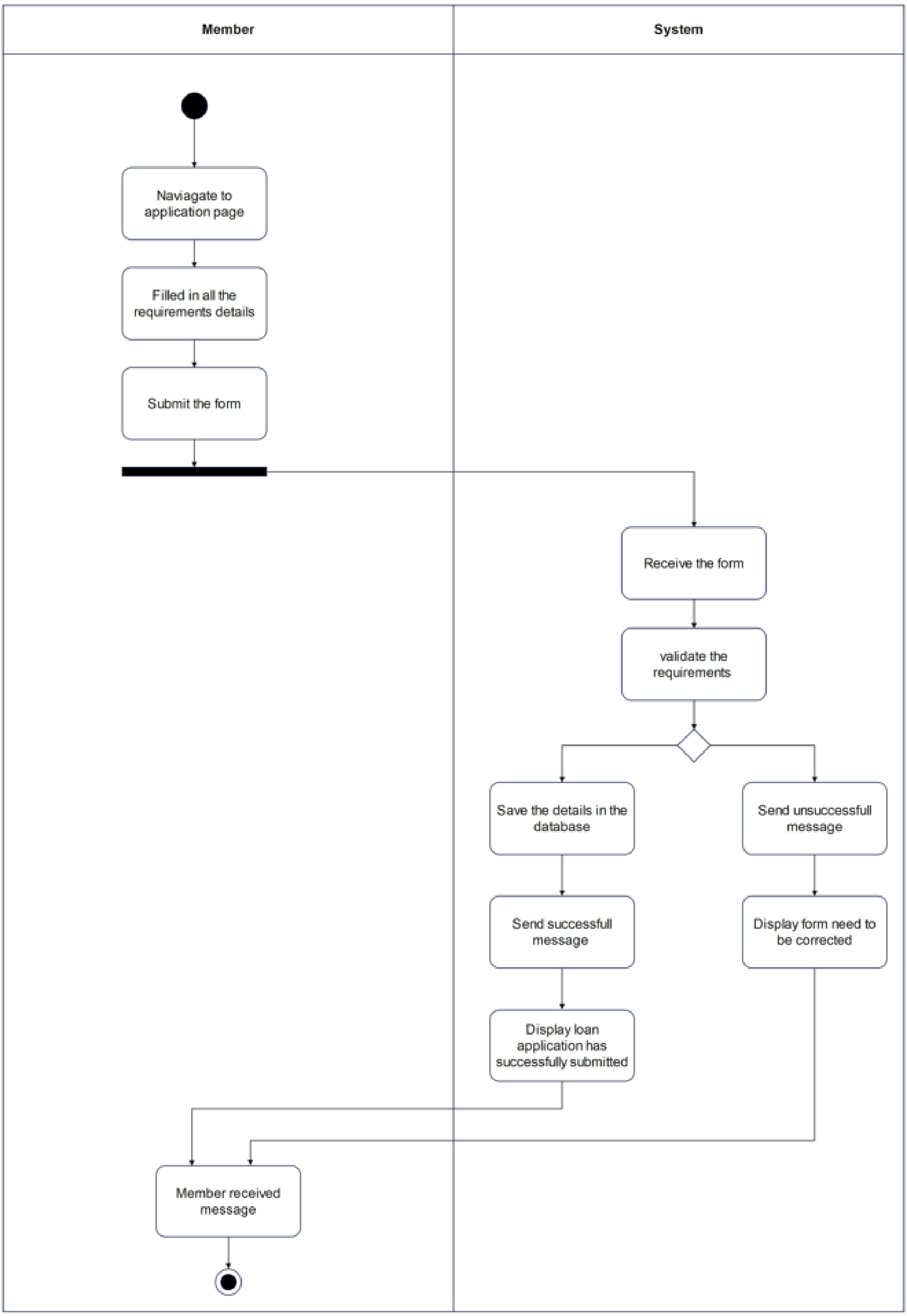


Figure 2.4.2 Activity Diagram for <Applying Loan>

2.4.3 **US003: User Story <Check Loan Status>**

Table 2.3: User Story Description for <Check Loan Status>

User story: <Check Loan Status>
ID: US003
User Story Description **As a** cooperative member of KADA **I want** to check my loan status after I have submitted the loan application **So that** I can acknowledge the current update regarding to the submission
Flow of events: 1. The member navigates to the profile page 2. Member click on the 'Semak Status' section 3. The system will appear the current loan application status to the member 4. The status will appear that the loan is successful, when the loan has been approved from ALK …
Alternative flow *n*: If the loan application is unsuccessful: 1. Loan is not approved 2. Display that there is some error in the submission 3. The member is suggested to resubmit the form
Acceptance Criteria Postcondition - Loan application will appear the loan status is being processed, successful, or unsuccessful precondition - Member must log in to check the loan application status
Exception flow: If the status is not available due to a system error, the ALK member receives a notification to retry later

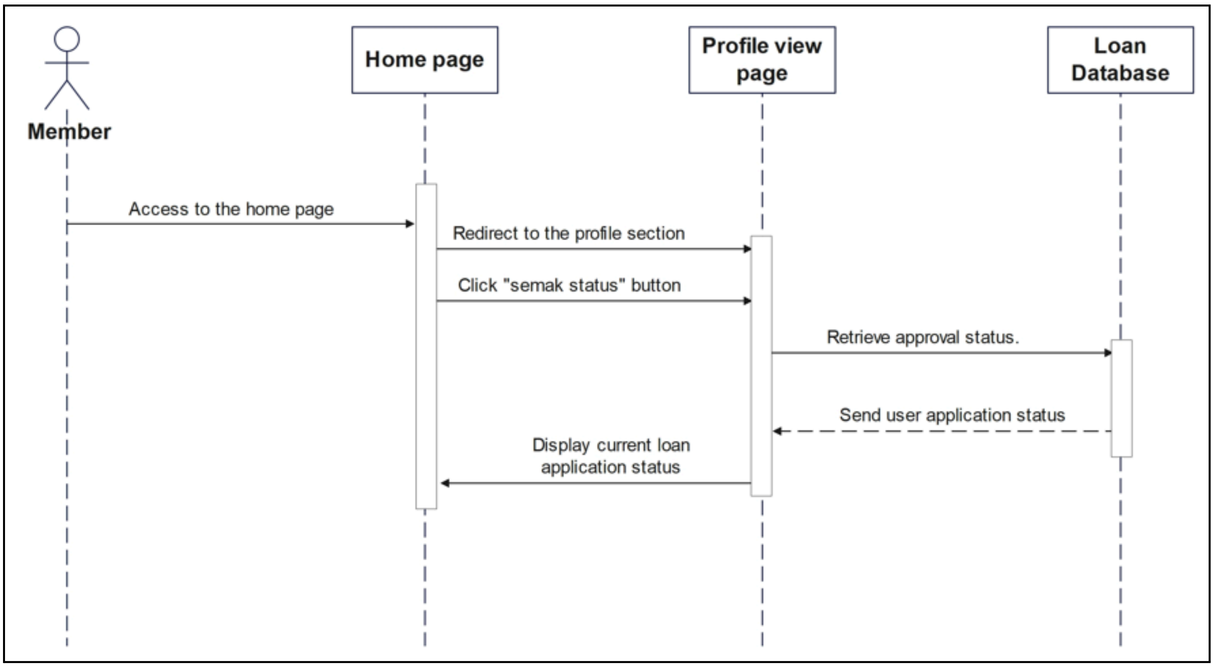


Figure 2.5.1: Sequence Diagram for <Check Loan Status>

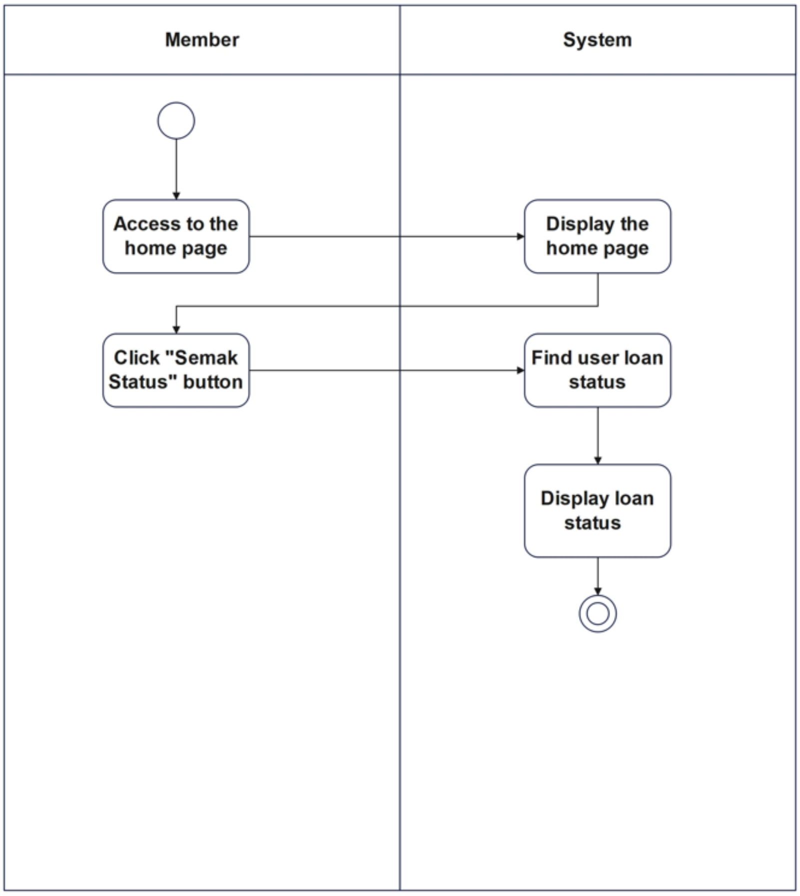
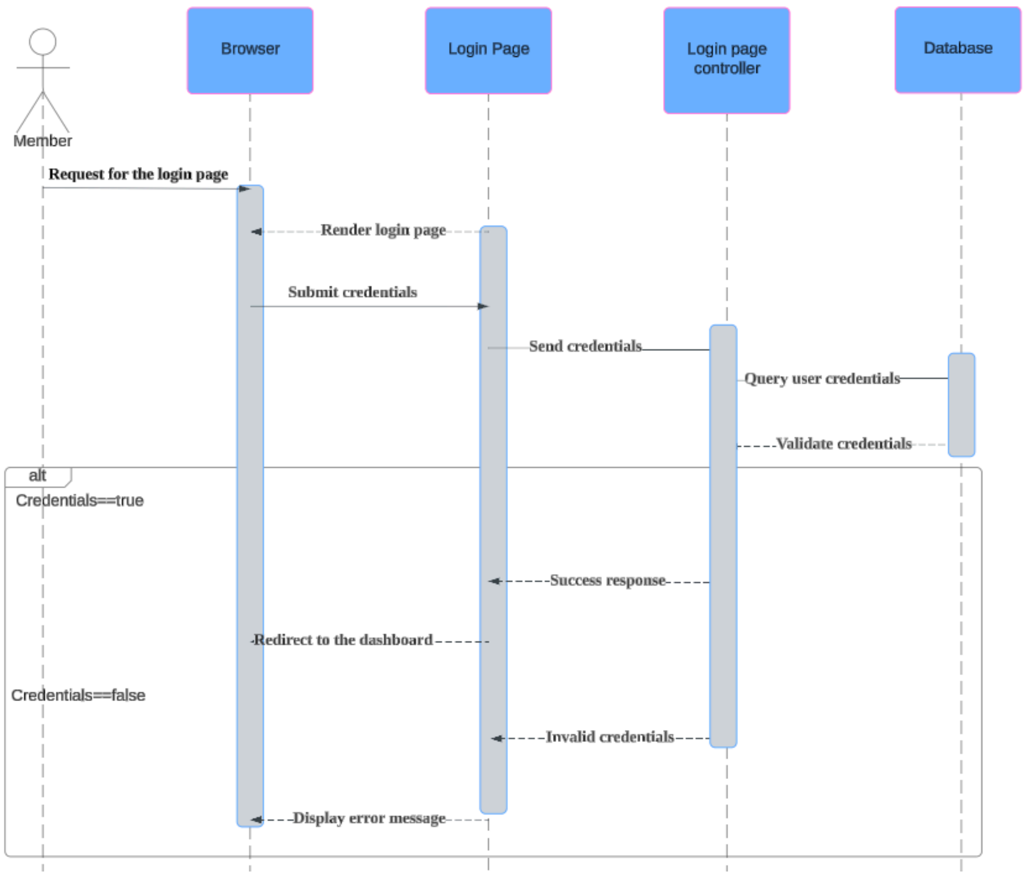


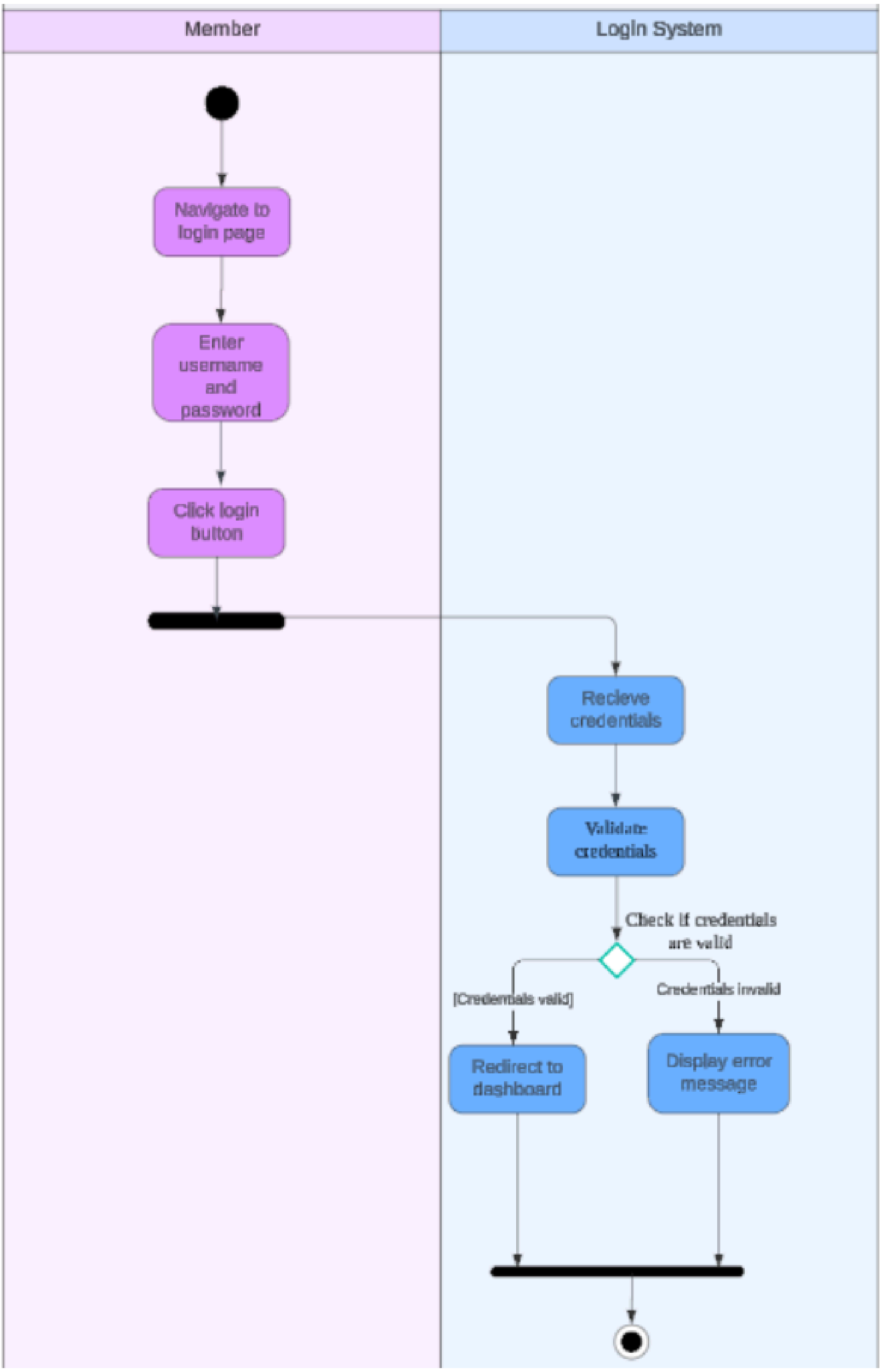
Figure 2.5.2: Activity Diagram for <check Loan Status>

2.4.4 **US004: User Story <Login>**

Table 2.4: User Story Description for <Login>

User story: <Login>
ID: US004
User Story Description **As a** member of the cooperative **I want** to be able to log into the system **So that** I can securely access my account and its features
Flow of events: 1. The member navigates to the login page 2. The member enters their username and password 3. The system validates their credentials 4. Upon successful validation, the member is redirected to their dashboard …
Alternative flow *n*: If the member enters invalid credentials: 1. The system displays an error message “Invalid username or password.” 2. The member is prompted to re-enter the credentials.
Acceptance Criteria Postcondition - The member is authenticated and given access to their account precondition - The member must have a registered account with the cooperative
Exception flow: If the system experiences server downtime during login: 1. An error message is displayed such as “The system is temporarily unavailable, please try again later.”


Browser
Login Page
Login page controller
Database
Member
Request for the login page
Render login page
Submit credentials
Send credentials
Query user credentials
Validate credentials
alt
Credentials==true
Success response
Redirect to the dashboard
Credentials==false
Invalid credentials
Display error message

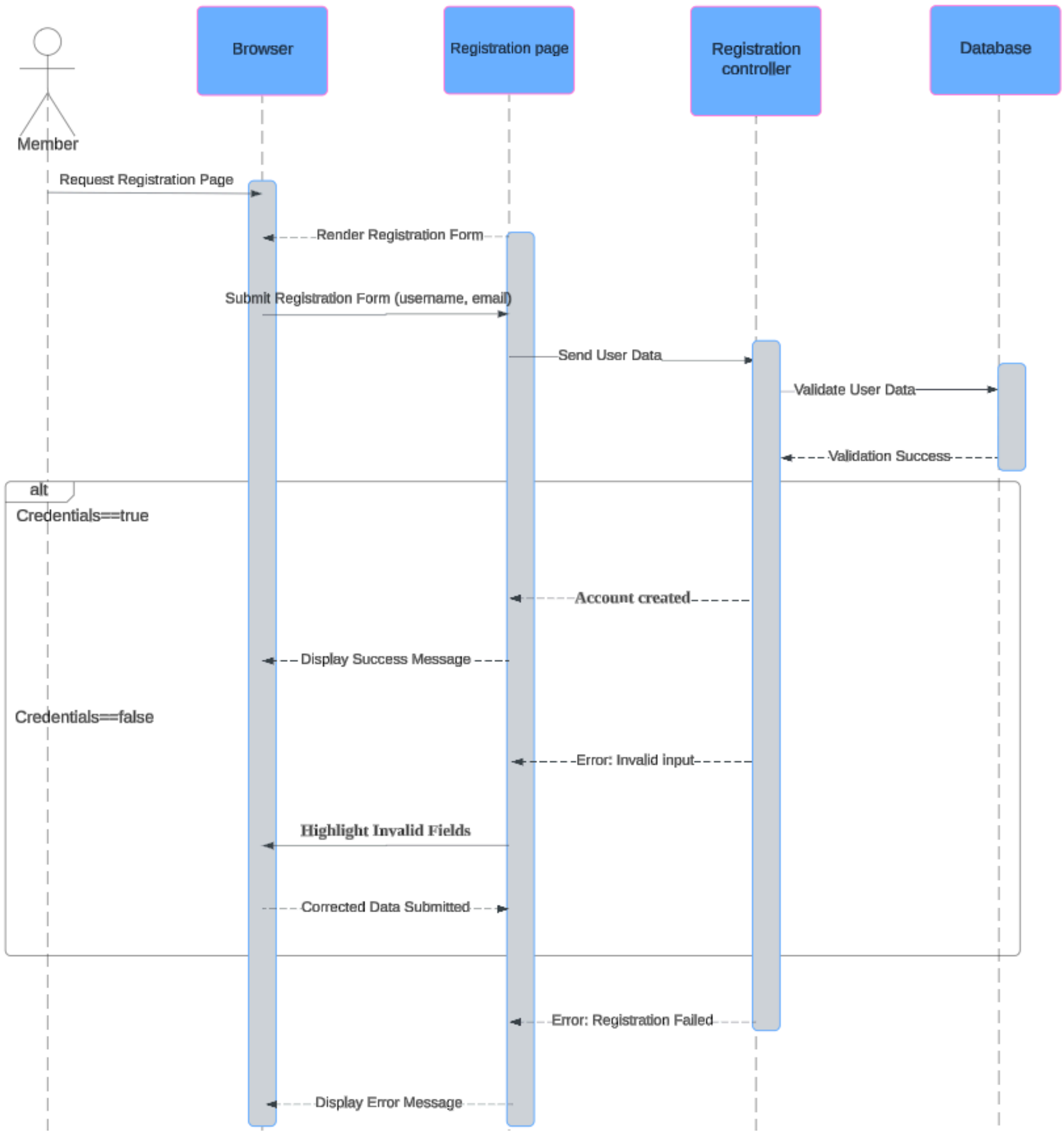

Member
Login System
Navigate to login page
Enter username and password
Click login button
Recieve credentials
Validate credentials
Check if credentials are valid
[Credentials valid]
Credentials invalid
Redirect to dashboard
Display error message

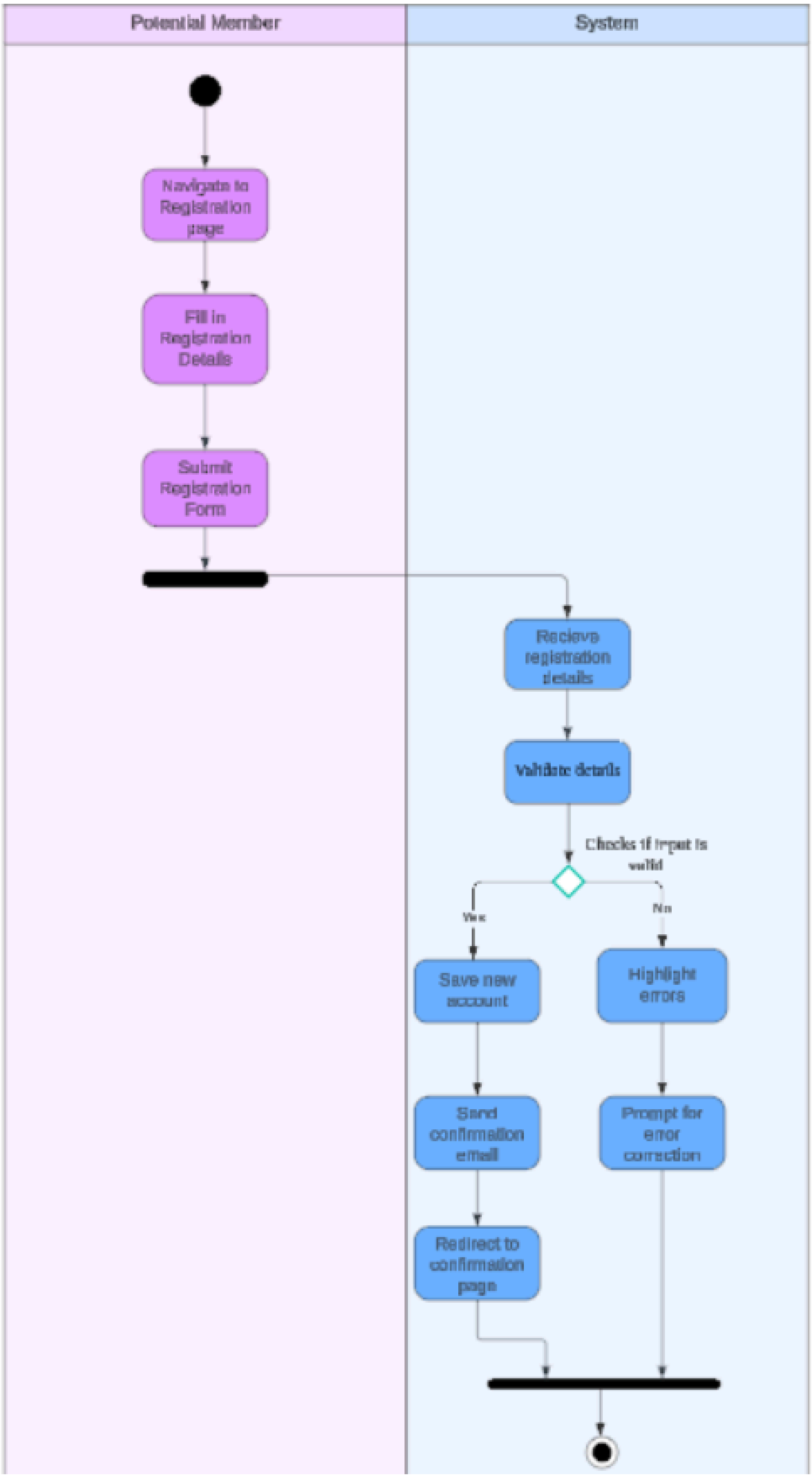
login

2.4.5 US005: User Story <Register>

Table 2.5: User Story Description for <Register>

User story: <Register>
ID: US005
User Story Description **As a** potential member of the cooperative **I want** to register for an account **So that** I can apply for loan
Flow of events: 1. The user navigates to the registration page 2. The user fills in the required details (username and email) given by the clerk 3. The customer submits the registration form. 4. The system validates the input and creates the account. 5. A confirmation email is sent to the customer. …
Alternative flow *n*: **If the customer provides invalid information:** **1. The system highlights the invalid or incomplete fields with error messages.** **2. The customer corrects the information and resubmits.**
Acceptance Criteria Postcondition - A new account is created in the system and the customer is notified precondition - The customer must receive the designated username and password given by the clerk through email
Exception flow: If the system fails during registration submission - An error message is displayed: “Registration failed, please try again later.”

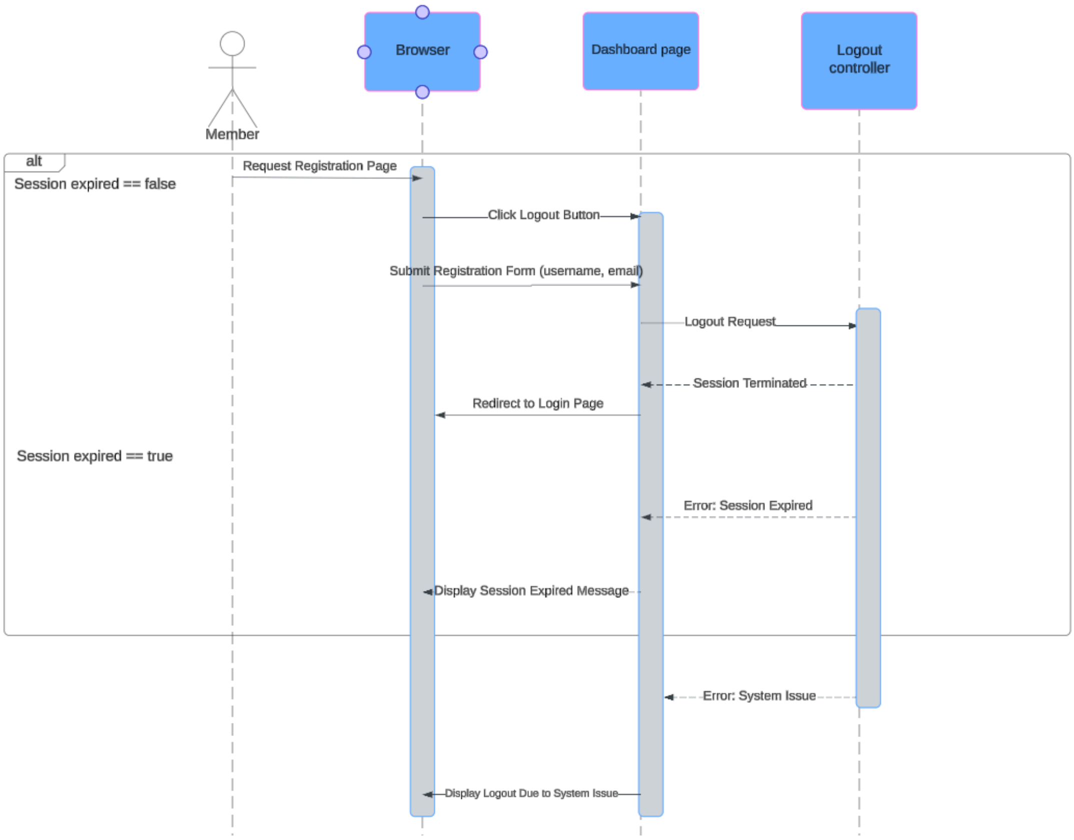

Browser
Registration page
Registration controller
Database
Member
Request Registration Page
Render Registration Form
Submit Registration Form (username, email)
Send User Data
Validate User Data
Validation Success
alt
Credentials==true
Account created
Display Success Message
Credentials==false
Error: Invalid input
Highlight Invalid Fields
Corrected Data Submitted
Error: Registration Failed
Display Error Message

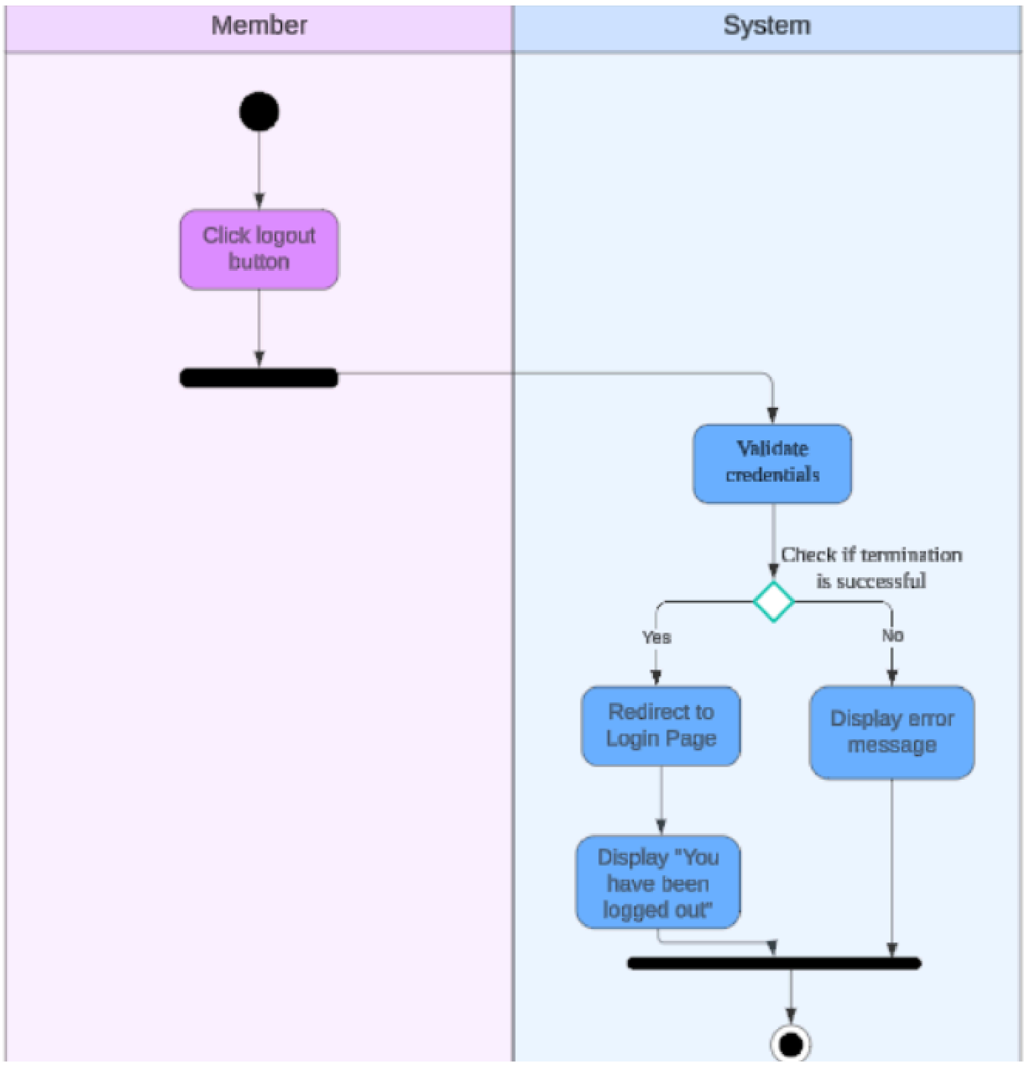

Potential Member
System
Navigate to Registration page
Fill in Registration Details
Submit Registration Form
Recieve registration details
Validate details
Checks if input is valid
Yes
No
Save new account
Highlight errors
Send confirmation email
Prompt for error correction
Redirect to confirmation page

2.4.6 US006: User Story <Logout>

Table 2.6: User Story Description for <Logout>

User story: <Logout>
ID: US006
User Story Description **As a** member of the cooperative **I want** to be able to log out of the system **So that** I can ensure the security of my account
Flow of events: 1. The customer clicks the "Logout" button in the dashboard. 2. The system ends the session.
Alternative flow *n*: If the customer's session has already expired: - The system displays a message: Your session has expired. Please log in again."
Acceptance Criteria Postcondition - The user session is terminated, ensuring account security precondition - The customer must be logged into the system
Exception flow: If the system encounters a server issue during logout: - The session ends, but the user is notified: "You have been logged out due to a system issue."


Member
Browser
Dashboard page
Logout controller
alt
Session expired == false
Request Registration Page
Click Logout Button
Submit Registration Form (username, email)
Logout Request
Session Terminated
Redirect to Login Page
Session expired == true
Error: Session Expired
Display Session Expired Message
Error: System Issue
Display Logout Due to System Issue


Member
System
Click logout button
Validate credentials
Check if termination is successful
Yes
No
Redirect to Login Page
Display error message
Display "You have been logged out"

2.4.7 US007: User Story <verify loan application >

Table 2.7: User Story Description for <verify loan application>

User story: <verify loan application>
ID: US007
User Story Description **As a** ALK **I want** verify loan application **So that** I can approve or reject loan applications
Flow of events: 1. ALK logs into the system. 2. ALK navigates to the "Loan Applications" section. 3. ALK selects a loan application from the list. 4. The system displays the loan application details. 5. ALK verifies the application and decides to approve or reject it. 6. The system updates the loan application status to **"Approved"** or **"Rejected"**
Alternative flow *n*: If the application has missing or invalid details: 1. The ALK is notified of the issue and can reject the application directly.
Acceptance Criteria Postcondition -The loan application status is updated to "Approved" or "Rejected" precondition -ALK must be logged into the system. ALK must have access to the loan applications section.
Exception flow: (if any in the event of error)

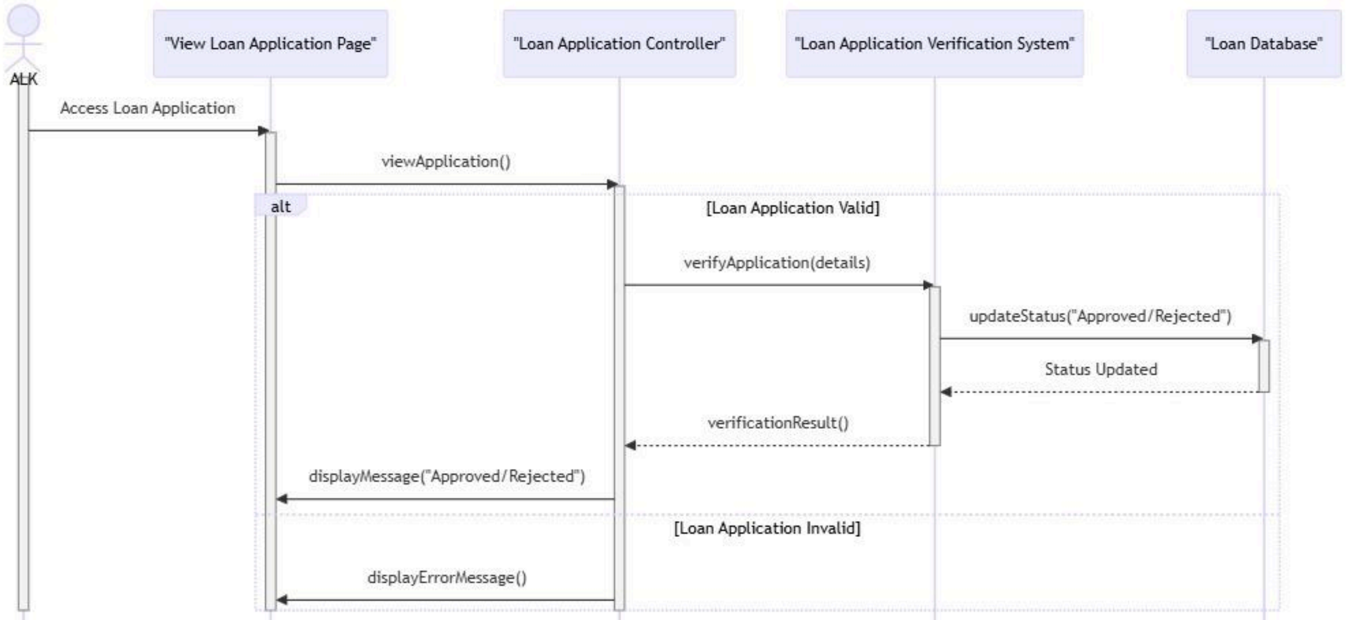


Figure 2.7.1: Sequence Diagram for <verify loan application>

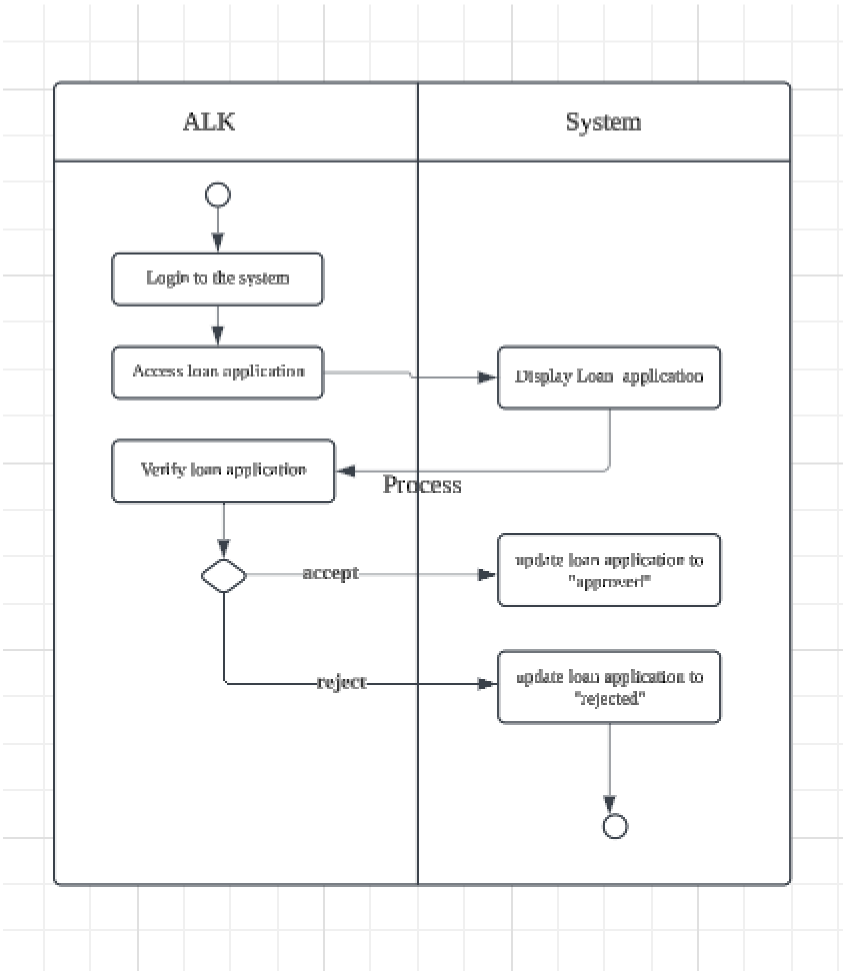


Figure 2.7.2: Activity Diagram for <verify loan application>

2.4.8 US008: User Story < view annual report >

Table 2.8: User Story Description for <View annual report>

User story: <View annual report>
ID: US008
User Story Description **As a** an ALK member **I want** view the annual report **So that** I can analyze the cooperative's financial performance.
Flow of events: 1. ALK members log into the system. 2. ALK members navigate to the reports section. 3. ALK members select "Annual Report" from the available reports. 4. The system displays the annual report, including financial summaries, profit and loss data, and loan statistics. 5. ALK members review the report for analysis.
Alternative flow *n*: If the ALK member needs specific data: 1. ALK member can filter the report by specific dates, loan types, or member categories. 2. System updates the report view based on the filters
Acceptance Criteria Postcondition -The ALK member successfully views the annual report. precondition -ALK member must be logged into the system and have access to financial reports. other conditions
Exception flow: 1. If the report is not available due to a system error, the ALK member receives a notification to retry later

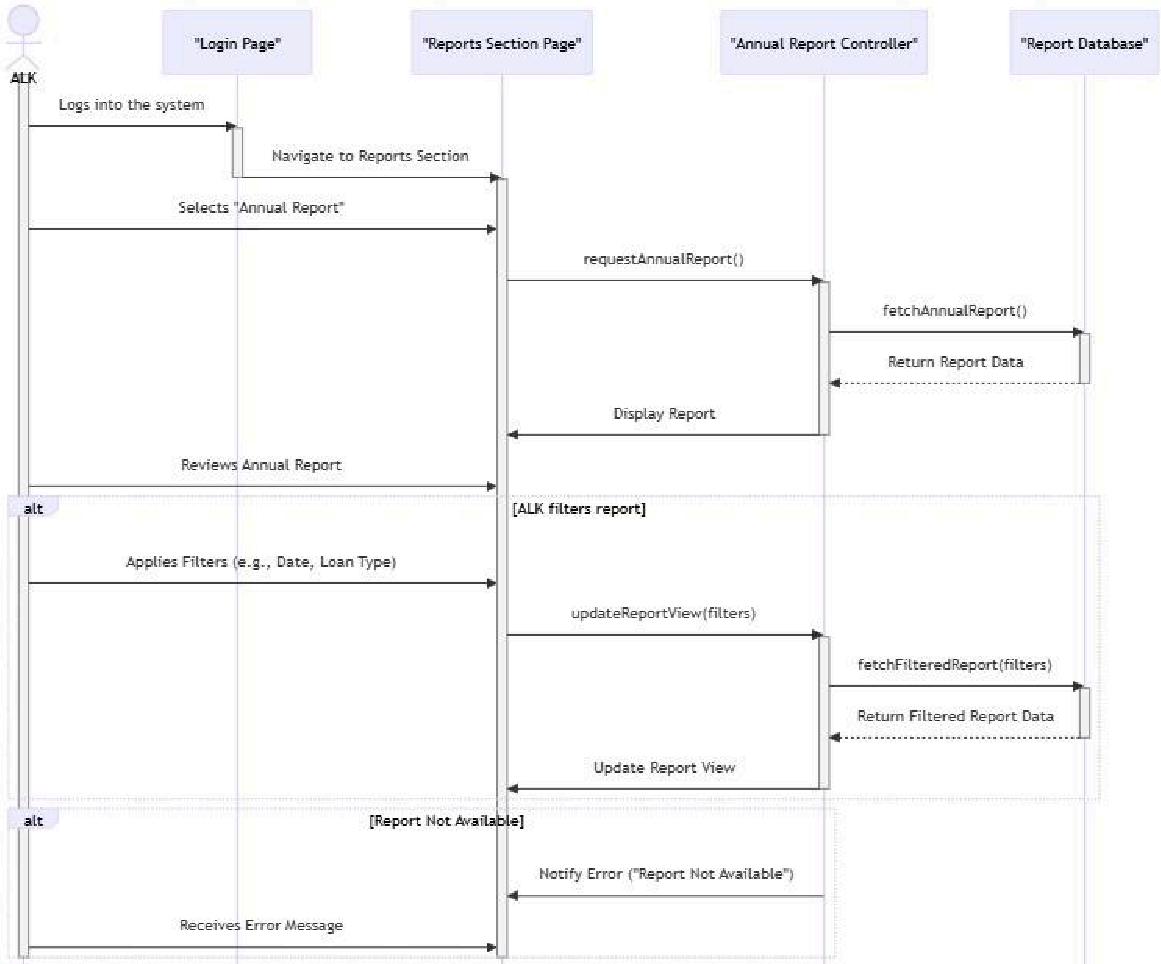


Figure 2.8.1: Sequence Diagram for <view annual report>

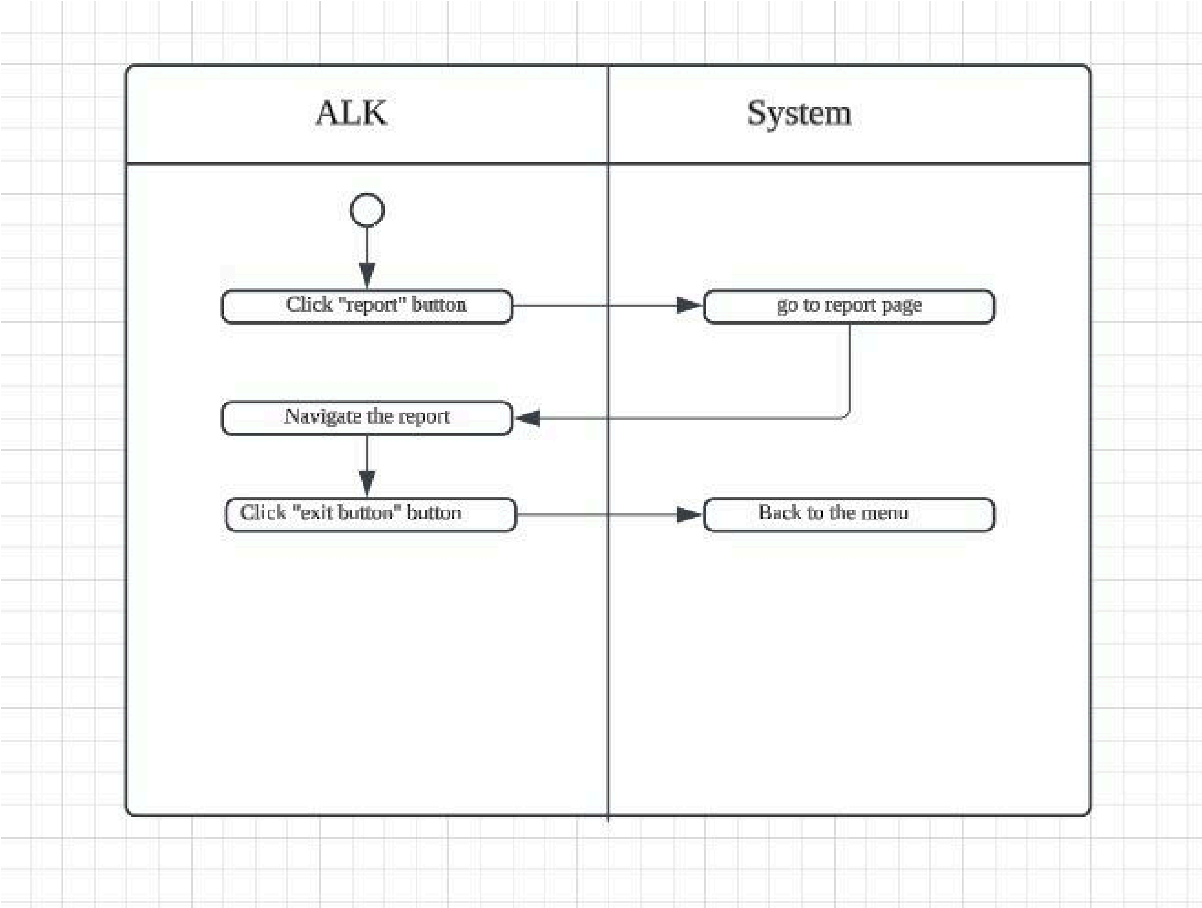


Figure 2.8.2: Activity Diagram for <view annual report>

2.4.9 US009: User Story <create an account>

Table 2.9: User Story Description for <Create an account>

User story: <Create an account>
ID: US009
User Story Description **As a** clerk of the KADA cooperative **I want to** create an account for the staff **So that** I can monitor who is the one requested to be a KADA cooperative member
Flow of events: 1. the clerk will check the list of the new register member 2. the clerk will make sure the that the one who is register is from KADA 3. the clerk will click the button ‘accept’ and it will lead to the create account page 4. the clerk will create an account for them by using their email, staff id , name. 5. the clerk will click the button ‘submit’ to indicate they are officially a member and it will notify and give a account username and password to the applicant by email
Alternative flow *n*: **1. the new register member are not from KADA** **1.1. the clerk check the staff id and it wrong** **1.2. the clerk will push the button ‘reject’ to clear up the list of new register member and it will automatically send an email to the applicant to indicate that the application has been decline**
Acceptance Criteria Postcondition - the account has been successfully create precondition - the clerk must verify the staff id whether it's true or not
Exception flow:

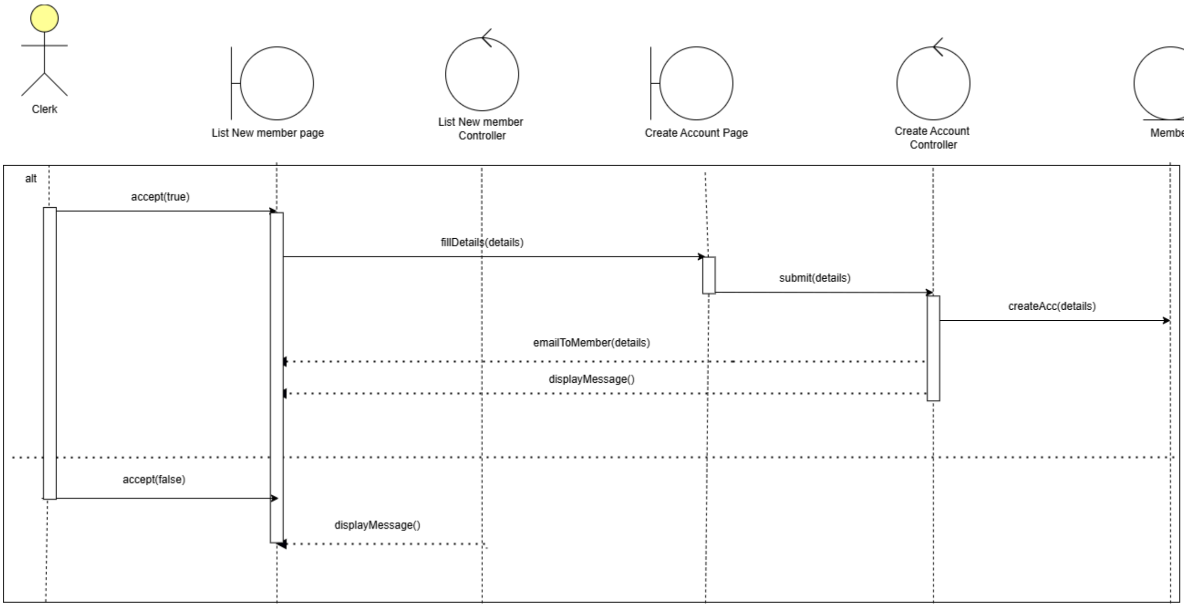


Figure 2.9.1: Sequence Diagram for <create an account>

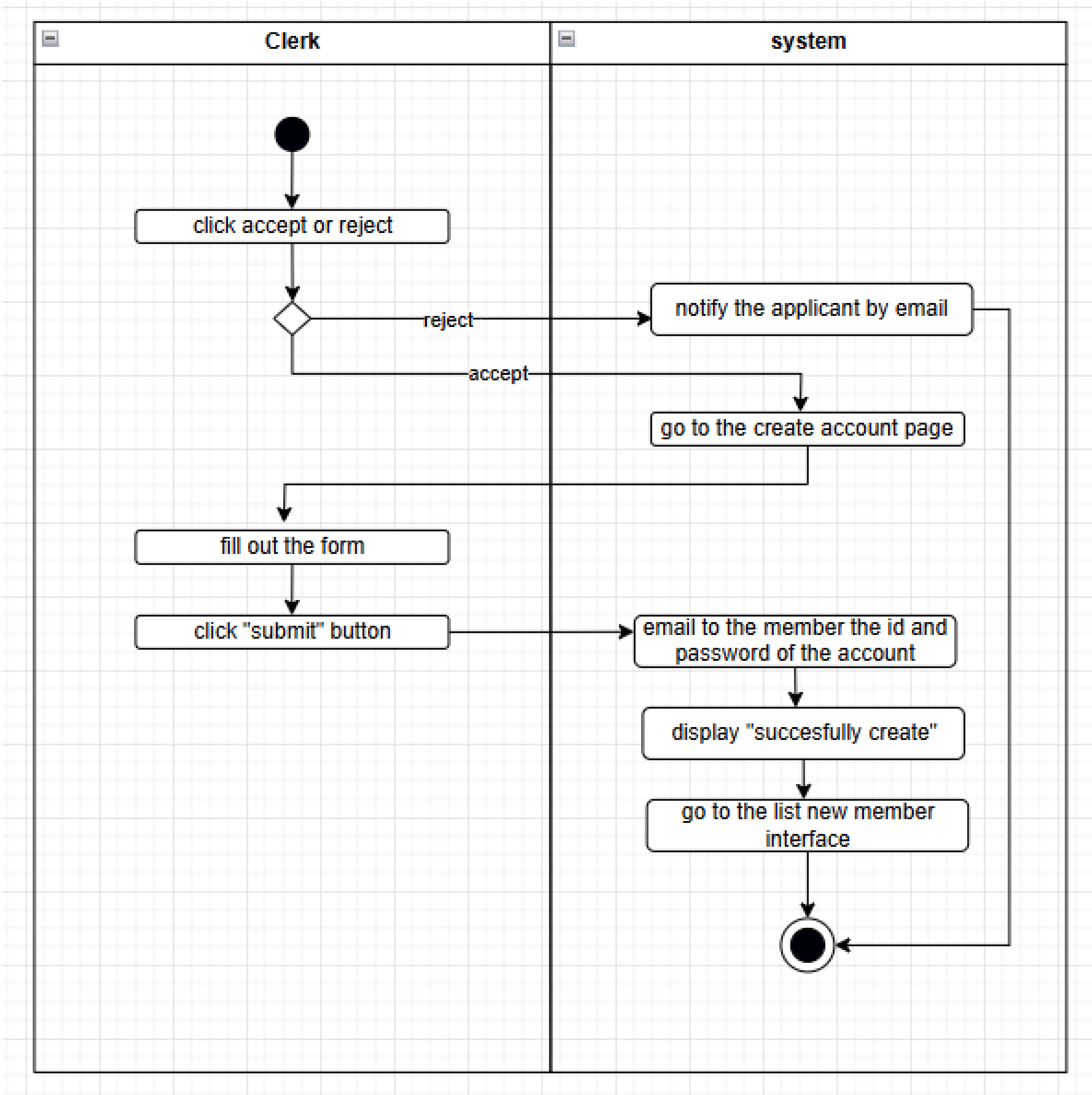


Figure 2.9.2: Activity Diagram for <create an account>

2.4.10 US010: User Story <manage list of member>

Table 2.10: User Story Description for <Manage list of member>

User story: <Manage list of member>
ID: US010
User Story Description **As a** clerk of the KADA cooperative **I want** to be able to manage the list of the member that has been successfully register as a member of KADA cooperative **So that** I can update, delete, view their personal information and ect
Flow of events: 1. the clerk can monitor overall information of the members
Alternative flow *n*: **1. members that already retired or want quit** **1.1. the clerk will push button 'delete' to delete the information of the members**
Acceptance Criteria Postcondition - the clerk must login as a clerk with clerk ID and password in login section precondition - the clerk can monitor overall information of the cooperative members
Exception flow:

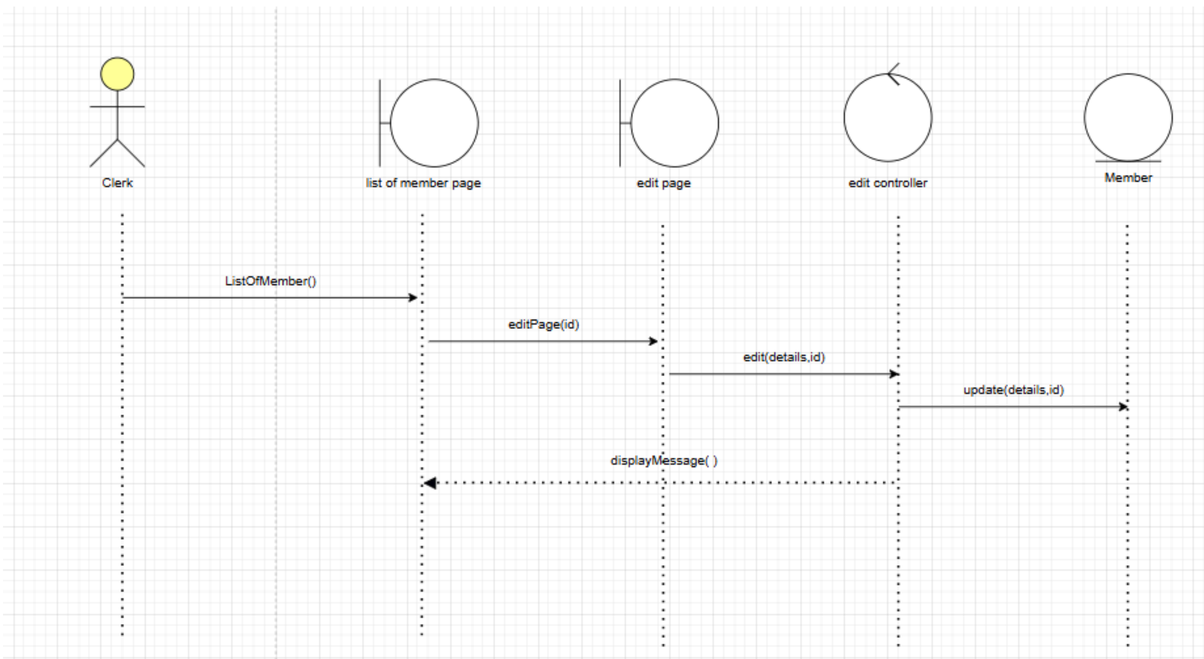


Figure 2.10.1: Sequence Diagram for <manage list of member (edit)>

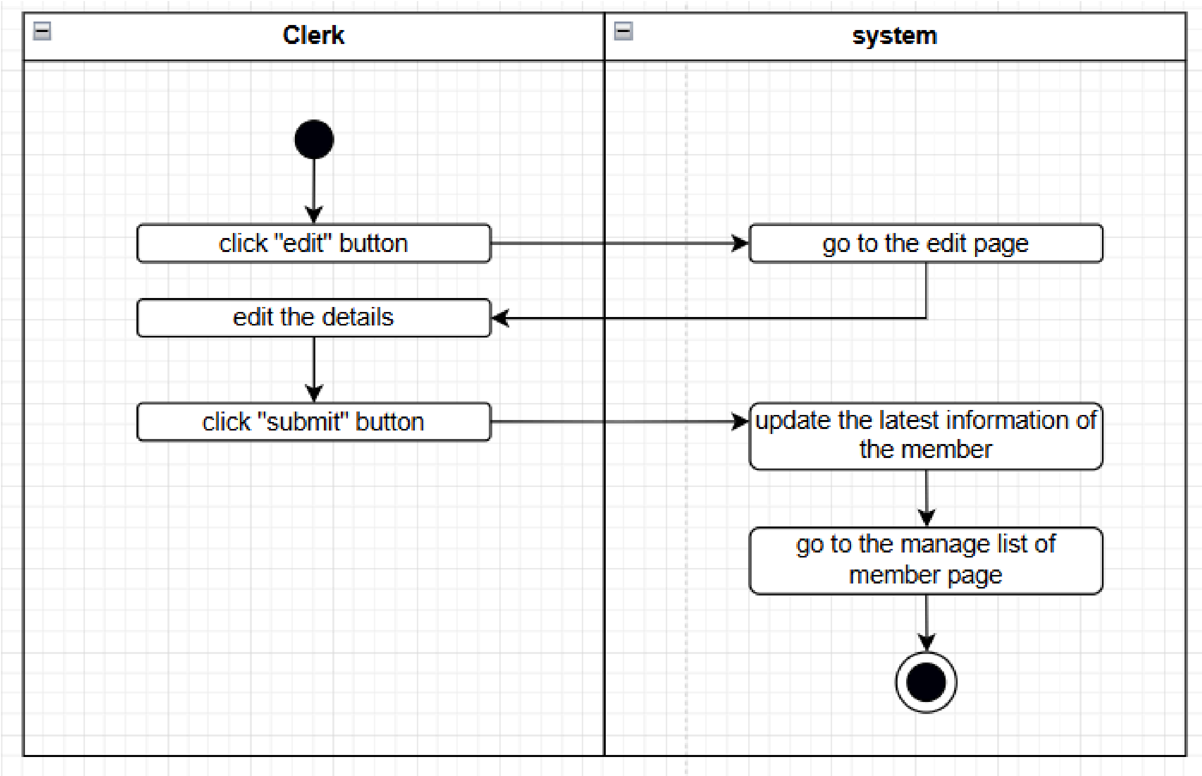


Figure 2.10.2: Activity Diagram for <manage list of member (edit)>

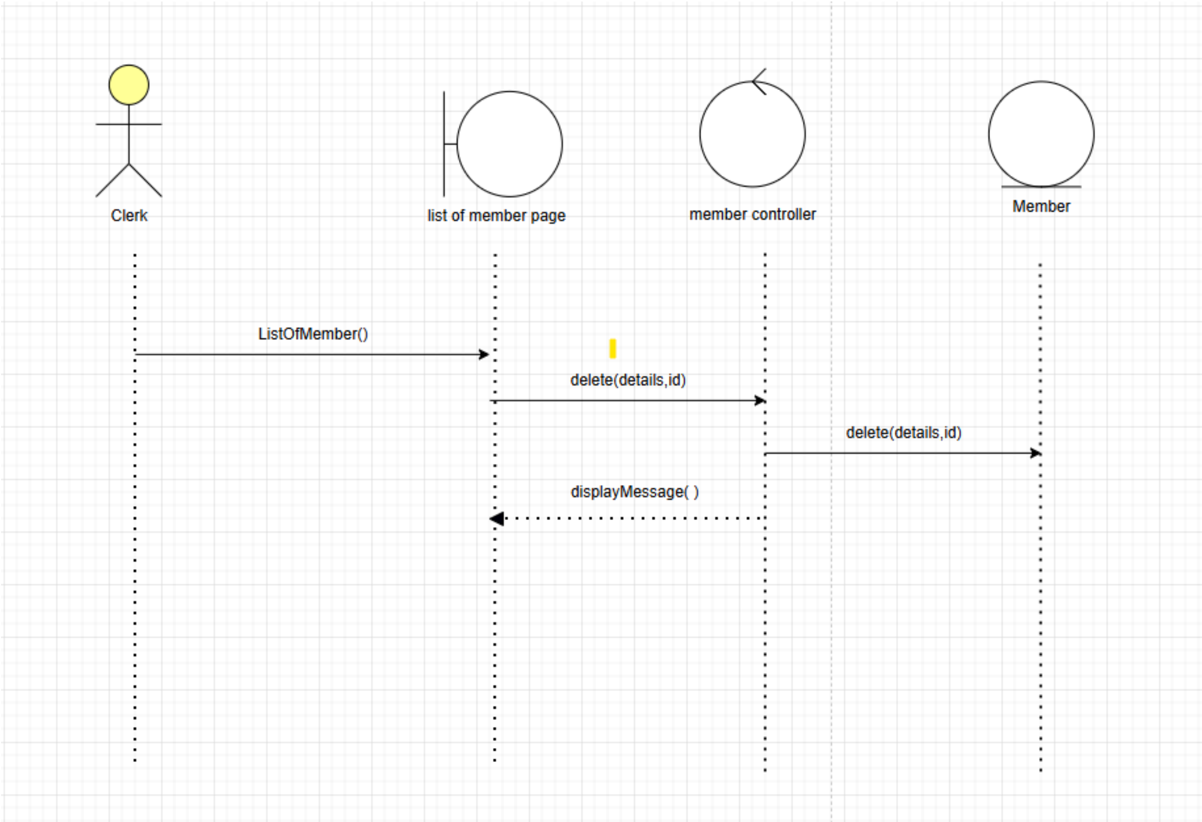


Figure 2.10.3: Sequence Diagram for <manage list of member (delete)>

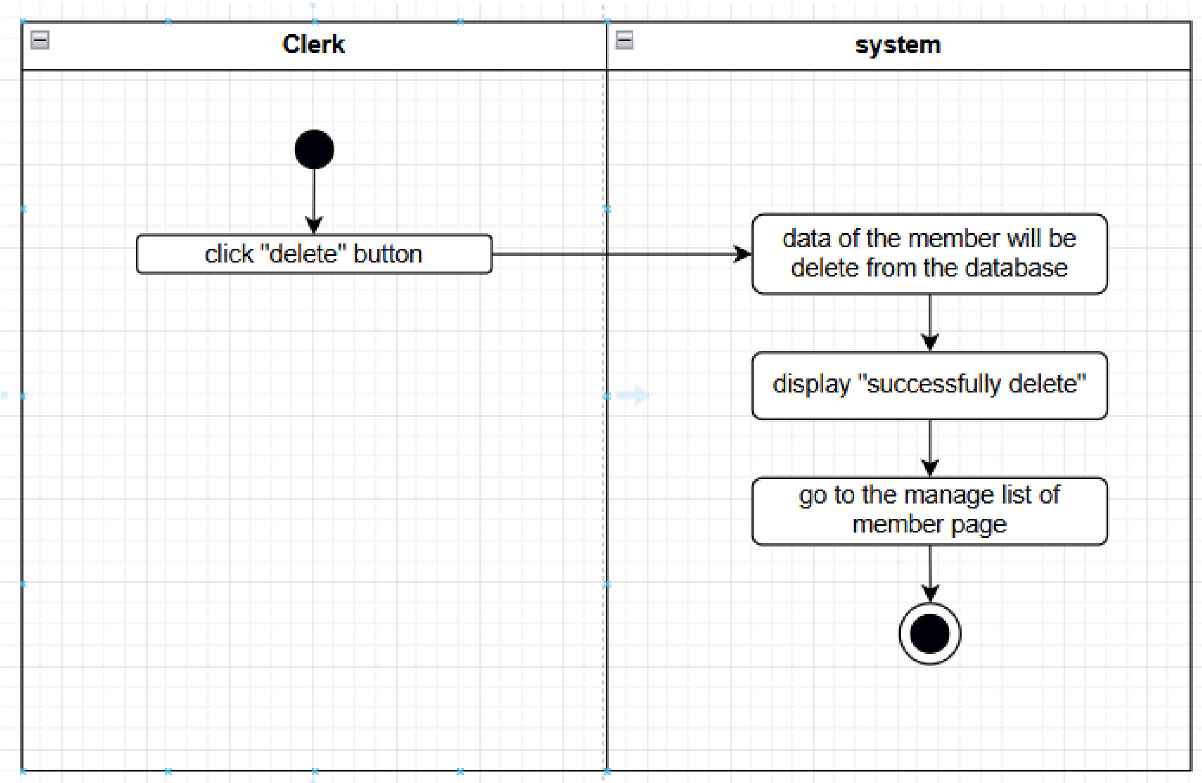


Figure 2.10.4: Activity Diagram for <manage list of member (delete)>

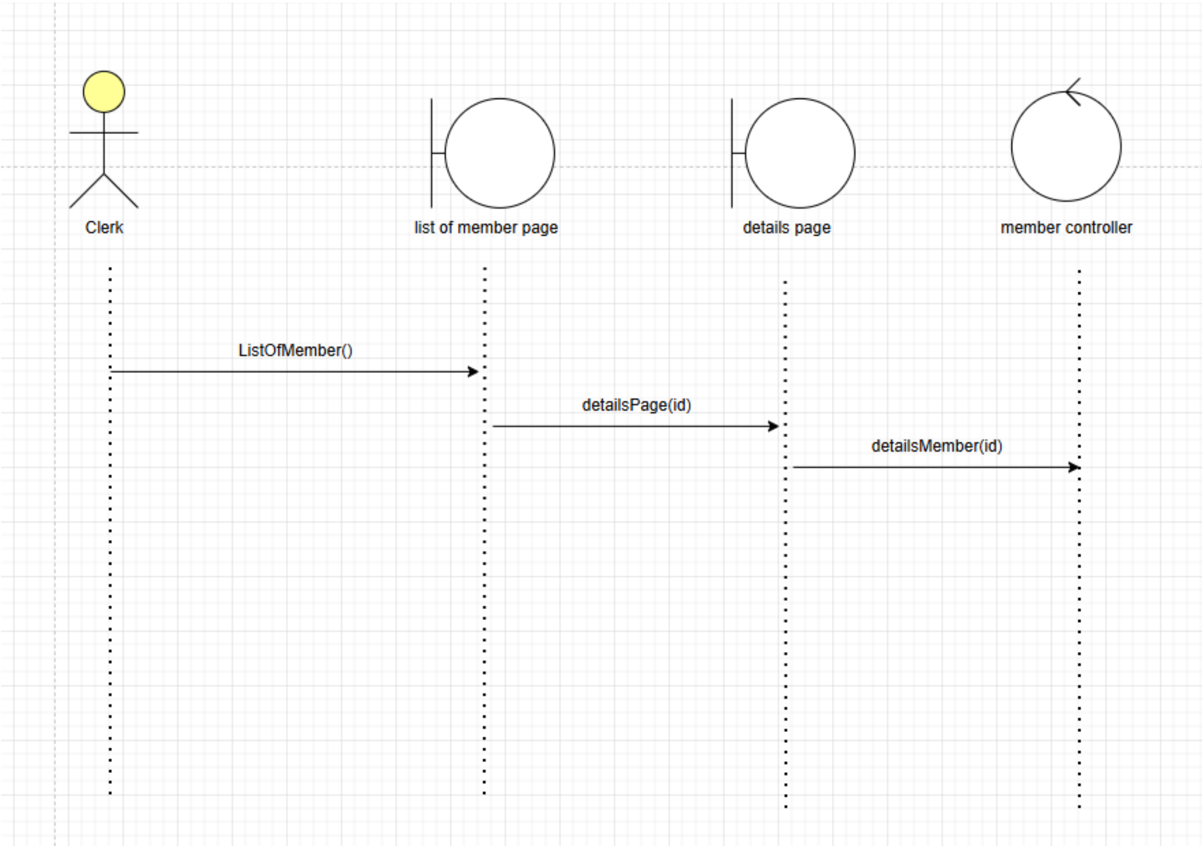


Figure 2.10.5: Sequence Diagram for <manage list of member (view details)>

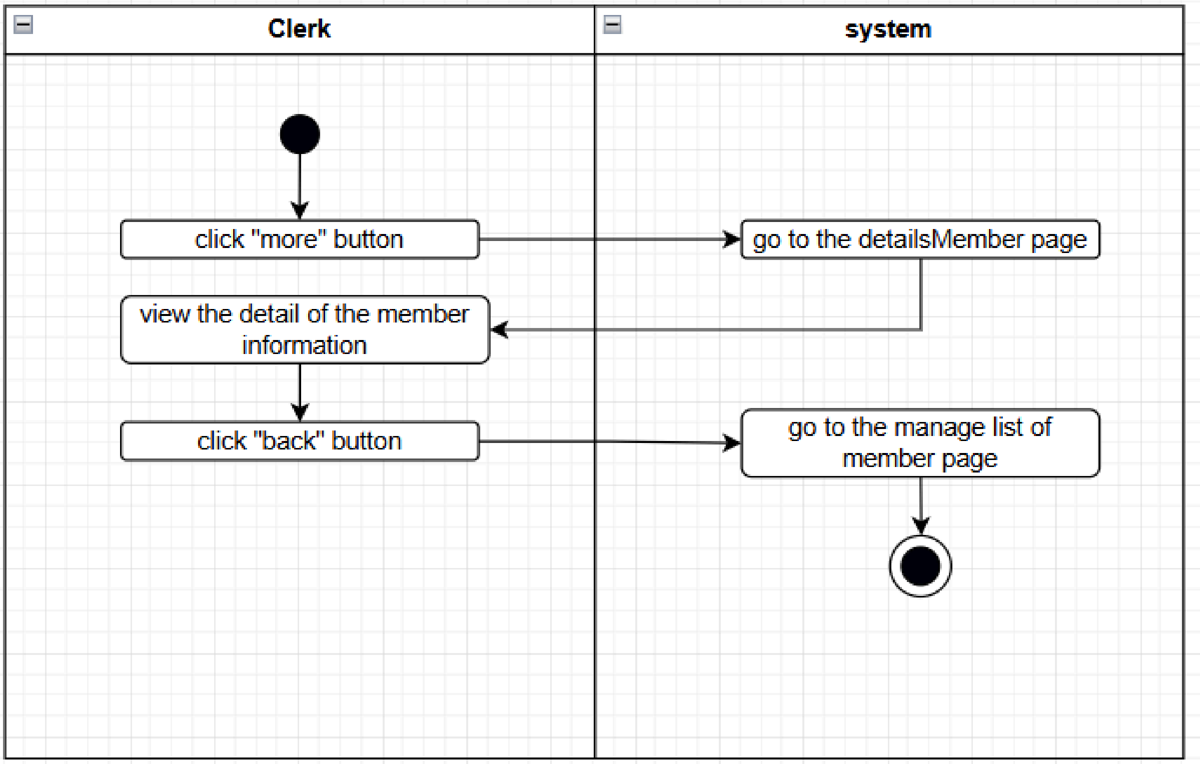


Figure 2.10.6: Activity Diagram for <manage list of member (view details)>

2.4.11 US011: User Story <manage loan application>

Table 2.11: User Story Description for <Manage loan application>

User story: <Manage loan application>
ID: US011
User Story Description **As a** clerk of the KADA cooperative **I want** to be able to manage the loan applicant that been request by the cooperative member **So that** I can check whether the applicant are qualify or not to apply the loan
Flow of events: 1. the clerk will go to the manage loan application 2. the clerk will go through all of the loan applicant details one by one by push the button “more” to see the overall information 3. the clerk than will tick at the button “red” or “green” that indicate whether the loan applicant are qualify or not
Alternative flow *n*:
Acceptance Criteria Postcondition - the clerk must login as a clerk with clerk ID and password in login section precondition - the clerk can monitor overall information of the loan applicant
Exception flow:

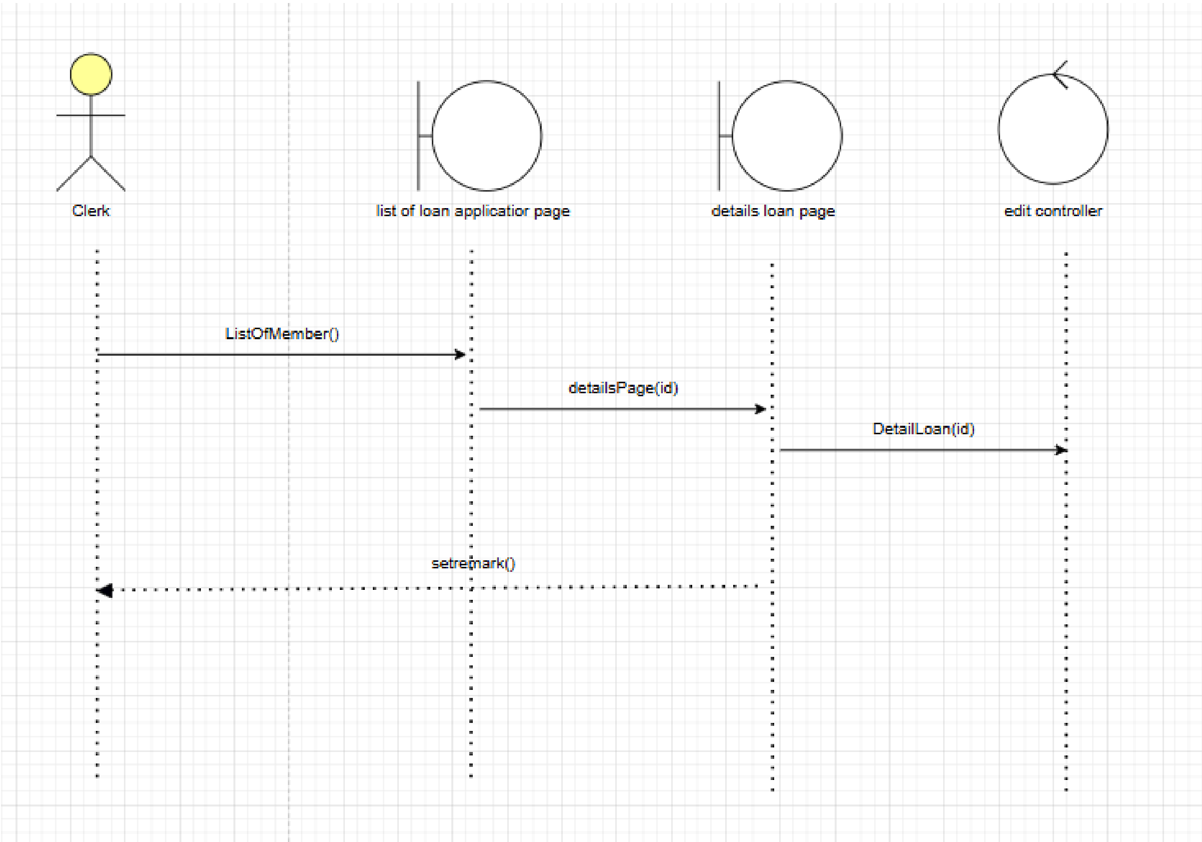


Figure 2.11.1: Sequence Diagram for <manage loan application>

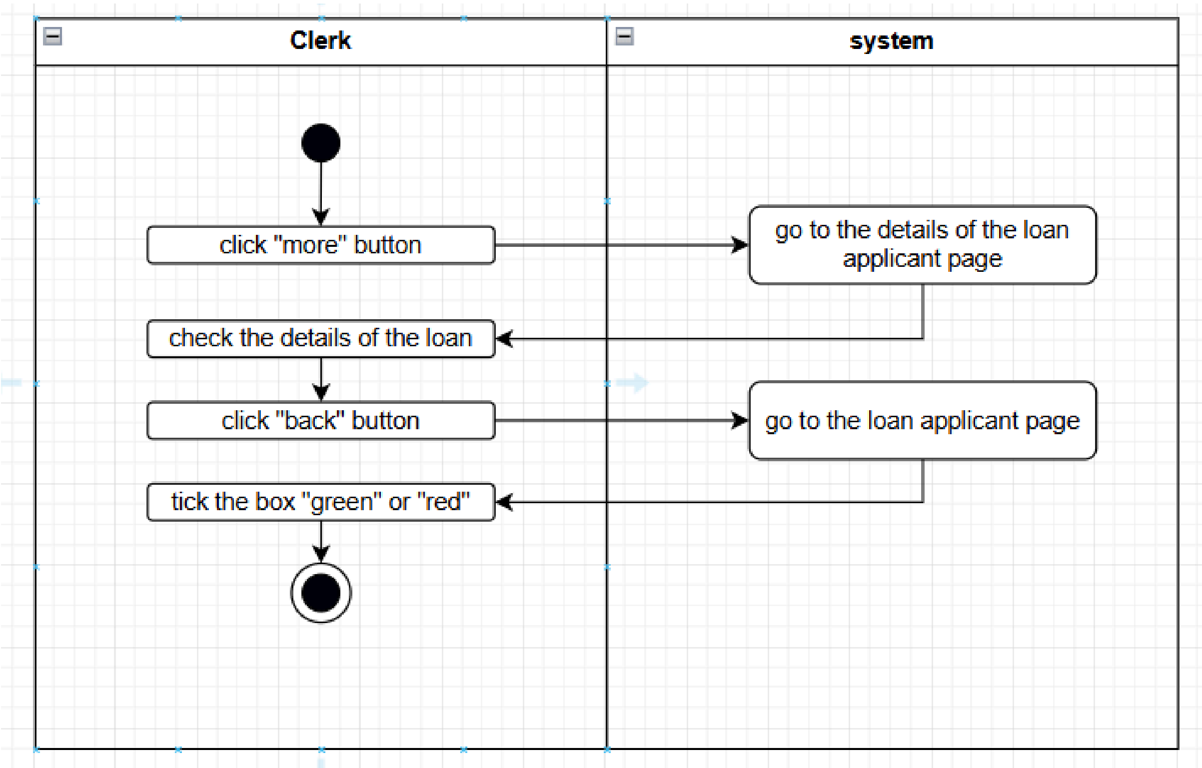


Figure 2.11.2: Activity Diagram for <manage loan application>

2.4.12 US012: User Story < *calculate loan payment*>

Table 2.12: User Story Description for <Calculate Loan Payment>

User story: <Calculate loan payment>
ID: US012
User Story Description **As a** member of KADA **I want** to be able to calculate my loan payment before applying it **So that** I know how much money I will get, the amount of repayment. and the duration of it.
Flow of events: 1. Member log in into the cooperative system 2. Member navigate to the loan calculator section 3. Member enter the loan type, loan amount, and the duration of repayment 4. Member click the "calculate" button 5. The system will calculate the loan and display the total repayment amount and the monthly repayment within the time duration provided. 6. Member reviews the calculated information
Alternative flow *n*:
Acceptance Criteria Postcondition: - Members acknowledge the loan information such as the repayment amount, and the duration of paying back. precondition: - Must be a member of KADA and log in
Exception flow:

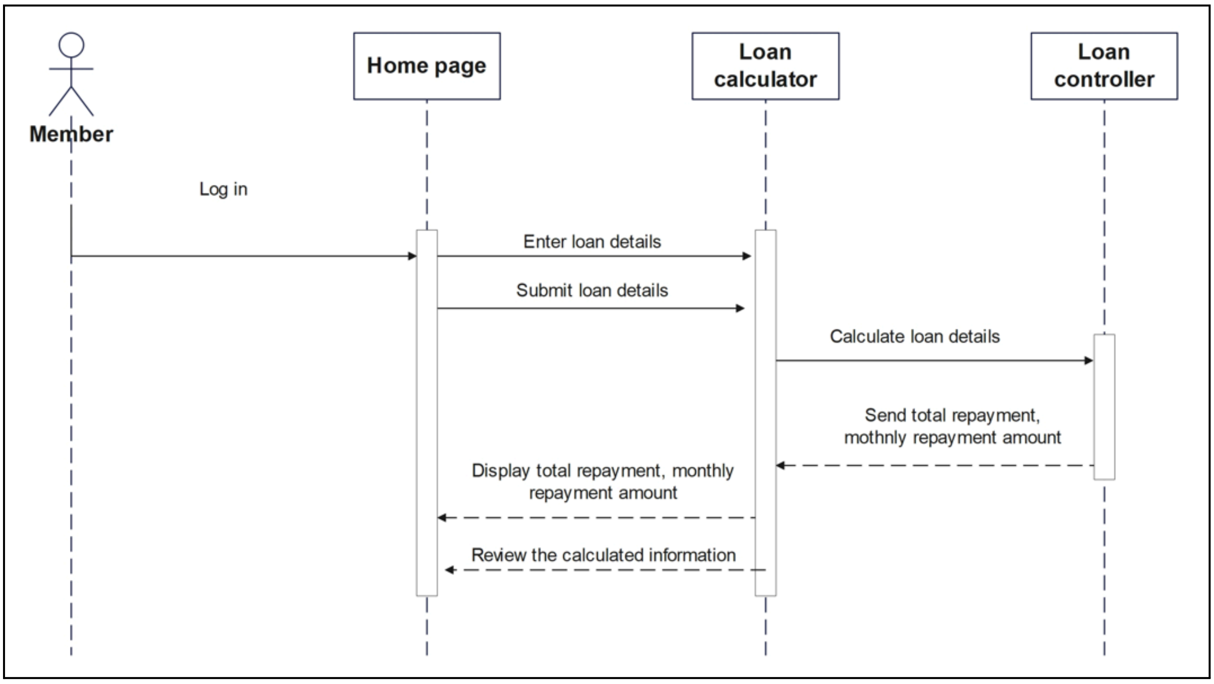


Figure 2.12.1: Sequence Diagram for <Calculate Loan Payment>

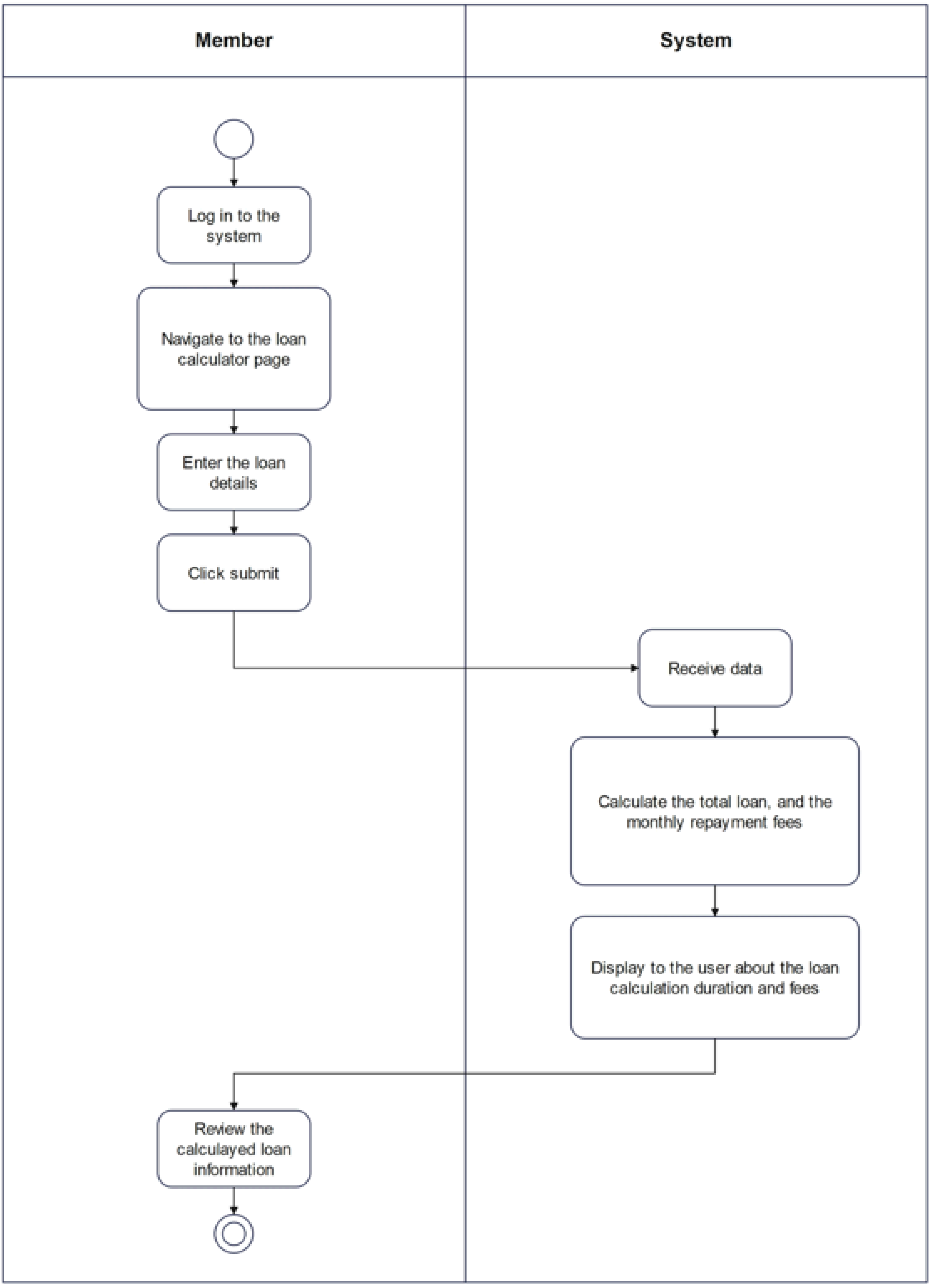


Figure 2.12.2: Activity Diagram for <Calculate Loan Payment>

2.4.13 US013: User Story <*payment*>

Table 2.13: User Story Description for <payment>

User story: <payment>
ID: US013
User Story Description **As a** clerk of KADA **I want** to update payment that the user already make **So that** I the system will have the data about the user payment
Flow of events: 1. clerk login to the cooperative system 2. clerk check the payment for the user in the current month 3. clerk update the payment status of the user
Alternative flow *n*:
Acceptance Criteria Postcondition: - clerk can see the latest payment of the user and the balance of the payment precondition: - Must be a clerk of KADA and log in
Exception flow:

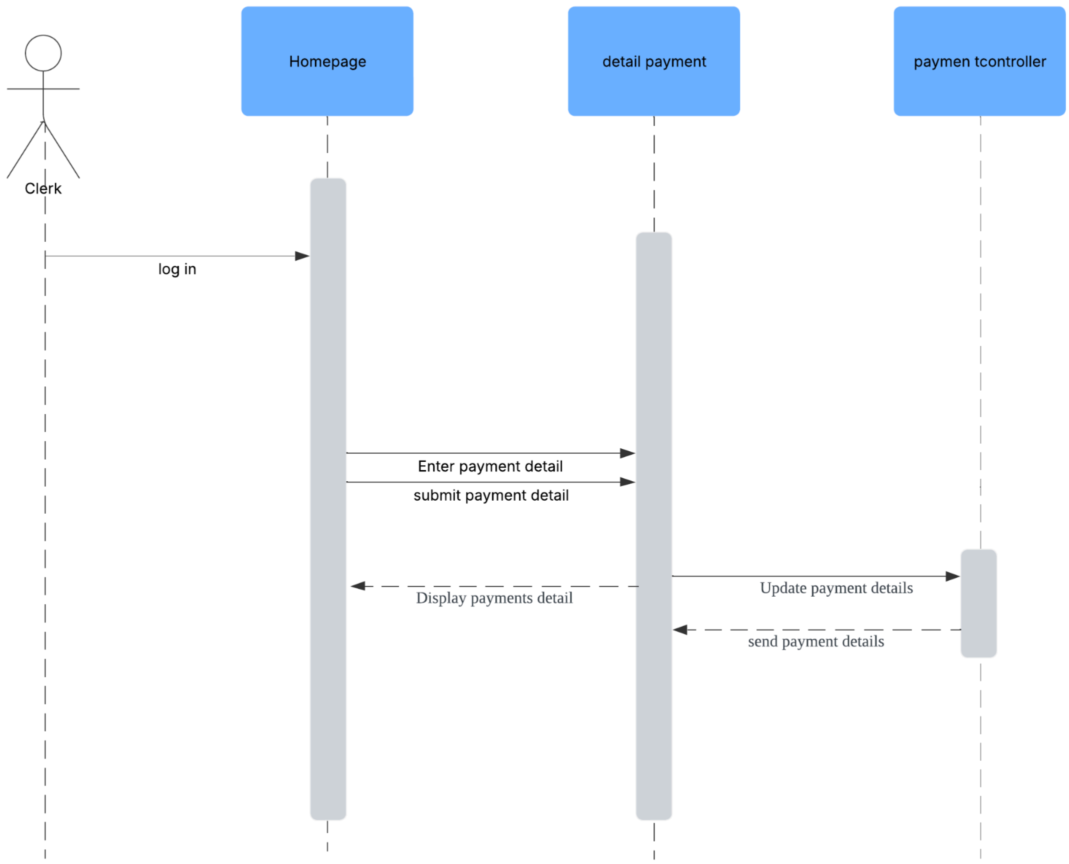


Figure 2.13.1: Sequence Diagram for <Payment>

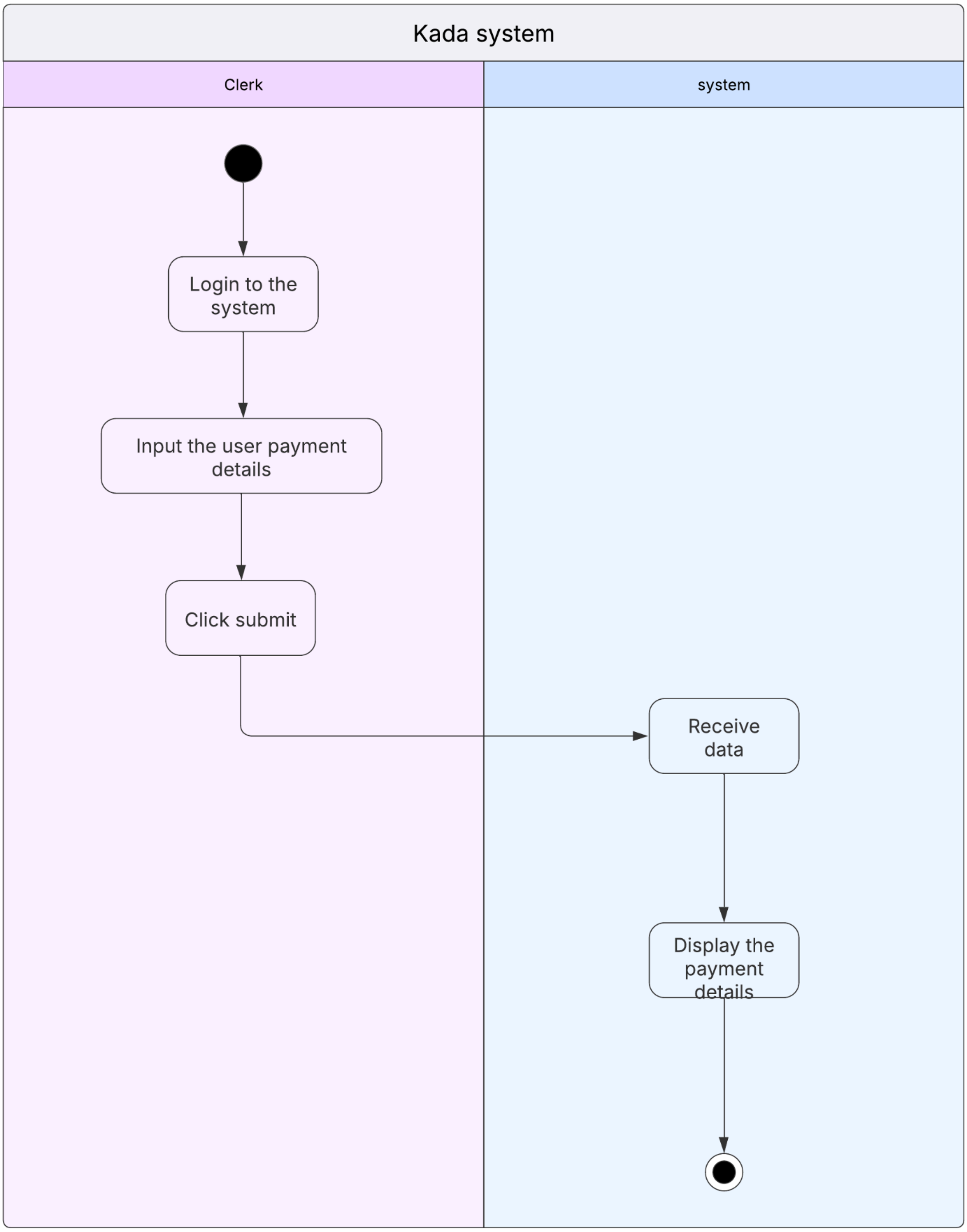


Figure 2.13.2: Activity Diagram for <Payment>

2.2.14 US014: User Story < *interest*>

Table 2.14: User Story Description for <Interest>

User story: <Interest>
ID: US014
User Story Description **As a** Clerk of the cooperative of KADA **I want** to be able to change the interest rate **So that** I can adjust it according to changes throughout the year.
Flow of events: 1. clerk login the into the cooperative system 2. clerk click the button 'faedah' to access the function 3. clerk key in the new interest rate of the company 4. clerk click "submit" to execute the process 5. the rate of interest has been change
Alternative flow *n*:
Acceptance Criteria Postcondition: - the rate of interest for the loan application will be update to the latest rate. precondition: - Must be a clerk of cooperative KADA and log in
Exception flow:

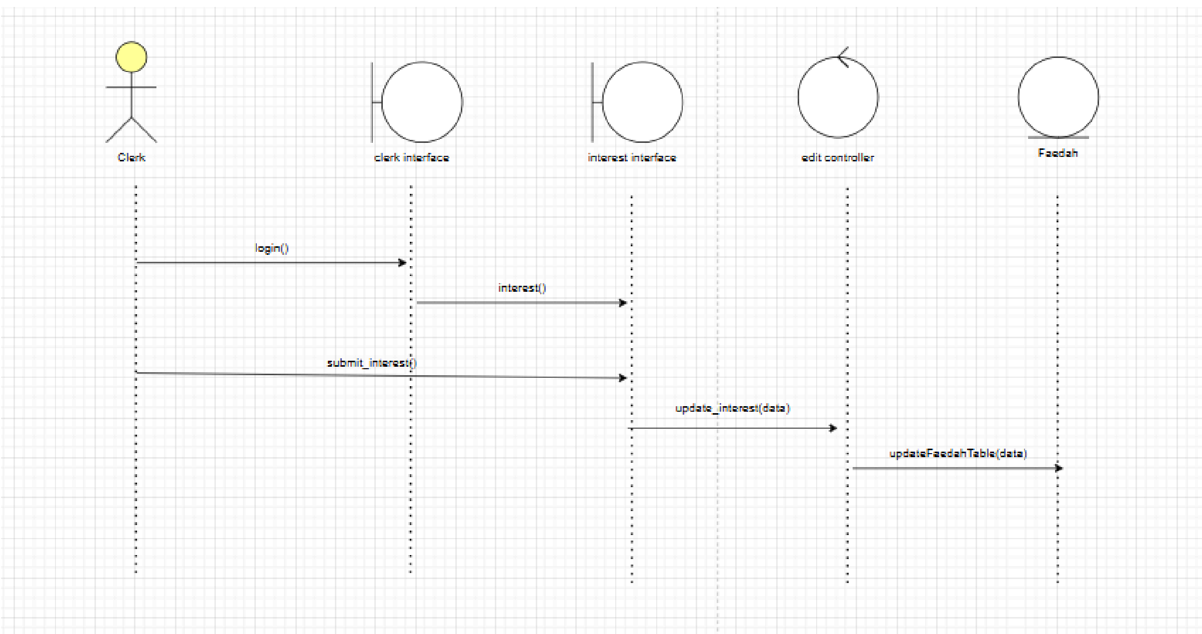


Figure 2.14.1: Sequence Diagram for <Interest>

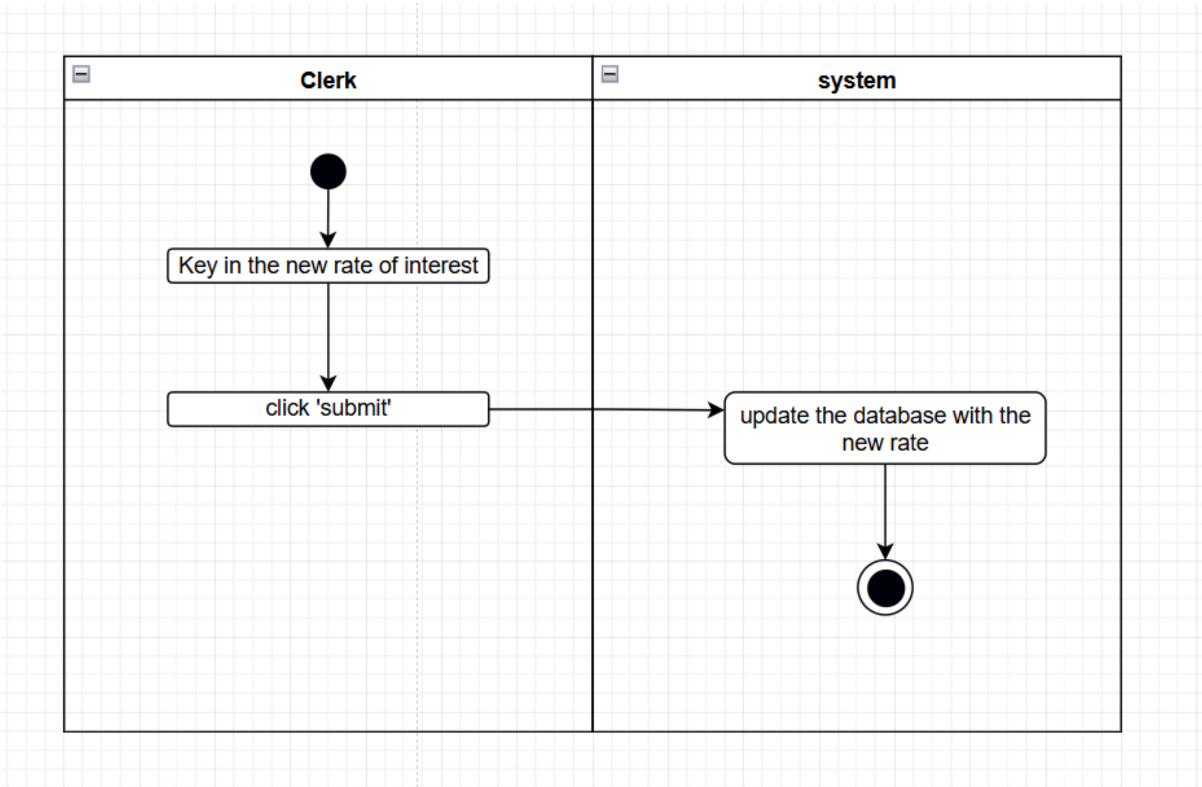


Figure 2.14.2: Activity Diagram for <Interest>

2.4.15 US015: User Story < *Add staff*>

Table 2.15: User Story Description for <Add staff>

User story: <payment>
ID: US013
User Story Description **As a** ALK of KADA **I want** to add new staff to the system **So that** I the system can have new staff that can be login
Flow of events: 1. Alk login to the cooperative system 2. Alk navigate to add staff section 3. Alk input their name, email and password 4. ALK click submit 5. system will add the staff to the database
Alternative flow *n*:
Acceptance Criteria Postcondition: - new added staff can login to the system precondition: - Must be a ALK of KADA and log in
Exception flow:

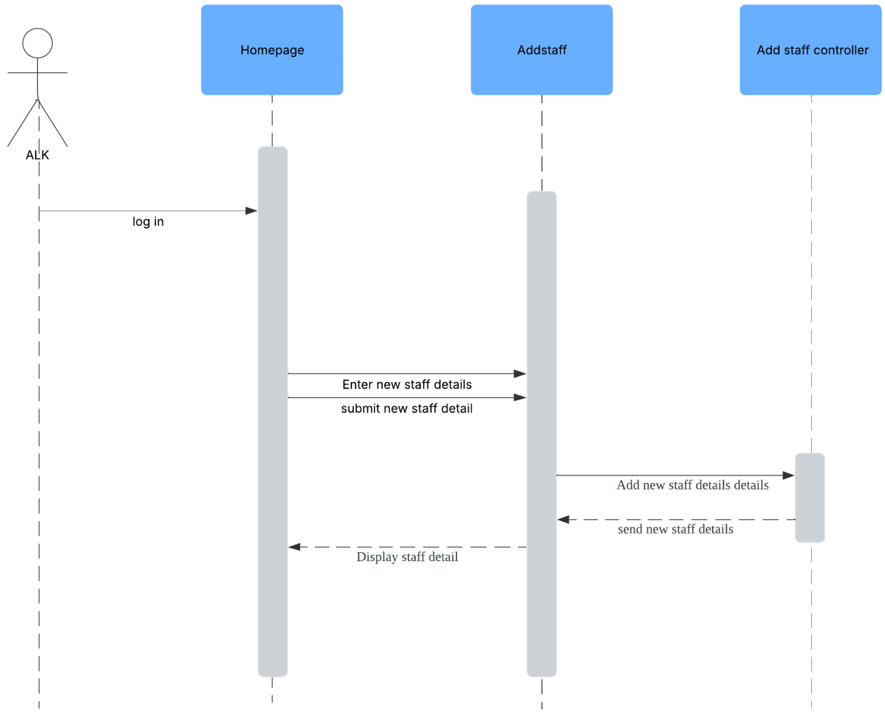


Figure 2.15.1: Sequence Diagram for <Add Staff>

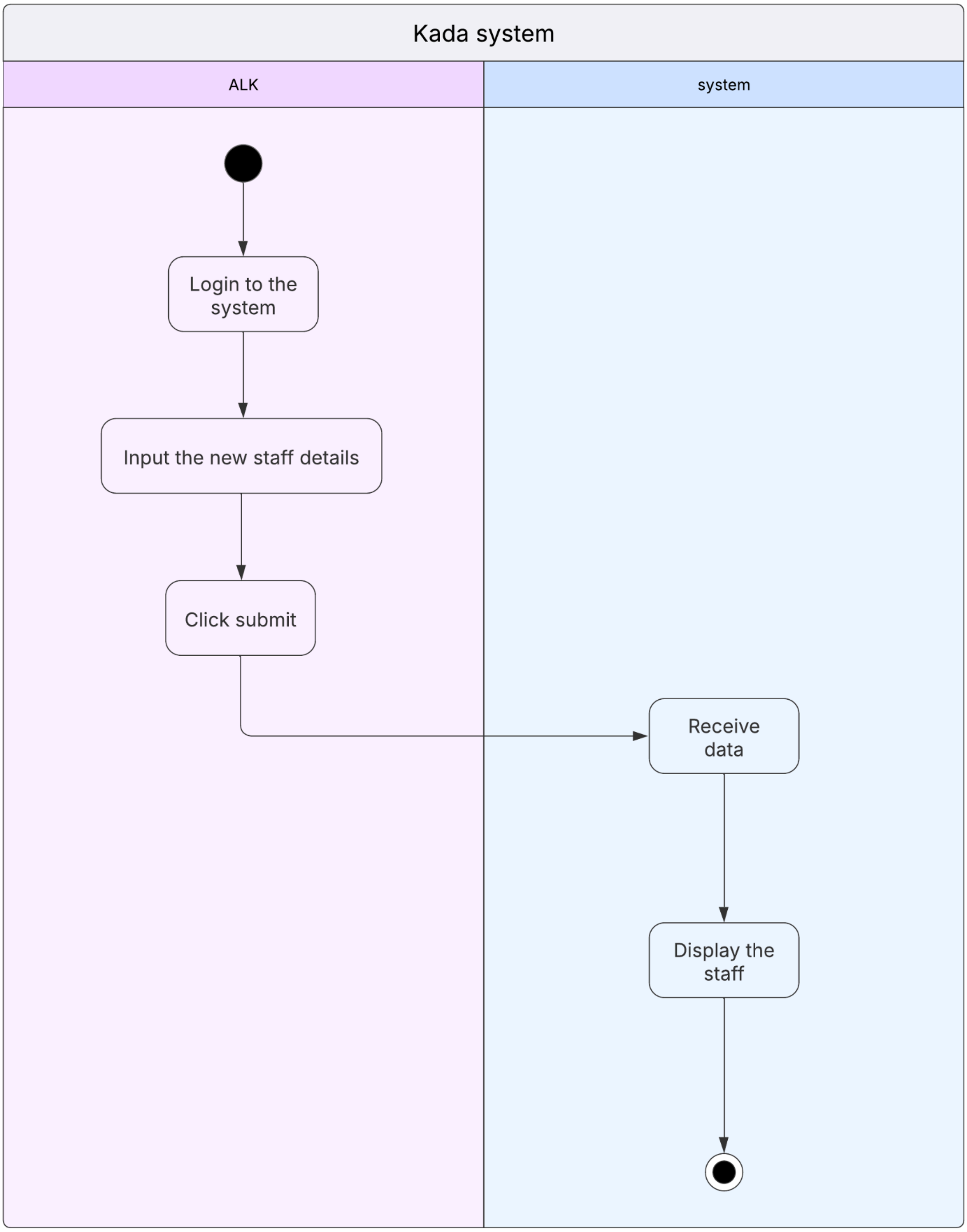


Figure 2.15.2: Activity Diagram for <Add staff>

2.5 Performance and Other Requirements

Performance requirements

Response time:
The system must provide timely responses to user interactions, such as submitting feedback, retrieving reports, and executing tasks. This ensures a seamless and efficient user experience, minimizing any frustration caused by delays.

Throughput:
The system must support a high number of simultaneous user requests, maintaining effective processing speeds and ensuring minimal wait times even during peak usage periods. This capability is critical to accommodate multiple users without performance degradation.

Capacity:
The system must efficiently manage and store a substantial volume of feedback submissions and related data. It should be designed to scale and support the entire university's student population, courses, and associated information without compromising performance.

Availability:
The system should maintain a consistently high uptime rate, ensuring it remains accessible and functional for users. It must minimize downtime and disruptions to provide reliable service at all times.

Other requirements

Security:
The system must implement comprehensive security measures, such as secure user authentication, robust data encryption, and strict access controls. These features are essential to safeguard student data and prevent unauthorized access or potential data breaches.

Safety:
The system must operate in a manner that prioritizes user safety and security, ensuring it does not present any risks or harm to users or the broader university environment. This includes avoiding technical vulnerabilities and ensuring system reliability.

Legal and regulatory:
The system must adhere to all applicable laws, regulations, and standards related to data privacy, confidentiality, and information security. Compliance ensures legal integrity and builds trust among users and stakeholders.

Maintainability:
The system is designed to be easily modified or enhanced due to the implementation of good coding practices. This includes thorough comments and the use of meaningful, descriptive variable names, ensuring maintainability and readability for developers.

Usability:
The system shall provide an intuitive, well-organized, and user-friendly interface to facilitate ease of use, enabling users to navigate and perform tasks without confusion.

2.6 Design Constraints

1. Environmental Constraints

The system needs to function well in environments with varying temperatures, ranging from freezing cold at 0°C to warm conditions of up to 40°C. This ensures it remains reliable even in locations without temperature-controlled server rooms. Additionally, the interface should be clear and usable whether users are in dimly lit areas or working under bright sunlight.

2. Hardware Constraints

The software should be lightweight enough to run on devices with basic specifications, like 4GB of RAM and dual-core processors, making it accessible to a wider range of users. For data storage, the database must have a minimum capacity of 1TB to accommodate the cooperative's expanding records, including member data and financial transactions.

3. Security Constraints

To protect sensitive information, such as members' personal and financial details, the system must use advanced encryption standards like AES-256. For added safety, key users like ALK members and clerks should use multi-factor authentication when accessing the platform. Moreover, the system should comply with data privacy regulations, including GDPR and Malaysia's Personal Data Protection Act (PDPA).

4. Compatibility Constraints

The platform should work seamlessly across popular web browsers like Chrome, Firefox, and Edge. It also needs to support both iOS and Android devices, ensuring accessibility for users on mobile. Integration with existing tools used by KADA and third-party services like email or SMS notification systems is essential to streamline operations.

5. Performance Constraints

The system should handle up to 500 users online at the same time, with responses delivered within 2 seconds on average. For bulk processing, such as reviewing a thousand loan applications at once, the process shouldn't take more than 10 minutes. Finally, to meet the cooperative's operational needs, the system must be highly reliable, maintaining an uptime of 99.9%.

3. Architectural Rationale

3.1 Architecture Style and Rationale

The architectural style chosen for this project is MVC (Model-View-Controller). This architecture is widely used in web and software due to its clear separation of concerns, hence making the system more maintainable and scalable.

Rationale for choosing MVC:

Separation of Concerns:
MVC divides the application into three distinct parts—Model, View, and Controller—each with a specific role: Model handles the data and business logic, View takes care of presenting the data to the user and Controller acts as the intermediary, processing user input, interacting with the Model, and updating the View accordingly.This separation allows each component to be developed, tested, and maintained independently, improving flexibility and reducing complexity.

Scalability and Maintainability:
Since the logic and the user interface are separated, changes to one component don't affect the other. For example, if the user interface needs to be updated, the Model and Controller remain unaffected. This makes the system easier to update and scale as requirements evolve.

Ease of Testing and Debugging:
The modularity of MVC enables each component to be tested in isolation. This approach simplifies debugging and quality assurance, making it easier to identify and resolve issues efficiently.

4. Architectural Views

This section presents and explains the architectural views of the KADA Cooperative System. Figure 4.1 illustrates these views, which include the scenario view, logical view, process view, development view, and physical view. Each view represents a specific stakeholder's perspective, a different stage in the system development life cycle, the concerns associated with each stage, and the types of diagrams and artifacts used in the system.

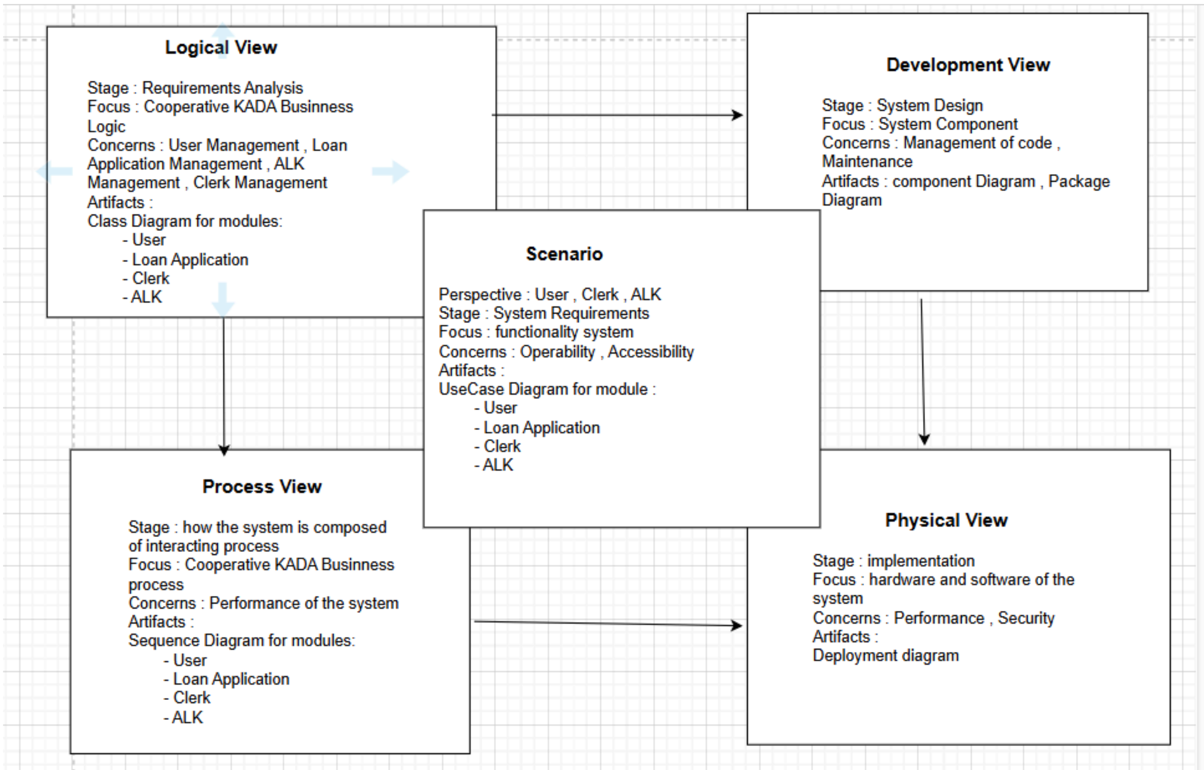


Figure 4,1 – Architectural Views

Table 4.1 Descriptions and Diagrams Used for Each Architectural View

Architectural Views	Description	Diagrams used
Scenario View	The scenario view illustrates how the user, clerk, and ALK, as users of the KADA Cooperative System, interact with and utilize the system's functions.	Use Case Diagram
Logical View	This view displays the attributes and methods required for each module in the KADA Cooperative System, as well as their relationships with one another.	Class Diagram
Process	The process view deals with the dynamic aspects of the KADA Cooperative system, explains the system processes and how they communicate, and focuses on the run time behavior of the system.	Sequence Diagram

Development (or Implementation) View	shows how the software is decomposed for development	Component Diagram Package Diagram
Physical (or Deployment) View	shows the system hardware and how software components are distributed across the processors in the KADA Cooperative system	Deployment DIagram

4.1 Use Case View

Figure 4.2 provides an illustration of the use case view, which outlines all the entities within the KADA Cooperative System and how they interact with various use cases. It highlights the relationships between each entity and their corresponding use cases, offering a clear representation of the system's structure.

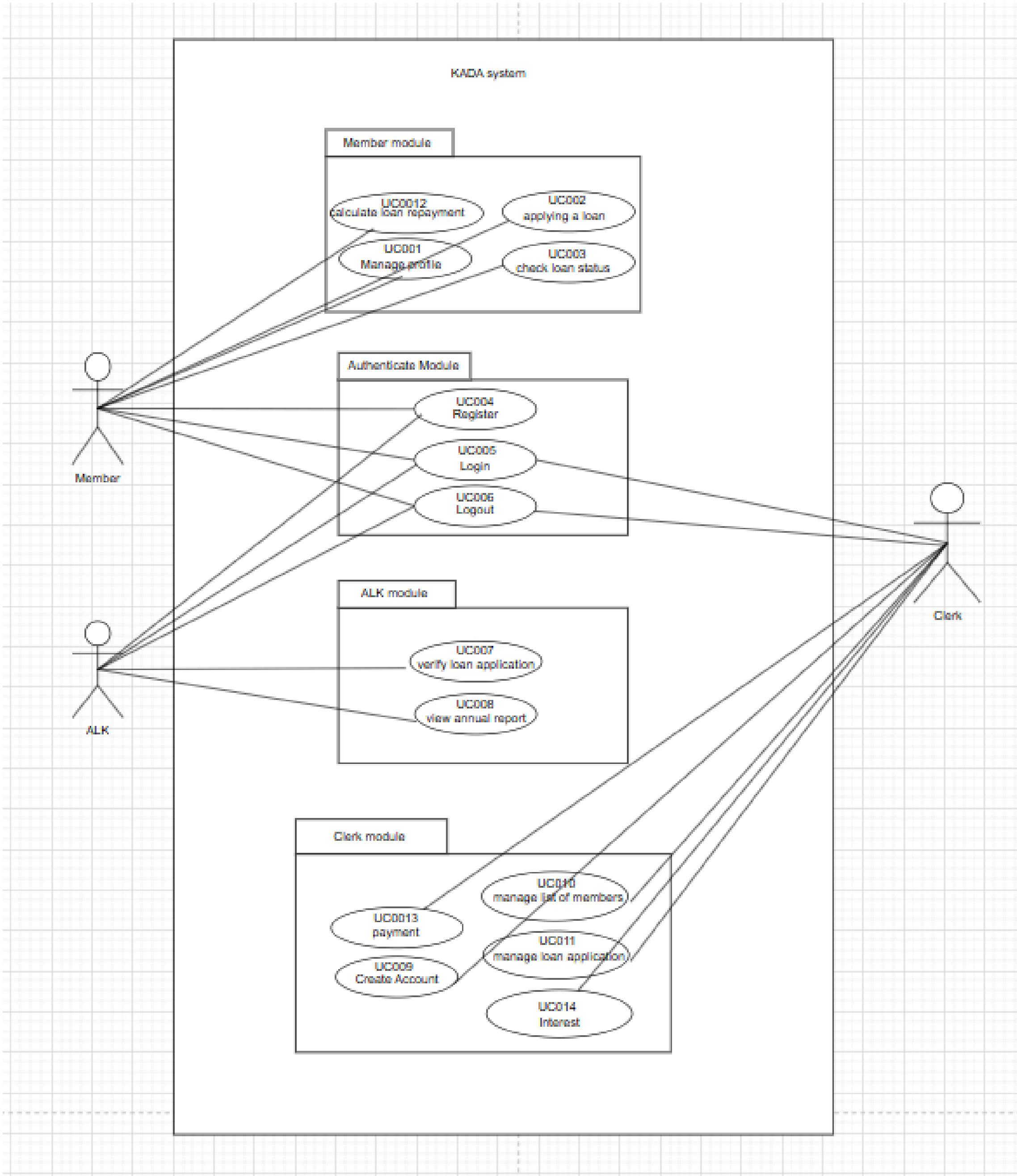


Figure 4.2: Use Case Diagram for <KADA system>

4.2 Implementation View

Figure 4.3 shows the implementation view of the KADA system offers a well-organized structure of software modules within the development environment. It outlines how the system is divided into high-level subsystems that work together to deliver the system's functionality. With a three-layer architecture, the system ensures modularity, scalability, and maintainability while following the MVC (Model-View-Controller) pattern. Each subsystem has specific responsibilities that contribute to the system's overall performance and efficiency.

The User Interface Layer serves as the front-end, enabling users to interact with the system. This layer consists of web pages built with HTML and Bootstrap, allowing users to enter data through forms like login credentials or task entries. Its main role is to display information and pass user input to the backend for processing. It sends requests to the Domain Layer, which handles business logic and processes user actions.

The Domain Layer, also known as the Business Logic Layer, is the heart of the system. It manages key functions such as authentication, user management, task creation, and event handling. This layer applies business rules to ensure that operations are carried out correctly before interacting with the database. Once user input is received from the UI Layer, the Domain Layer validates the data, applies the necessary logic, and communicates with the Data Access Layer to fetch or store information.

The Data Access Layer manages database operations, ensuring secure and efficient data handling. It stores and retrieves system data, including user accounts, task details, and progress tracking. The database ensures data consistency and security, providing structured access to the data. The Domain Layer interacts with the Data Access Layer whenever data needs to be stored or retrieved, ensuring smooth system operation.

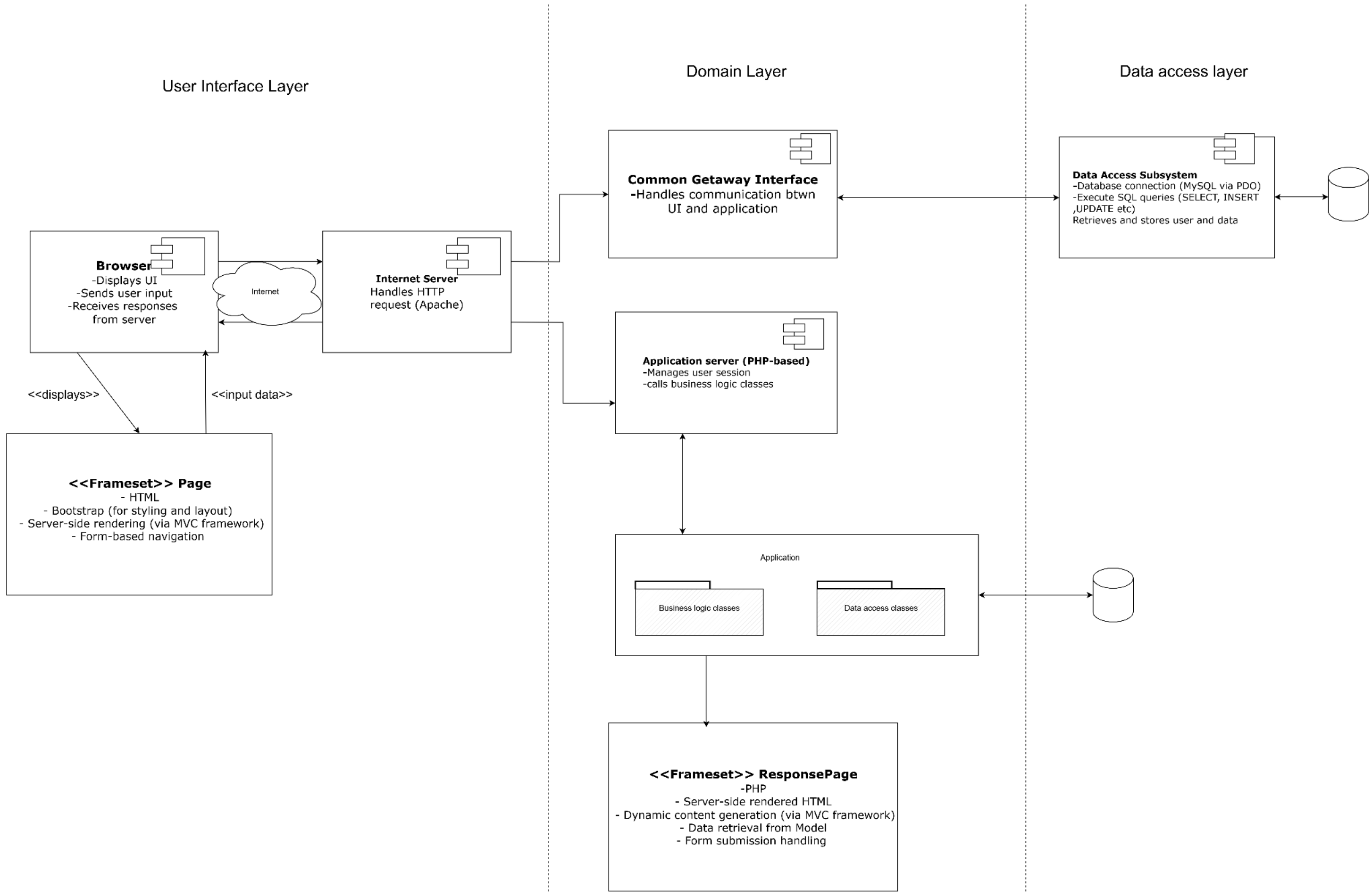


Figure 4.2: Component Diagram of <KADA Cooperative System>

Figure 4.3 shows the overall package diagram represents the architectural structure of the system, following the Model-View-Controller (MVC) pattern. It is divided into three main layers:

1. User Interface / Presentation Layer – Handles user interactions and displays relevant information through different pages.
2. Domain / Application Layer – Acts as the bridge between the user interface and data layers, processing business logic and user requests.
3. Data Layer – Manages data storage and retrieval, ensuring consistency and integrity across the system.

By organizing the system into distinct packages, the diagram ensures separation of concerns, making development, testing, and maintenance more efficient. This modular structure allows for scalability and flexibility, enabling updates to individual components without affecting the entire system.

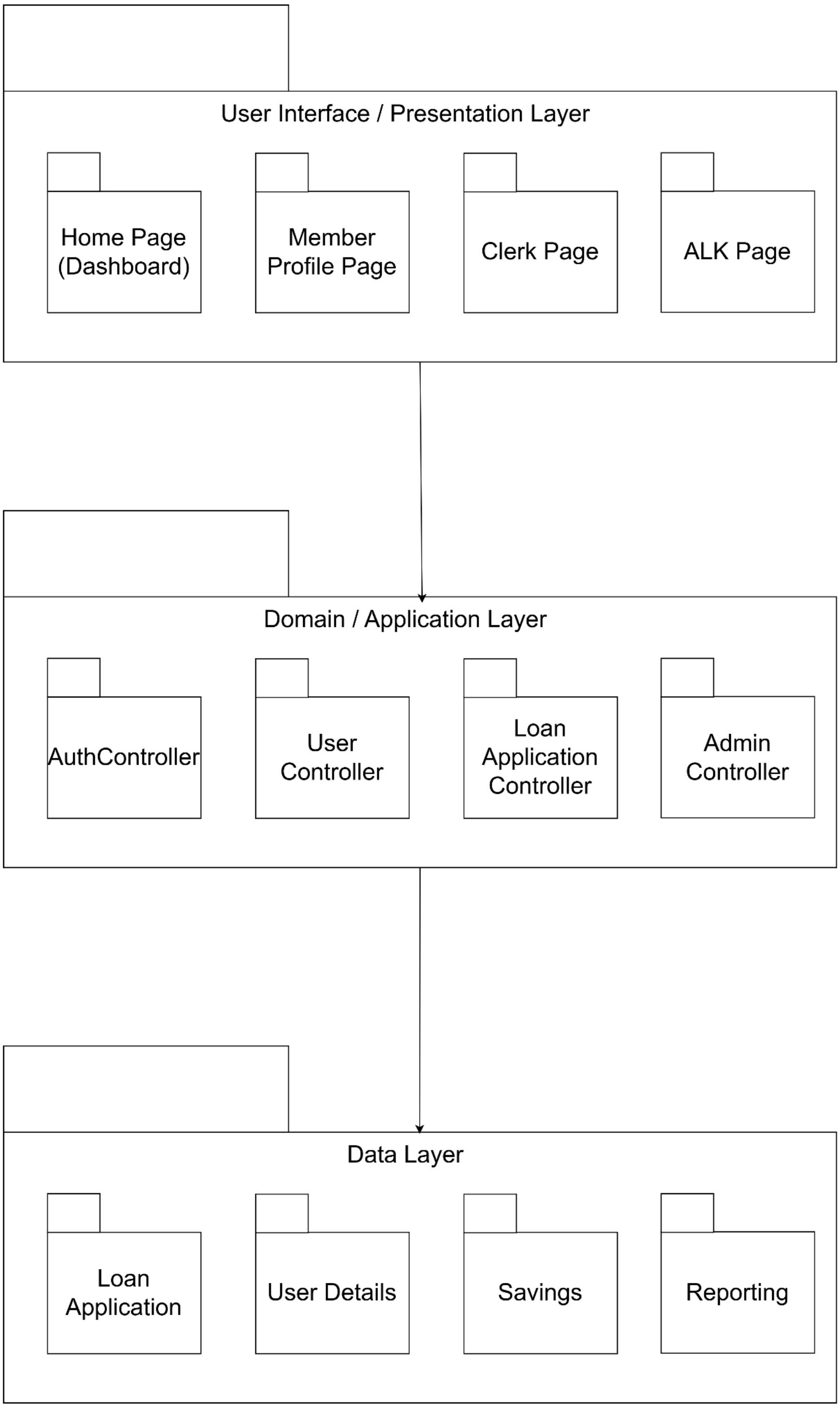


Figure 4.3 Overall Package Diagram of KADA Cooperative System

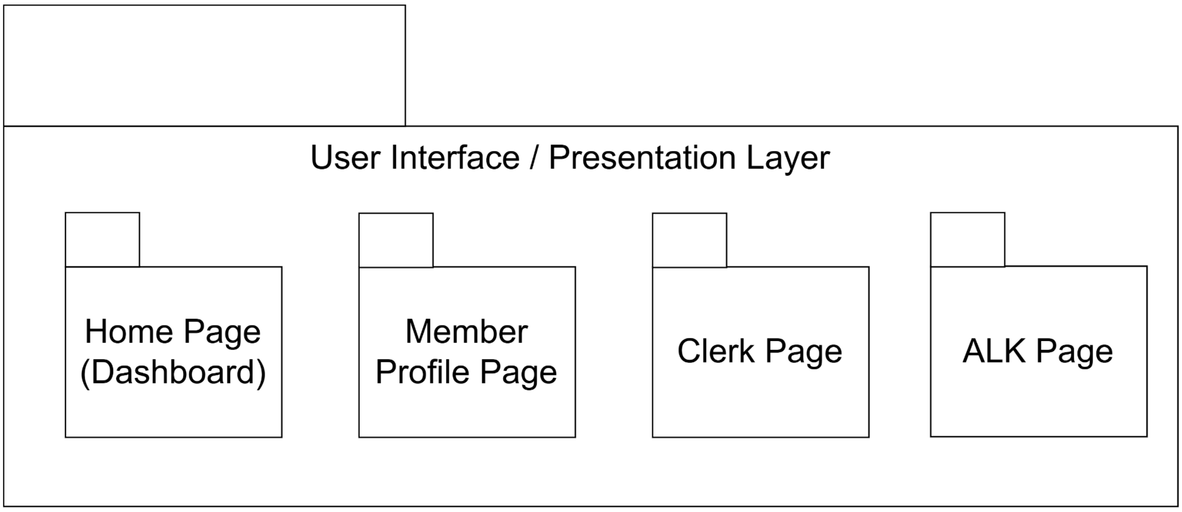


Figure 4.4: Package Diagram for <User Interface Layer (P001)> Subsystem
This layer is responsible for displaying the user interface and handling user interactions. It includes various pages that different users (members, clerks, ALKs) interact with.

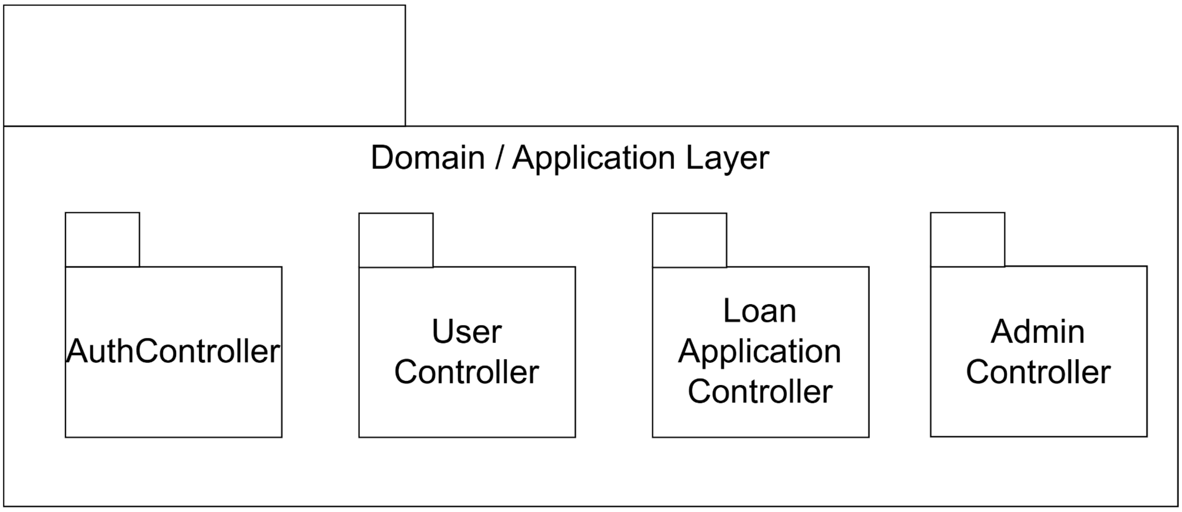


Figure 4.5: Package Diagram for <Domain Layer (P002)> Subsystem
This layer acts as the intermediary between the UI and the data layer. It processes user inputs, applies business logic, and directs requests to the appropriate model.

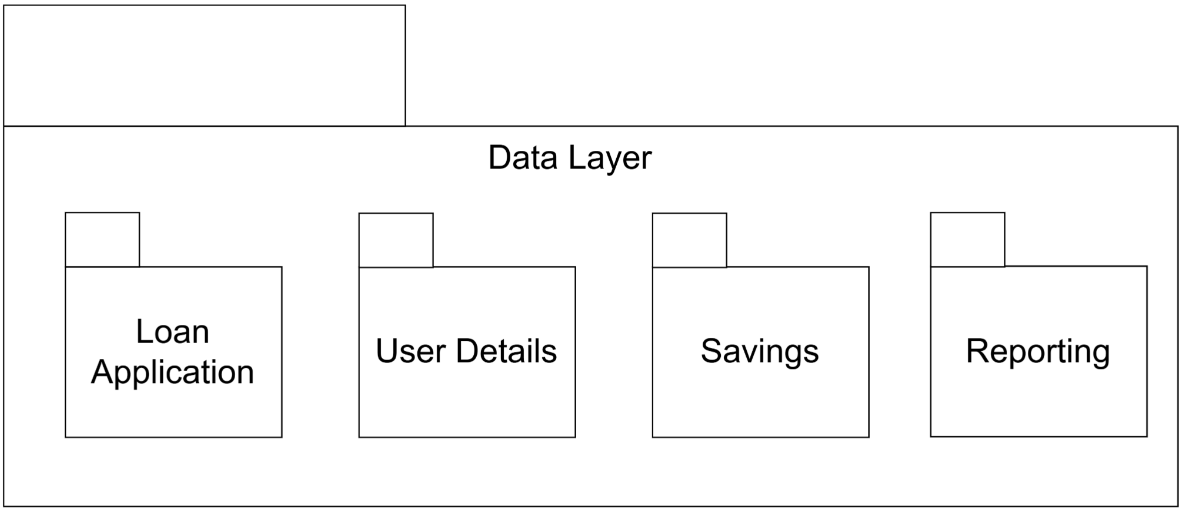


Figure 4.6: Package Diagram for <Data Layer (P003)> Subsystem

This layer is responsible for storing, retrieving, and managing system data. It ensures data persistence and integrity

4.3 Logical View

module user

This module focuses on managing user interactions within the cooperative system. It simplifies tasks like registering accounts, logging in, updating profiles, and securely logging out. By streamlining these operations, it ensures that members can access and manage their accounts with ease.

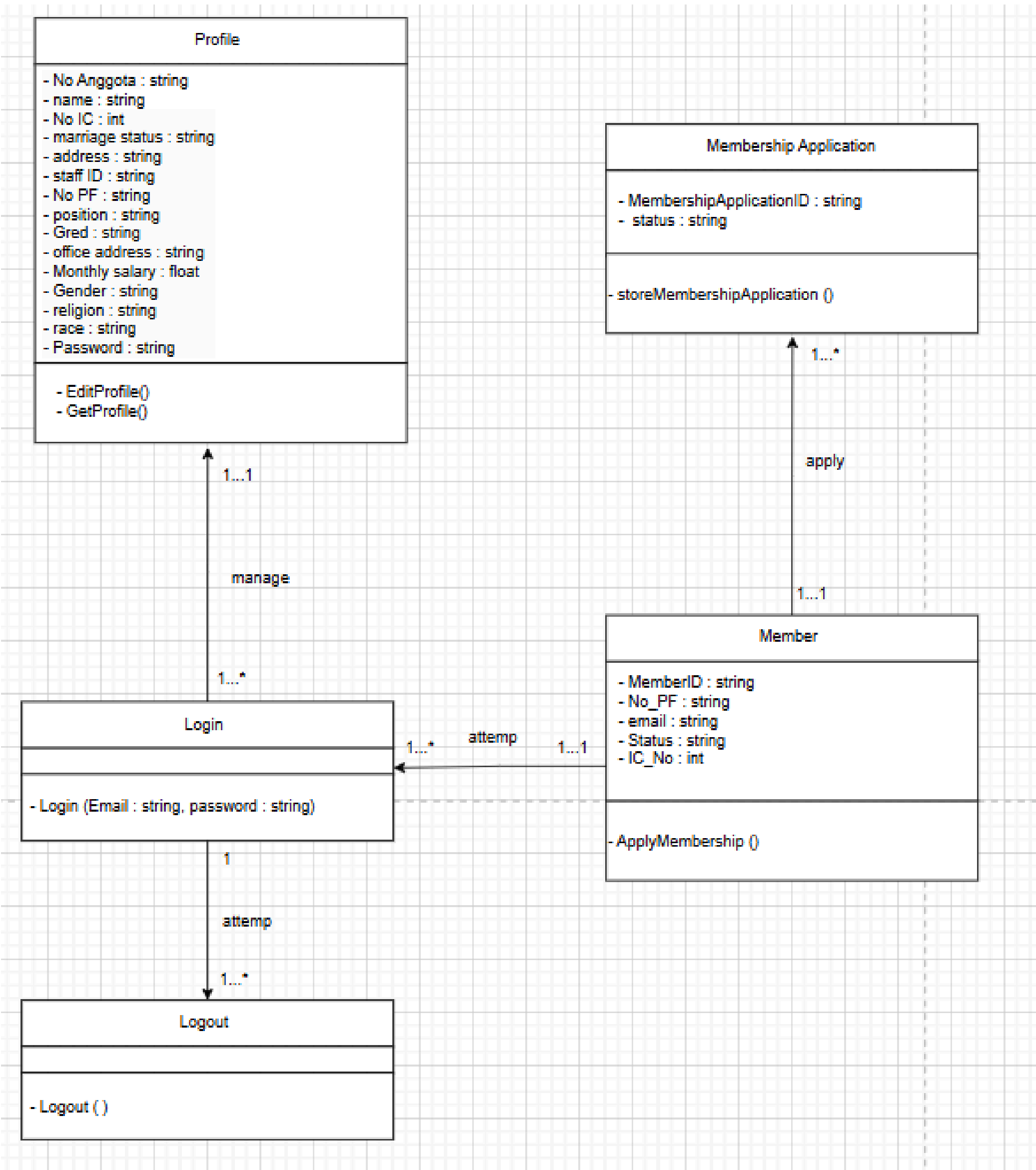


Figure 4.3.1: Class diagram for <User> Subsystem

module loan Application

This module will involve members of KADA only. There are 3 use case inside this module: UC002: applying a loan, UC003: check loan status, and UC0012: calculate loan repayment

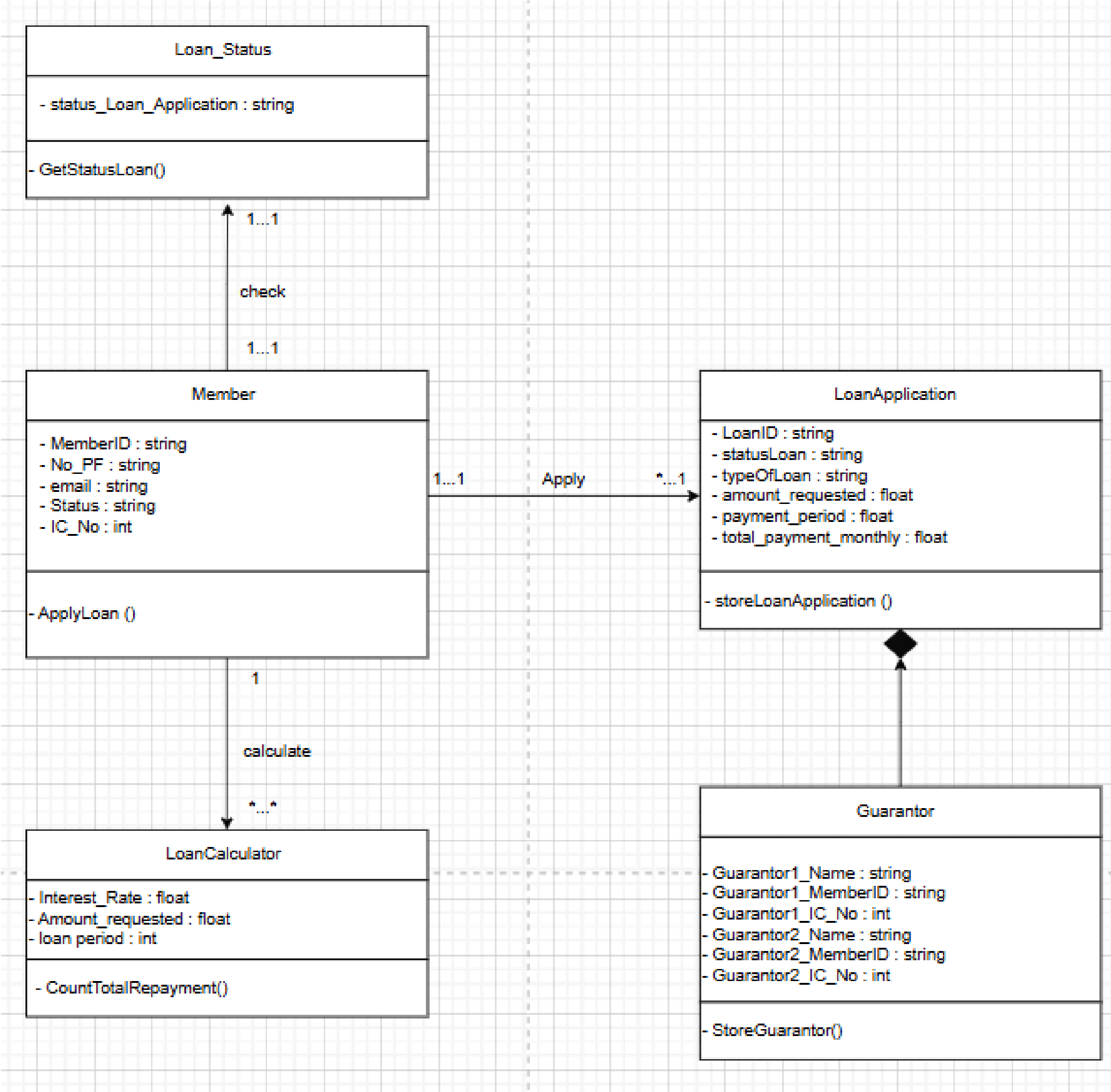


Figure 4.3.2: Class diagram for <loan Application> Subsystem

Module Clerk

This module is dedicated to the clerk's responsibilities within KADA. It includes four key use cases: creating an account (UC009), managing the list of members (UC010), managing loan applications (UC011) and payment (UC013) and interest (UC014).

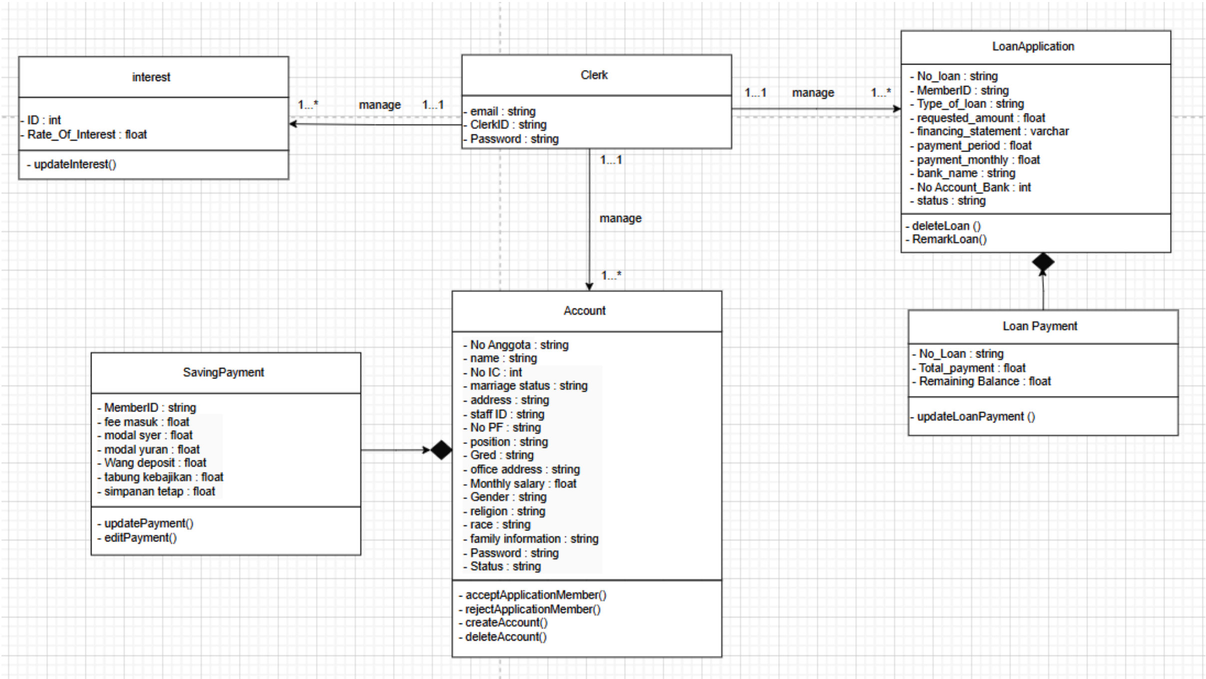


Figure 4.3.3: Class diagram for <Clerk> Subsystem

Module ALK

This module is responsible for ALK's duties within the KADA system, specifically focusing on verifying loan applications and viewing annual reports. It encompasses two key use cases: verifying loan applications (UC007) and viewing the annual report (UC008).

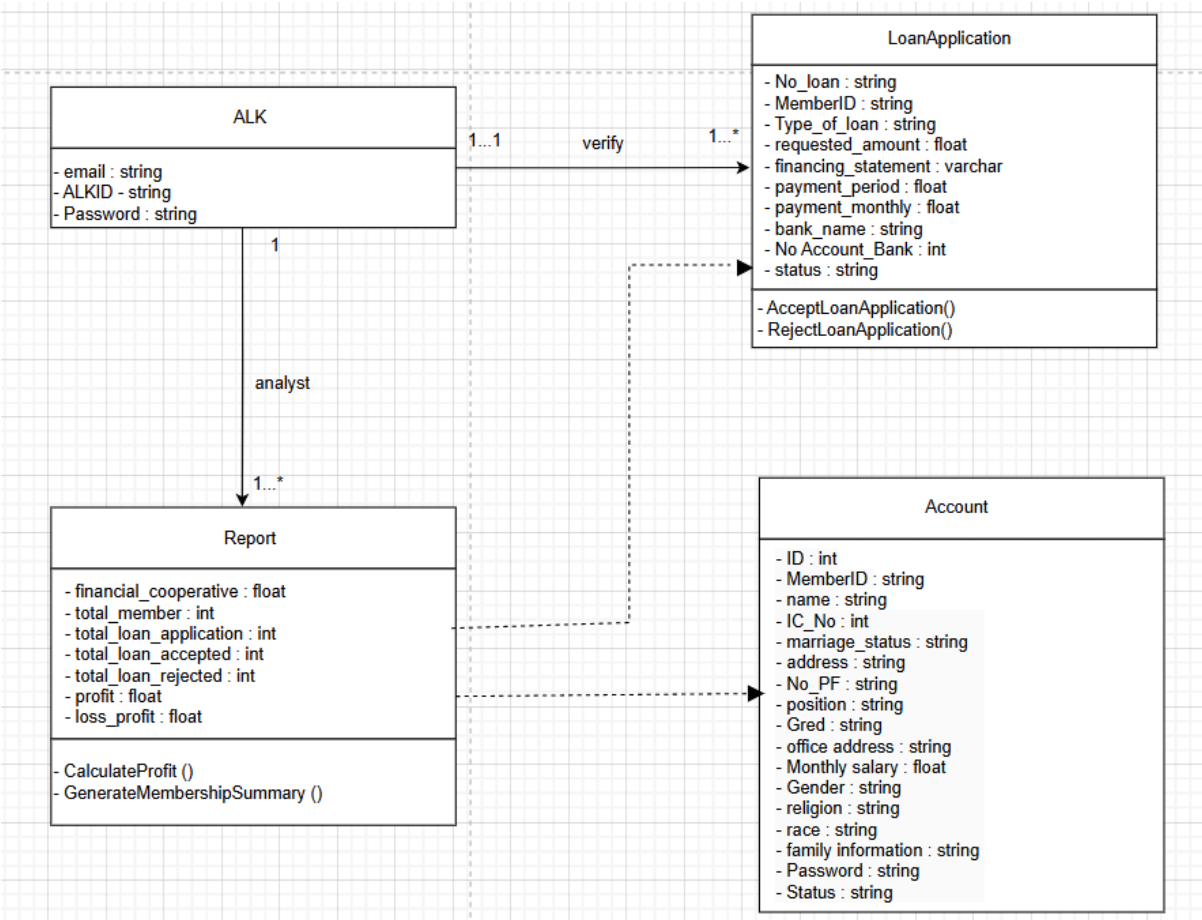


Figure 4.3.4: Class diagram for <ALK> Subsystem

4.4 Process View

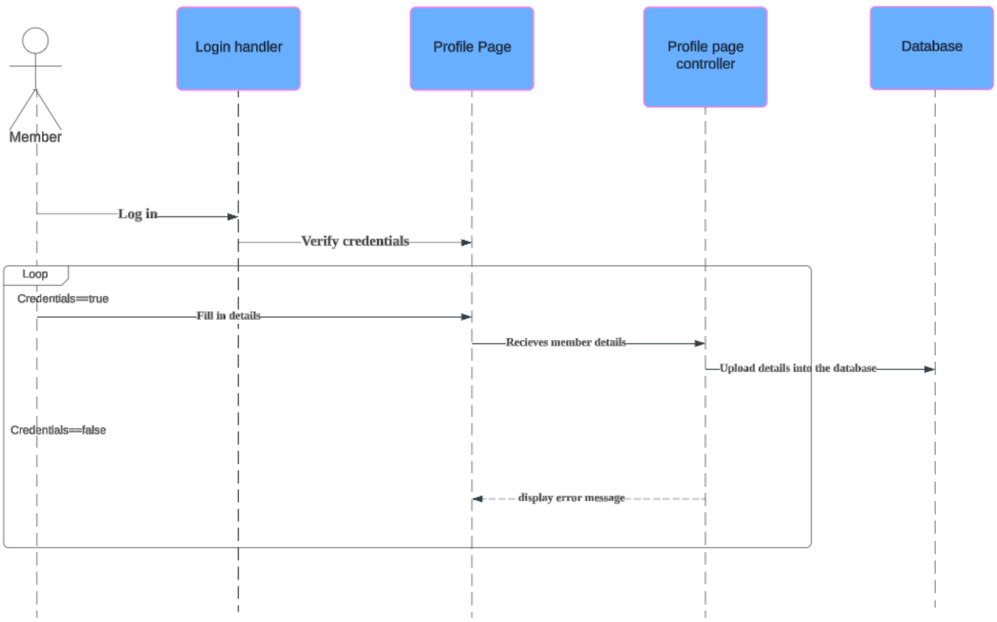


Figure 4.10: SD001 Sequence Diagram for <Manage Profile>

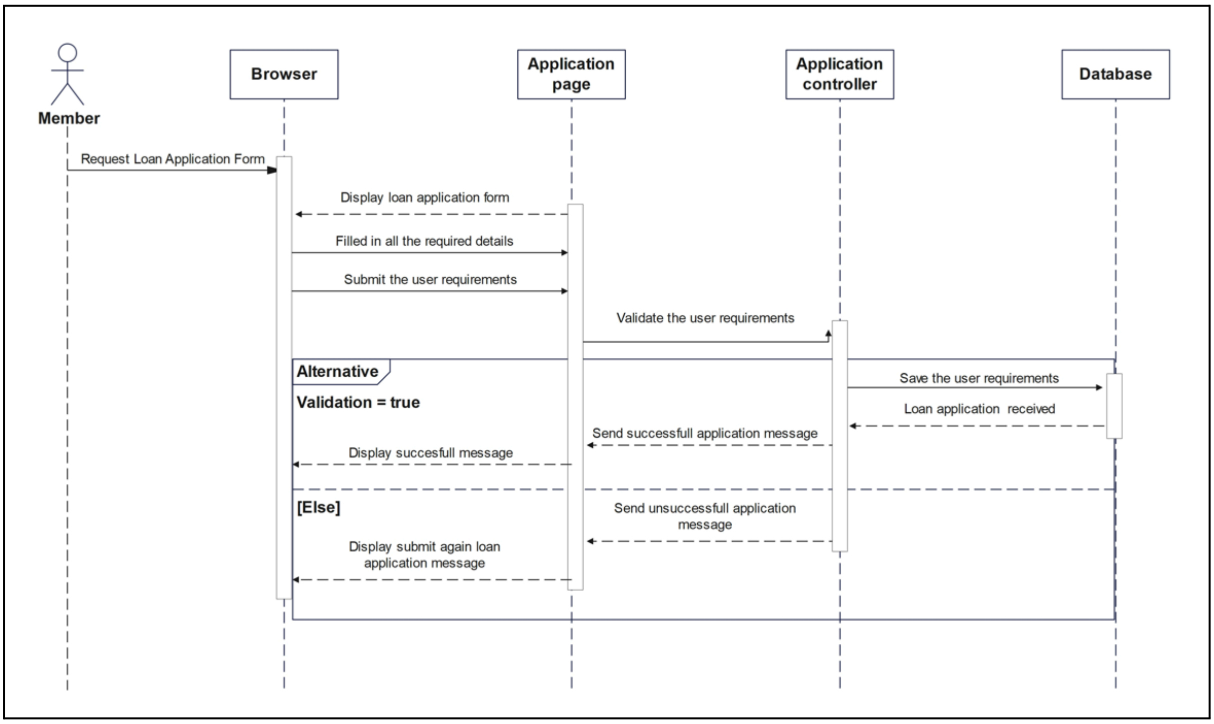


Figure 4.11: SD002 Sequence diagram for <Applying Loan>

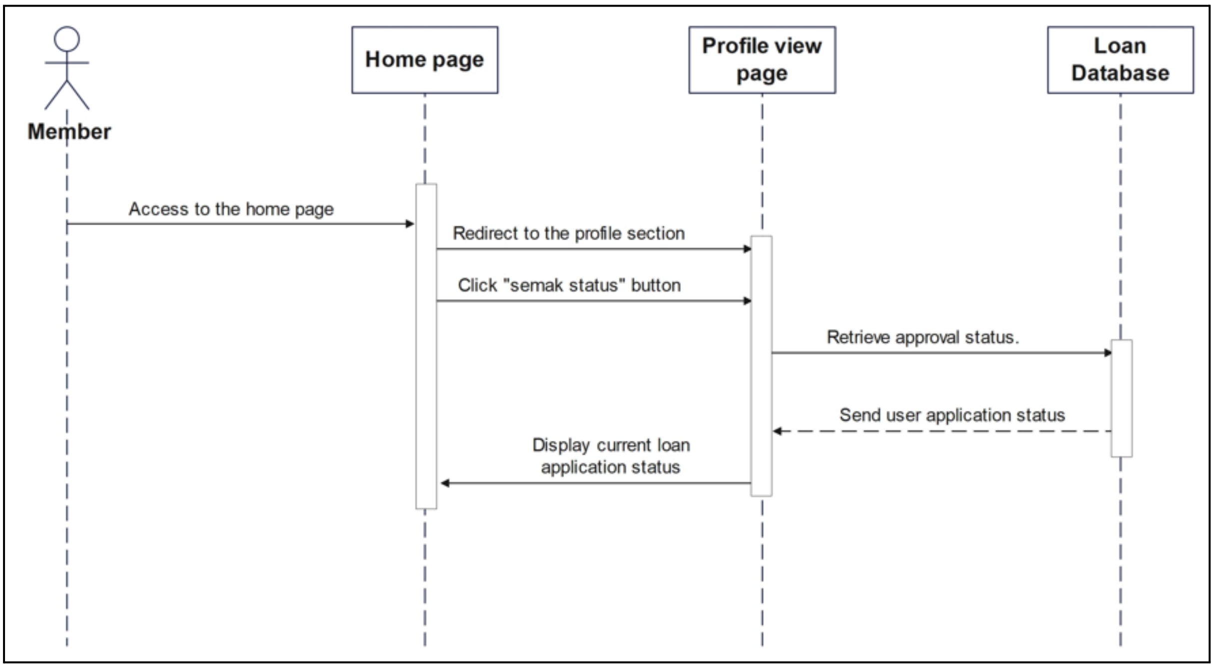


Figure 4.12: SD003 Sequence Diagram for <Check Loan Status>

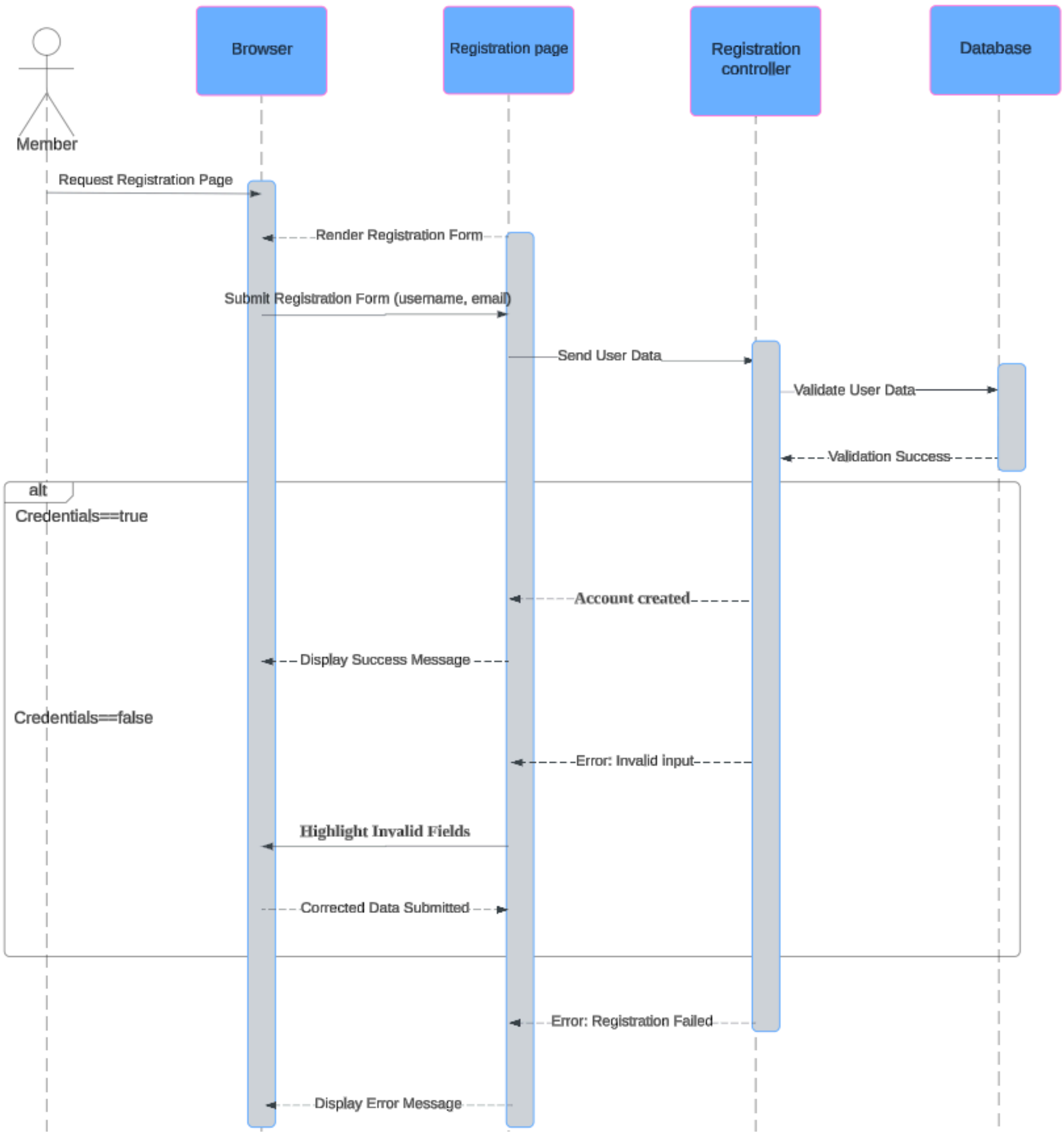


Figure 4.13: SD004 Sequence Diagram for <Register>

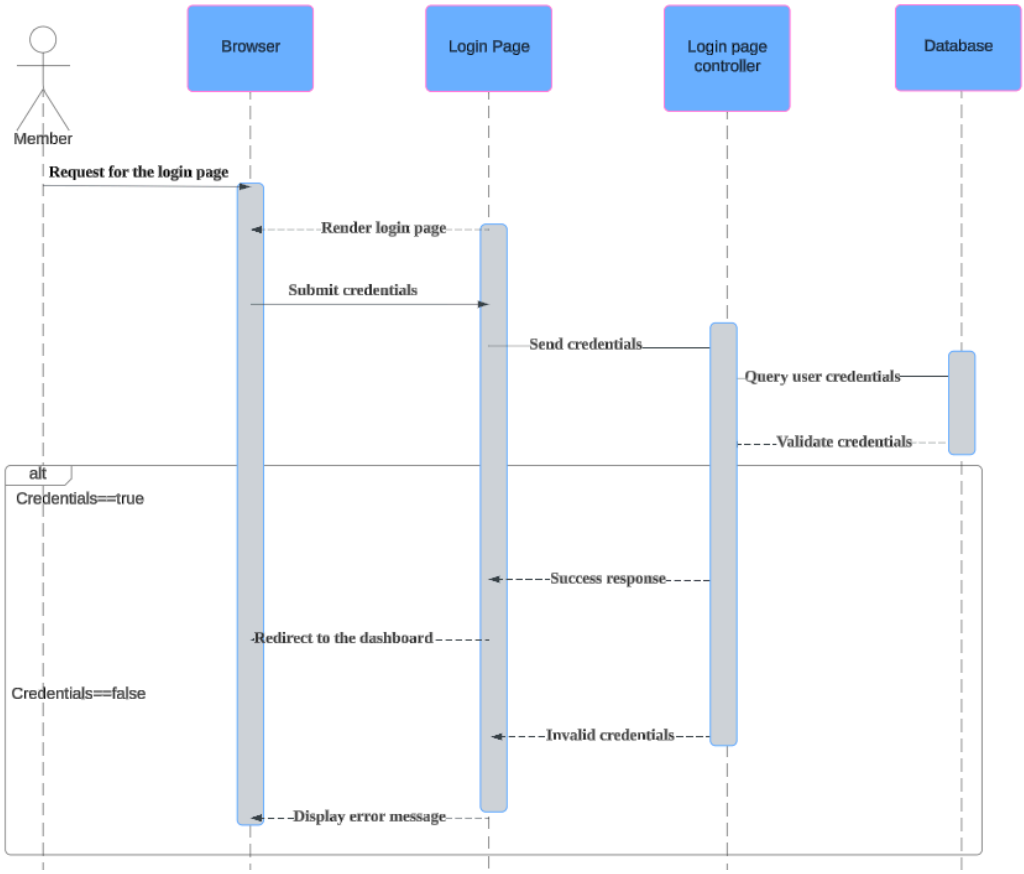


Figure 4.14: SD005 Sequence Diagram for <login>

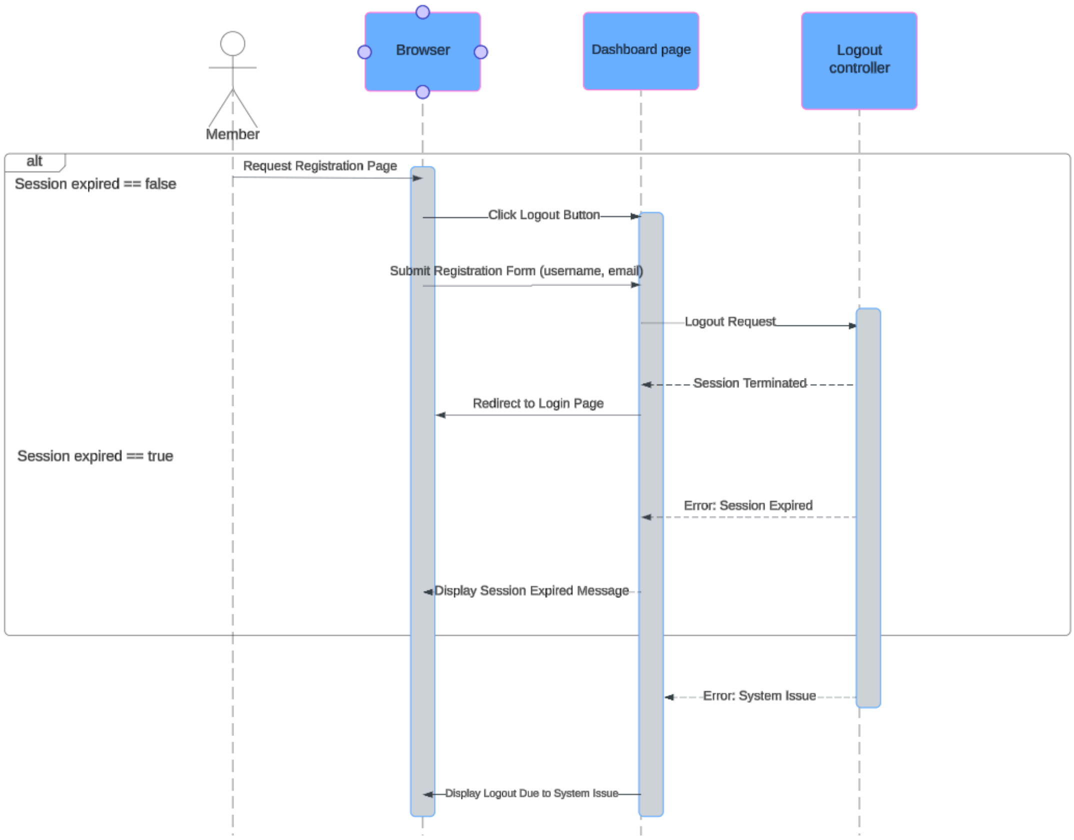


Figure 4.15: SD005 Sequence Diagram for <Logout>

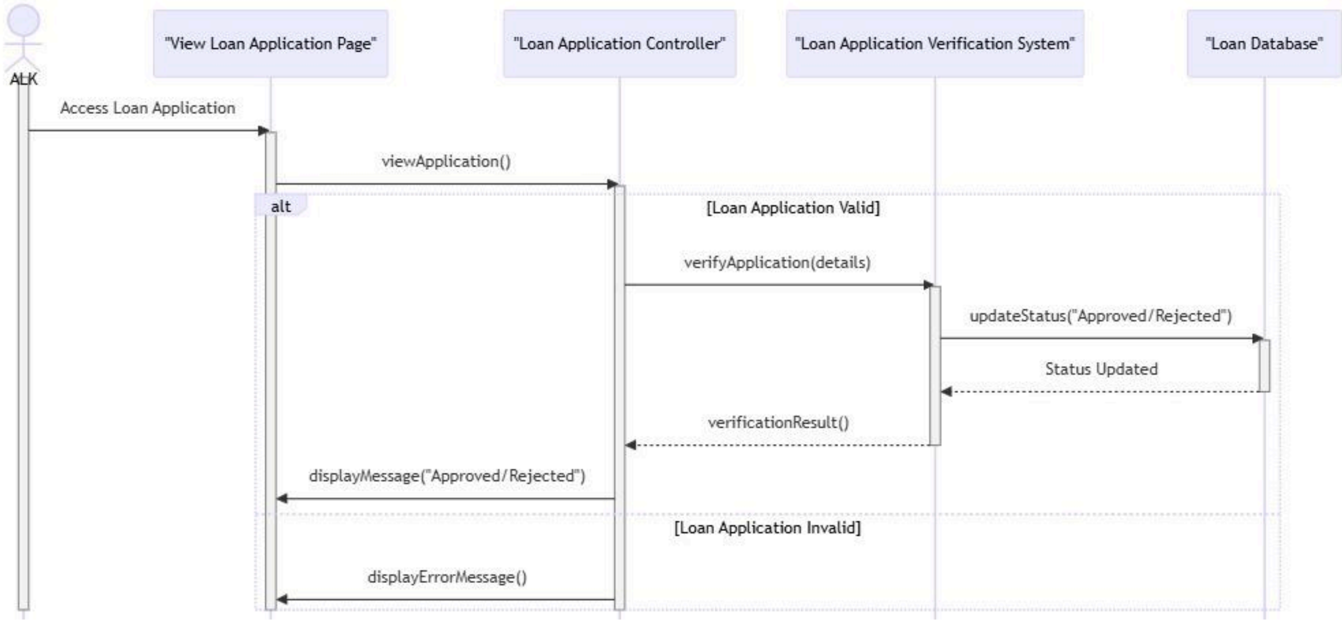


Figure 4.16: SD007 Sequence Diagram for <verify loan application>

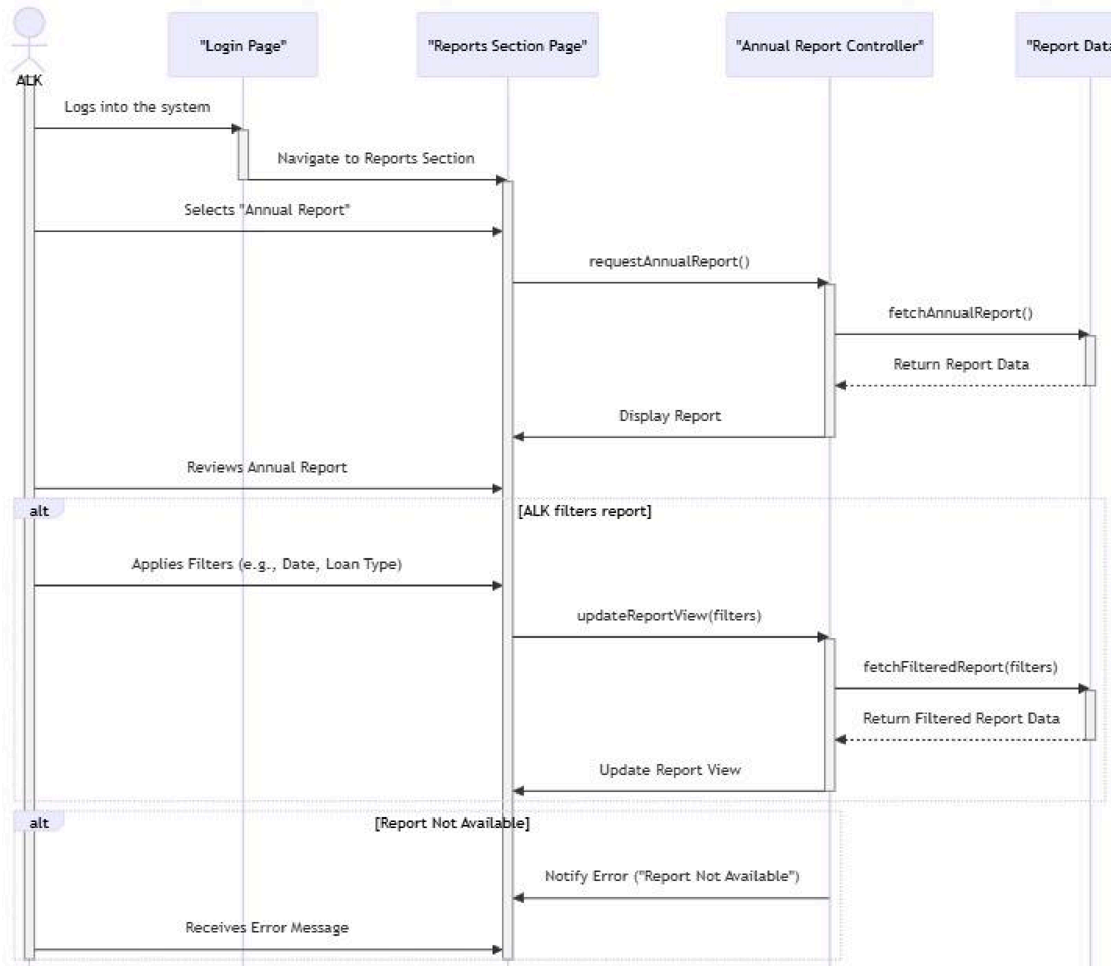


Figure 4.17: SD008 Sequence Diagram for <monthly report>

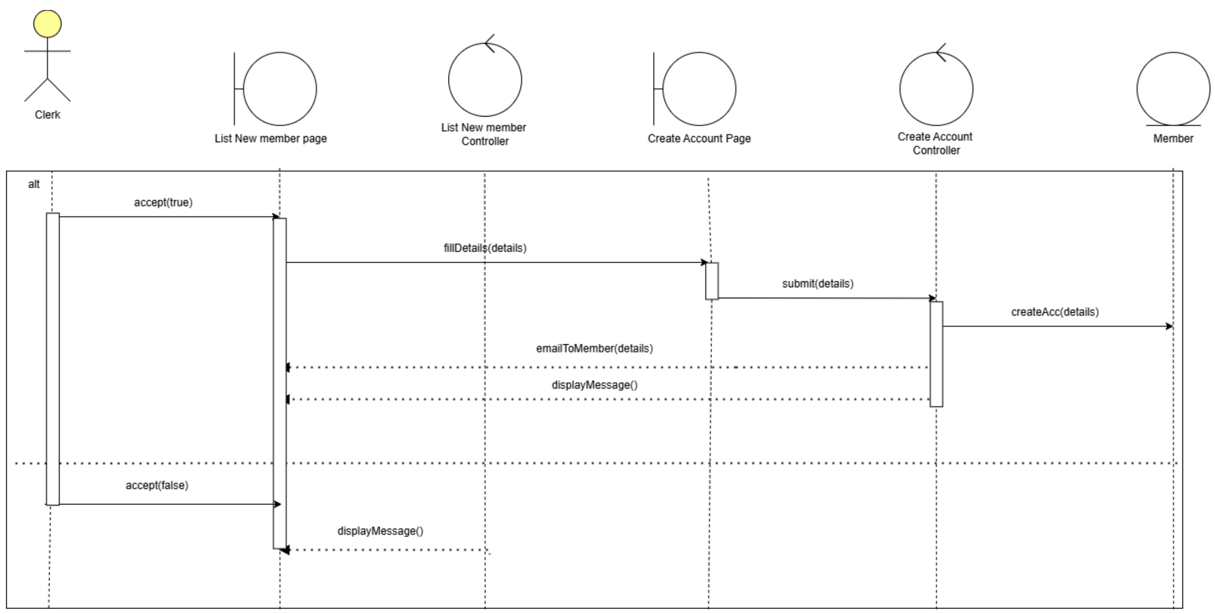


Figure 4.17: SD009 Sequence Diagram for <Create account>

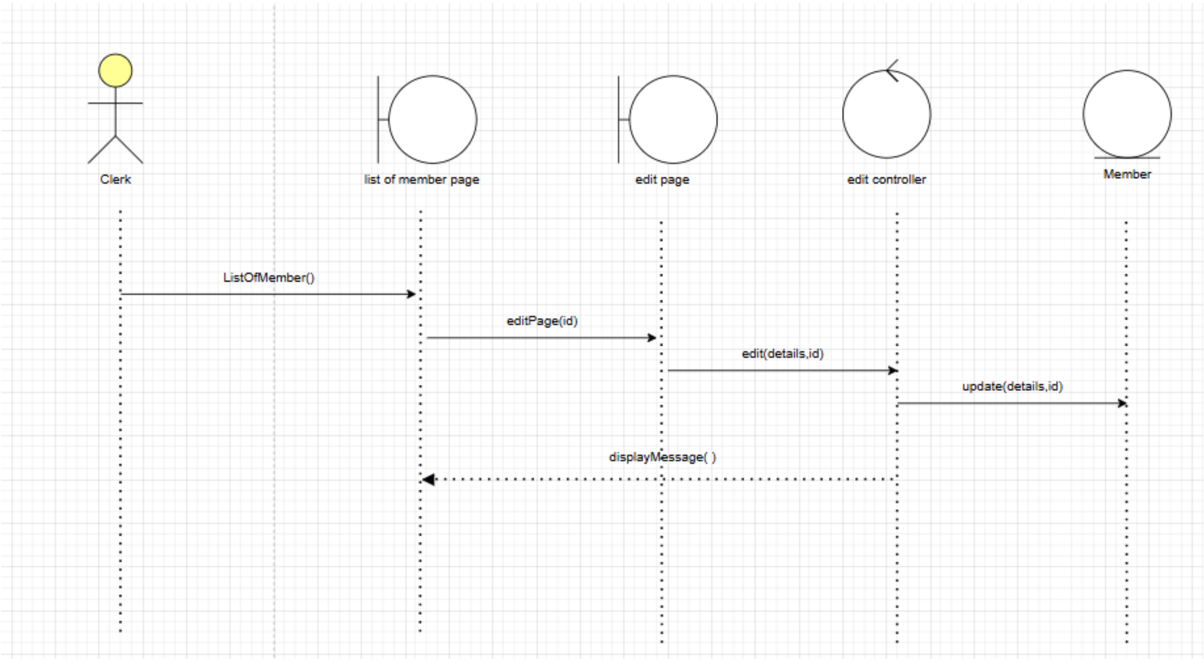


Figure 4.17: SD010 Sequence Diagram for <manage list of member (edit)>

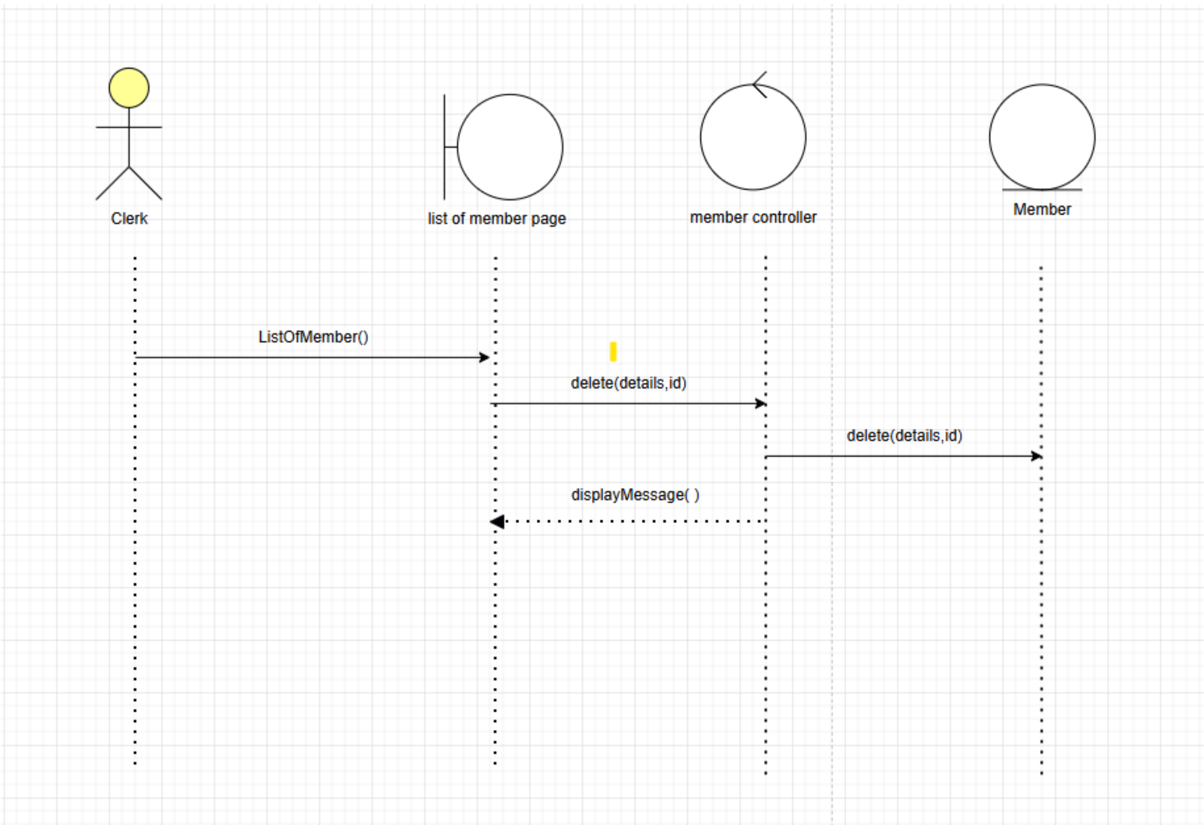


Figure 4.17: SD010 Sequence Diagram for <manage list of member (delete)>

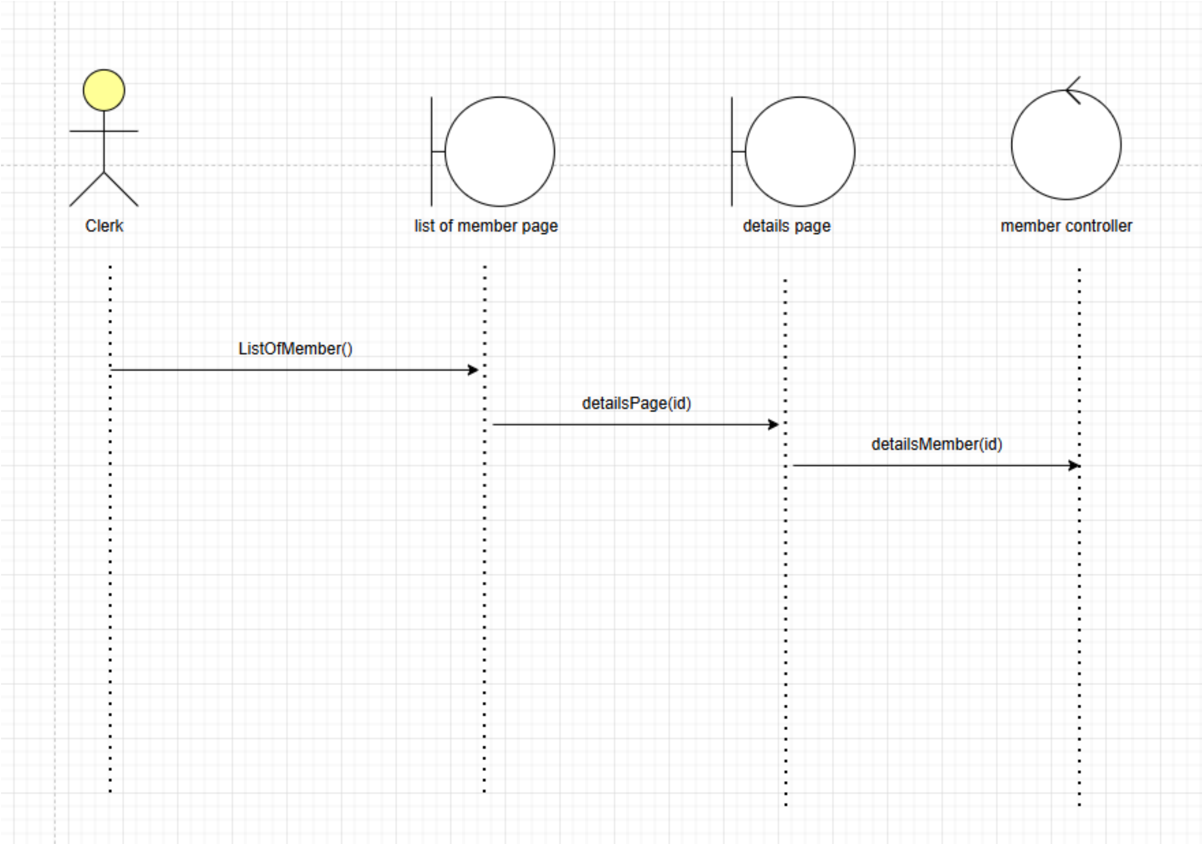


Figure 4.17: SD010 Sequence Diagram for <manage list of member (view details)>

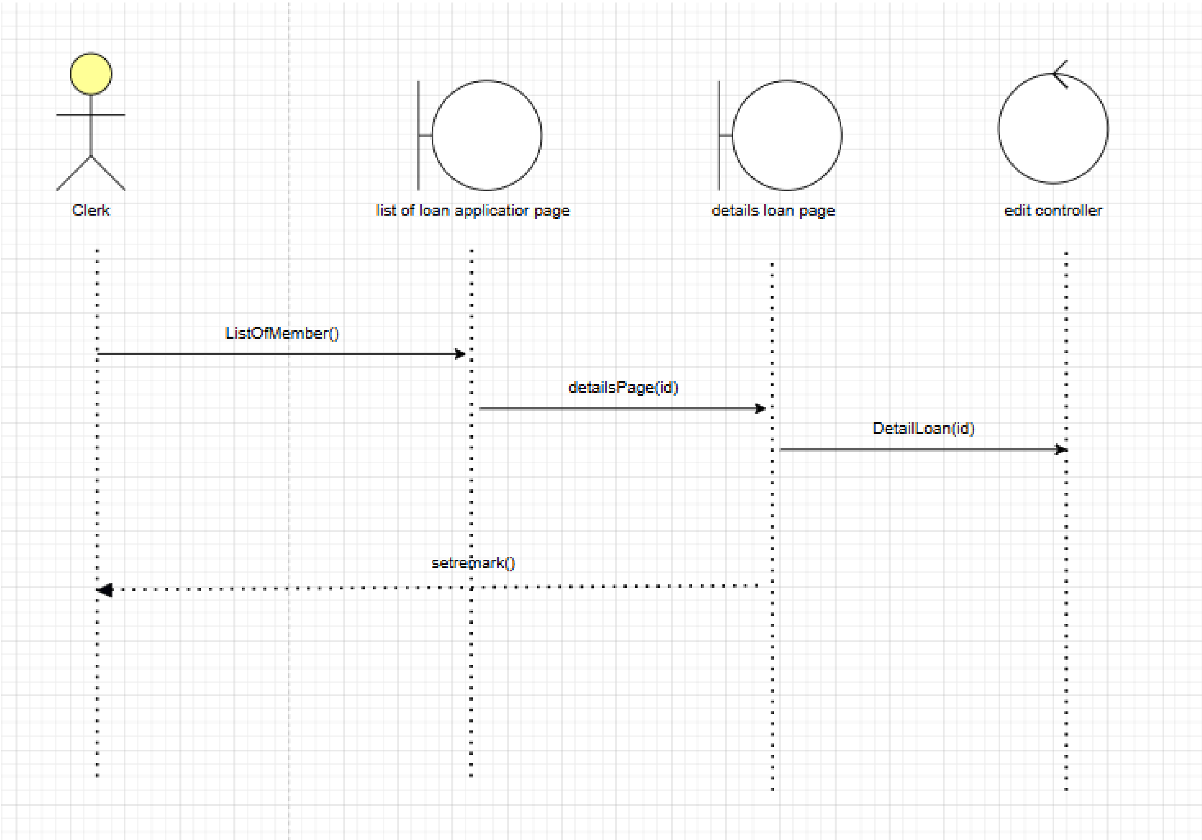


Figure 4.17: SD011 Sequence Diagram for <Manage loan application>

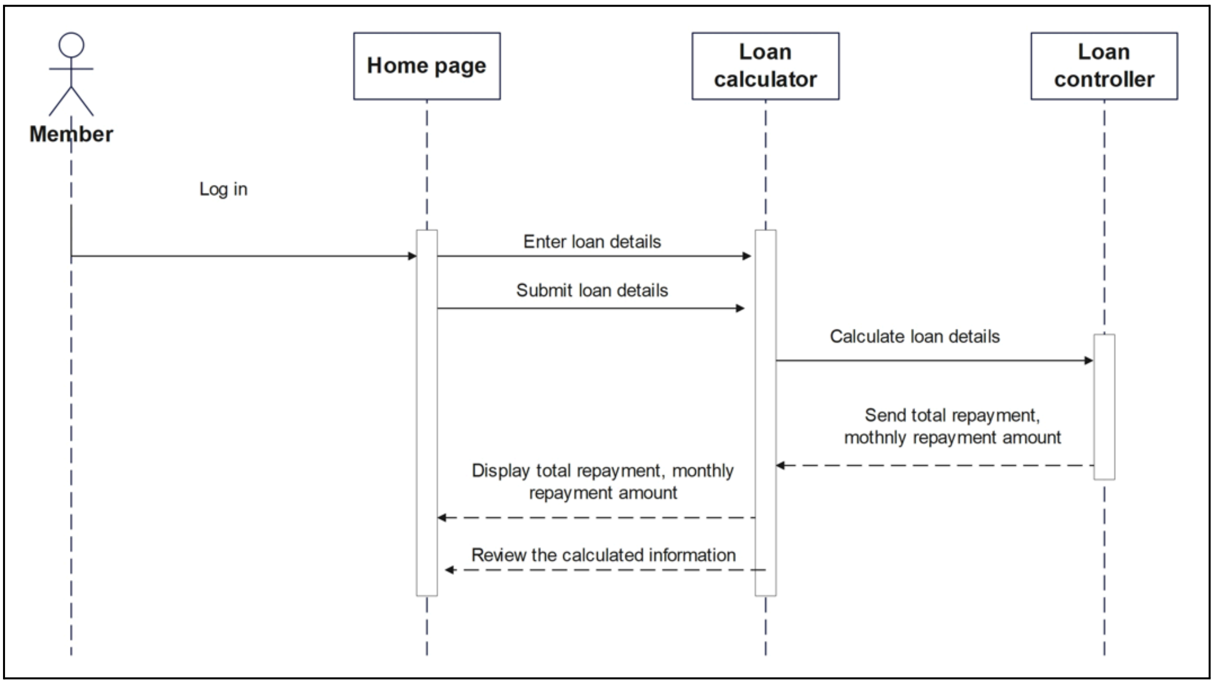


Figure 4.17: SD012 Sequence Diagram for <Calculate loan repayment>

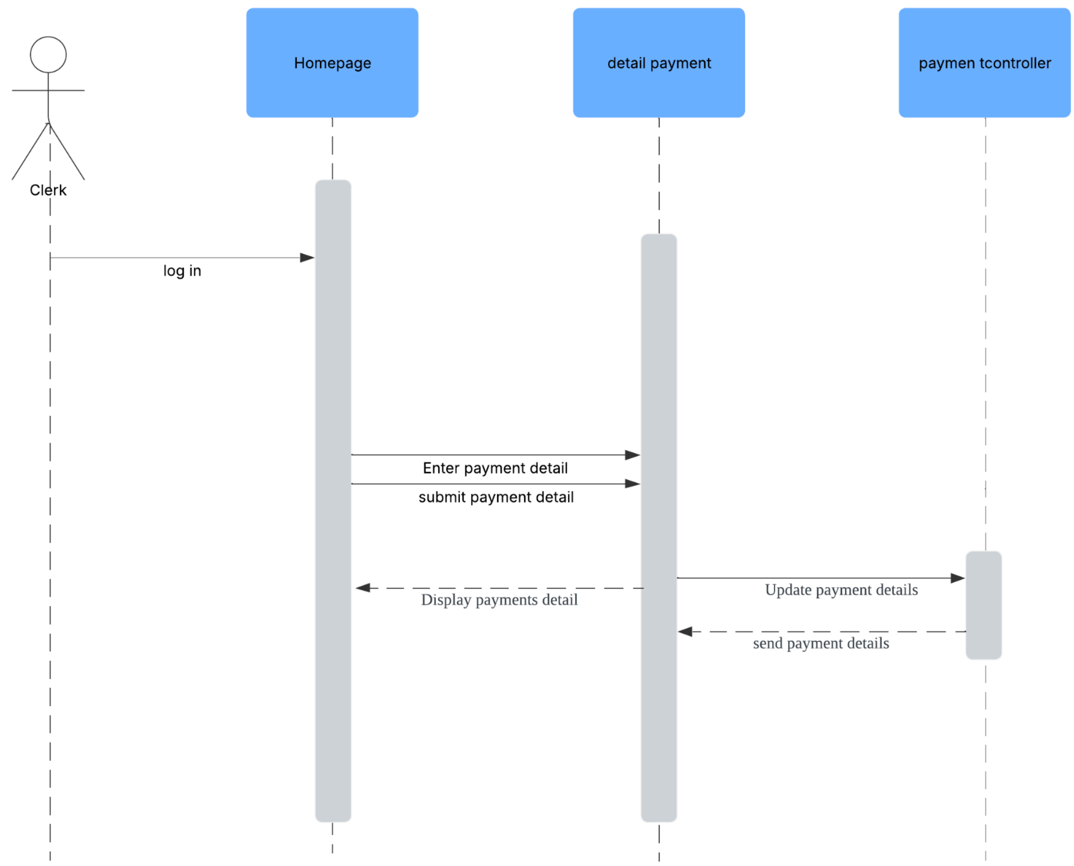


Figure 4.17: SD013 Sequence Diagram for <payment>

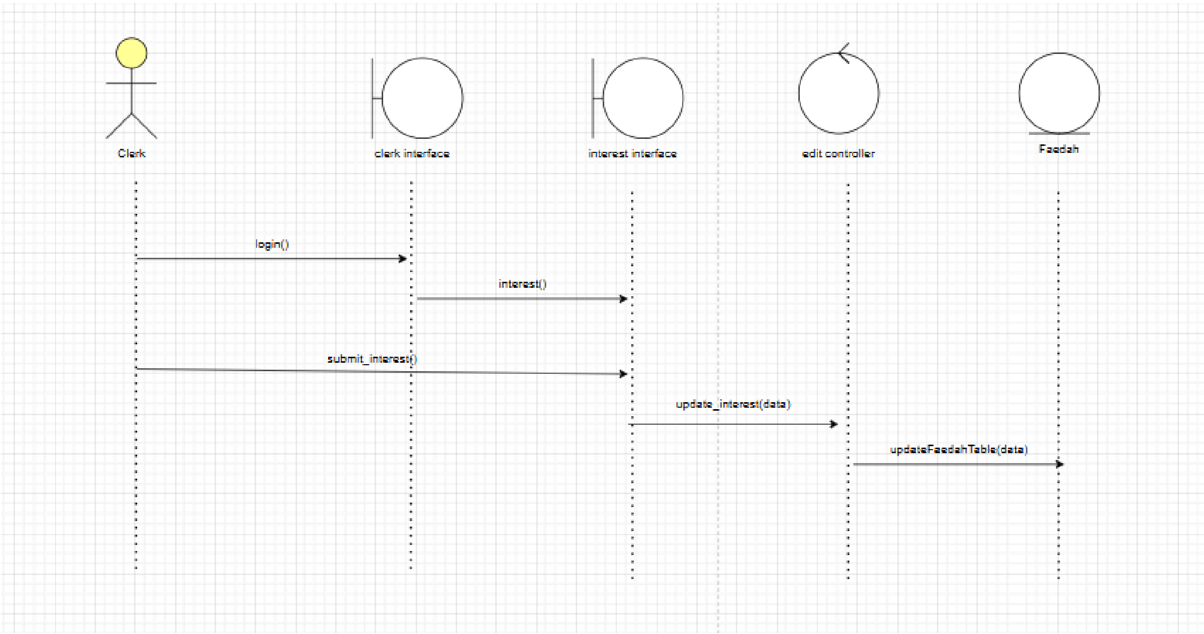


Figure 4.17: SD014 Sequence Diagram for <Interest>

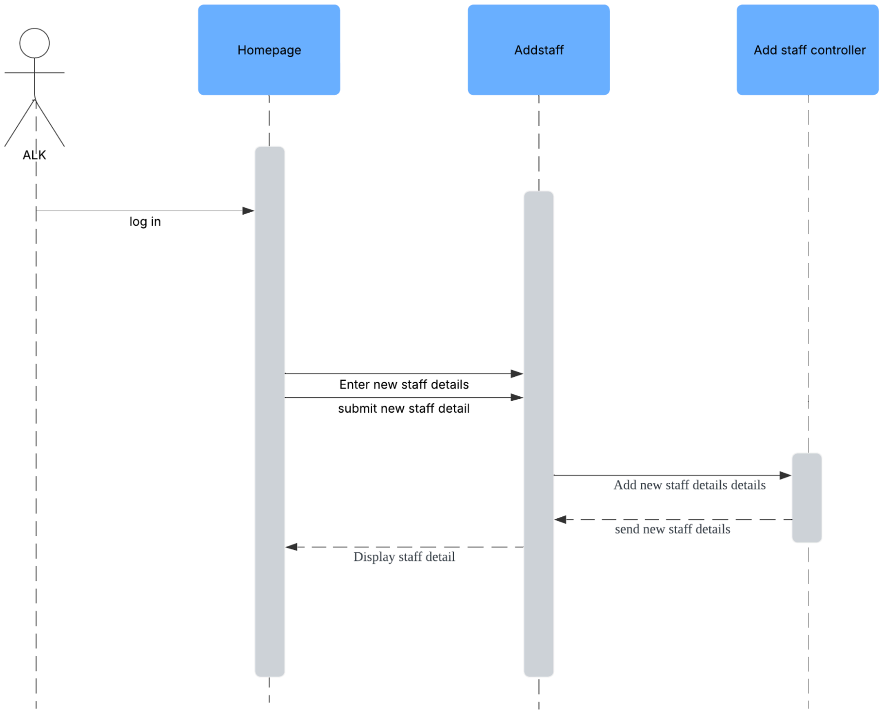


Figure 4.17: SD015 Sequence Diagram for <Add staff>

4.5 Physical/ Deployment View

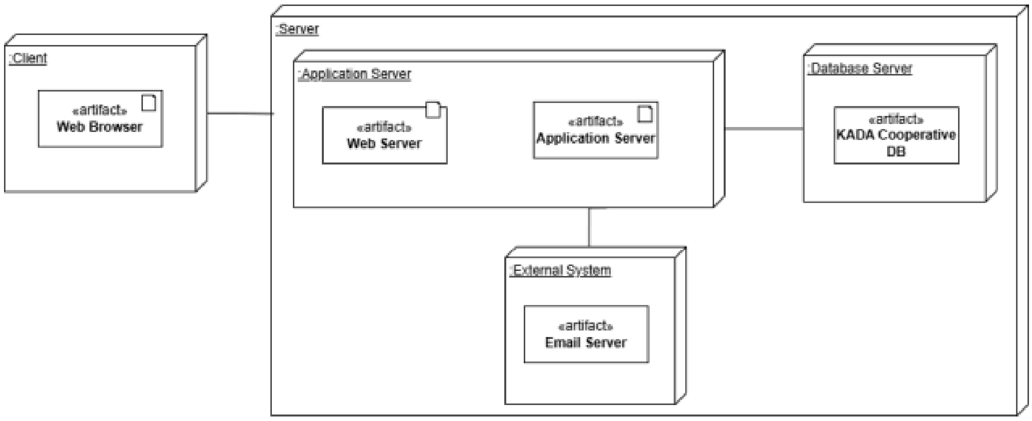


Figure 4.12: Deployment diagram for Internet-based System

5. Data Design

The major data or systems entities are stored into a relational database named as KADA cooperative management system , processed and organized into *n* entities as listed in Table 5.1.

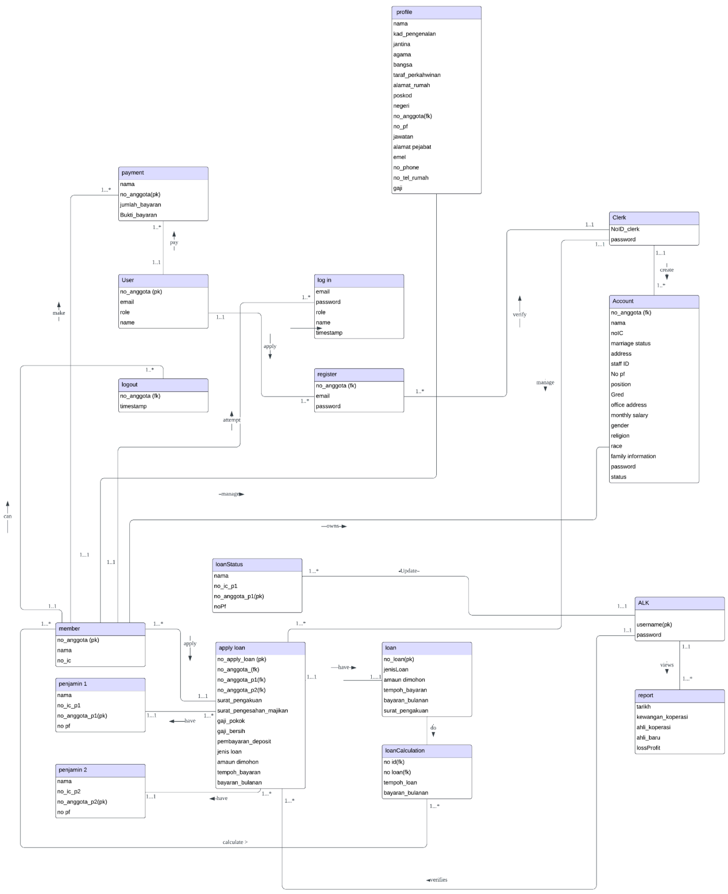


Table 5.1: Description of Entities in the Database

Entiti Name	Description
member	User who already owns an account
penjamin 1	Guarantor 1 is to verified the details accuracy for member in the loan application
penjamin 2	Guarantor 2 is to verified the details accuracy for member in the loan application
applyLoan	Loan application form for members
loanCalculation	give a rough view amount about the loan repayment calculation

loanStatus	give a current loan status to the member whether it has been approved, pending or not approved
Clerk	A user who manages the loan application and membership
Account	details of the applicant for membership in the KADA
ALK	A user who verify the loan application and view the report
report	give a overall report about the cooperation
payment	give a user service to make a loan payment

5.1 **Data Dictionary**

5.1.1 **Entity: <Member>**

Attribute Name	**Type**	**Description**
no_anggota (PK)	Varchar (255)	Uniquely identify each membership number
nama	Varchar (255)	Member's full name
no_IC	Varchar (20)	Member identification card number
email	Varchar (255)	Member's unique email address
role	Varchar (50)	Member's position in KADA

5.1.2 **Entity: <Penjamin 1>**

Attribute Name	**Type**	**Description**
nama	Varchar (255)	Guarantor's 1 full name
no_ic_p1	Varchar (20)	Guarantor's 1 identification card number
no_anggota_p1 (PK)	Varchar (255)	Guarantor's 1 unique membership number

no_pf1	Varchar (50)	Guarantor’s 1 registration number

5.1.3 **Entity: <Penjamin 2>**

Attribute Name	**Type**	**Description**
nama	Varchar (255)	Guarantor’s 2 full name
no_ic_p2	Varchar (20)	Guarantor’s 2 identification card number
no_anggota_p2 (PK)	Varchar (255)	Guarantor’s 2 unique membership number
no_pf2	Varchar (50)	Guarantor’s 2 registration number

5.1.4 Entity: <applyLoan>

Attribute Name	Type	Description
no_apply_loan (PK)	int (10)	after filled in all the details, the application will be provided with the loan application serial number
no_anggota (FK)	Varchar (255)	unique membership number by member
no_anggota_p1 (FK)	Varchar (255)	Guarantor 1 unique membership number by penjamin 1
no_anggota_p2 (FK)	Varchar (255)	Guarantor 2 unique membership number by penjamin2
jenis_loan	Varchar (255)	Type of loan wants apply
amaun_dimohon	decimal (10,2)	Amount to borrowed from KADA
tempoh_bayaran	int (10)	Duration of the loan repayment
bayaran_bulanan	decimal (10,2)	Monthly amount needs to be paid by the member
surat_pengakuan	Varchar (255)	members upload the declaration letter files
surat_pengesahan_majikan	Varchar (255)	members upload the employer verification letter
gaji_pokok	decimal (10,2)	basic salary of member

5.1.5 Entity: <loanStatus>

Attribute Name	Type	Description
no_anggota (FK)	Varchar (255)	unique membership number by member
no pf	Varchar (255)	member registration number
statusLoan	text	loan current state

5.1.6 Entity: <Clerk>

Attribute Name	Type	Description
email	varchar(255)	Account email for login

password	varchar(255)	Account password for login

5.1.7 **Entity: <Account>**

Attribute Name	**Type**	**Description**
No Anggota	varchar(255)	unique identifier for each member
Name	varchar(255)	member full name
No IC	VARCHAR(20)	unique identifier for each member
marriage status	varchar(255)	member's marriage status
address	varchar(255)	member's home address
No PF	VARCHAR(50)	unique identifier for each member
position and Gred	VARCHAR(100)	the position and the gred of the member
Office address	VARCHAR(100)	member's office address
monthly salary	INT	member's salary
Gender	VARCHAR(50)	member's gender
religion	VARCHAR(50)	member's religion
race	VARCHAR(50)	member's race
password	varchar(255)	a unique password for the account
Status	VARCHAR(50)	status of the member application

5.1.8 **Entity: <ALK>**

Attribute Name	**Type**	**Description**
username	varchar (255)	ALK username
password	varchar(255)	a uniques password for the account

5.1.9 **Entity: <report>**

Attribute Name	**Type**	**Description**

tarikh	date	specific date for the report
kewangan_koperasi	int	cooperative cash flow
ahli_koperasi	int	number of the cooperative member
ahli_baharu	int	number of new member of the cooperative
lossProfit	int	number of profit or loss that cooperative had

5.1.10 **Entity: <Payment>**

Attribute Name	**Type**	**Description**
nama	varchar(255)	user full name
no_anggota	varchar(255)	unique identifier for each member
jumlah_bayaran	int	payment make to the system
bukti_bayaran	Varchar (255)	user upload the transaction receipt

5.1.11 **Entity: <loanApplication>**

Attribute Name	**Type**	**Description**
no_anggota (FK)	Varchar (255)	unique membership number by member
no_loan (FK)	Varchar (10)	Unique loan number by loan
tempoh_loan	int (10)	Duration of the loan repayment
bayaran_bulanan	decimal (10,2)	Monthly amount needs to be paid by the member

6. User Interface Design

6.1 Overview of user interface

This section presents the system's interface as it appears to users based on their designated roles: User, Clerk, and ALK. It provides an overview of the different functionalities and features available to each role, ensuring that users can navigate the system efficiently according to their specific responsibilities and access levels.

6.2 Screen Images

6.2.1 Module Member

Figure 6.2.1 Home Interface

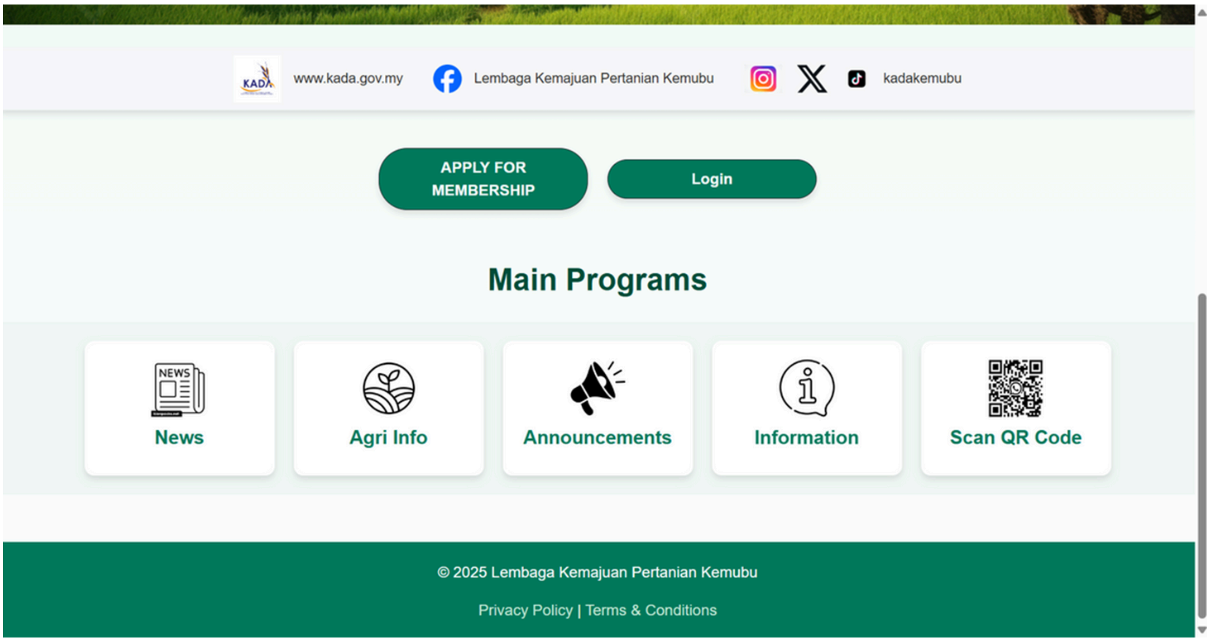


Figure 6.2.2 Home Interface

Figure 6.2.3 Interface for <Register>

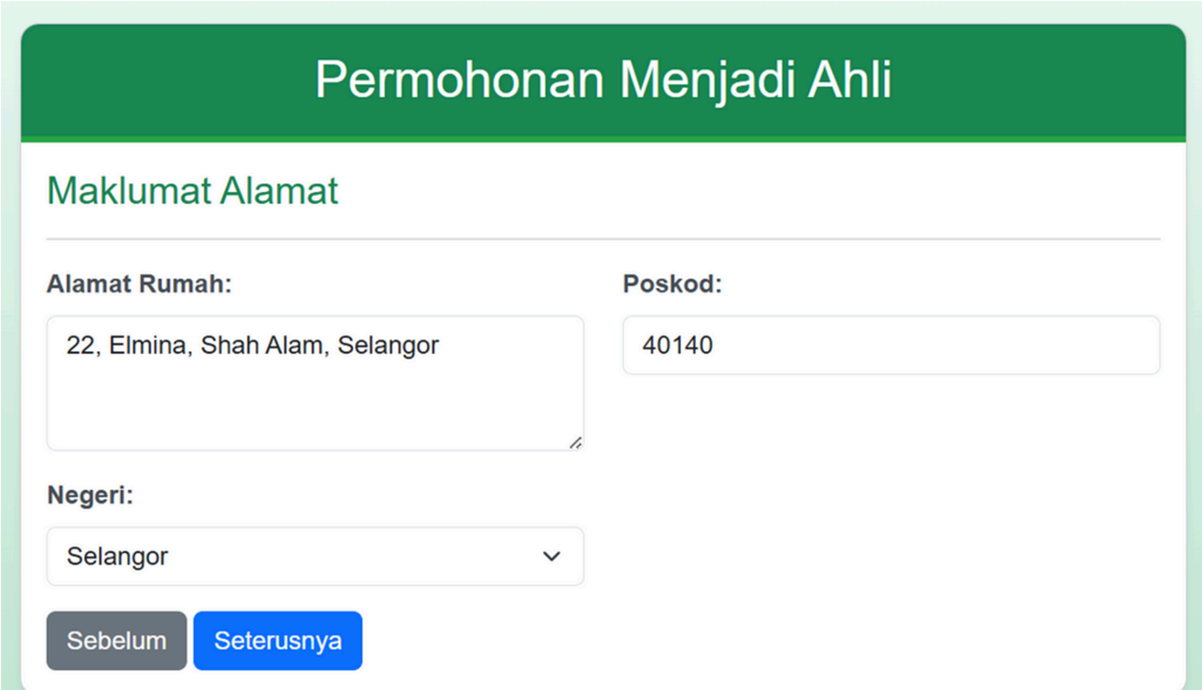


Figure 6.2.4 Interface for <Register>

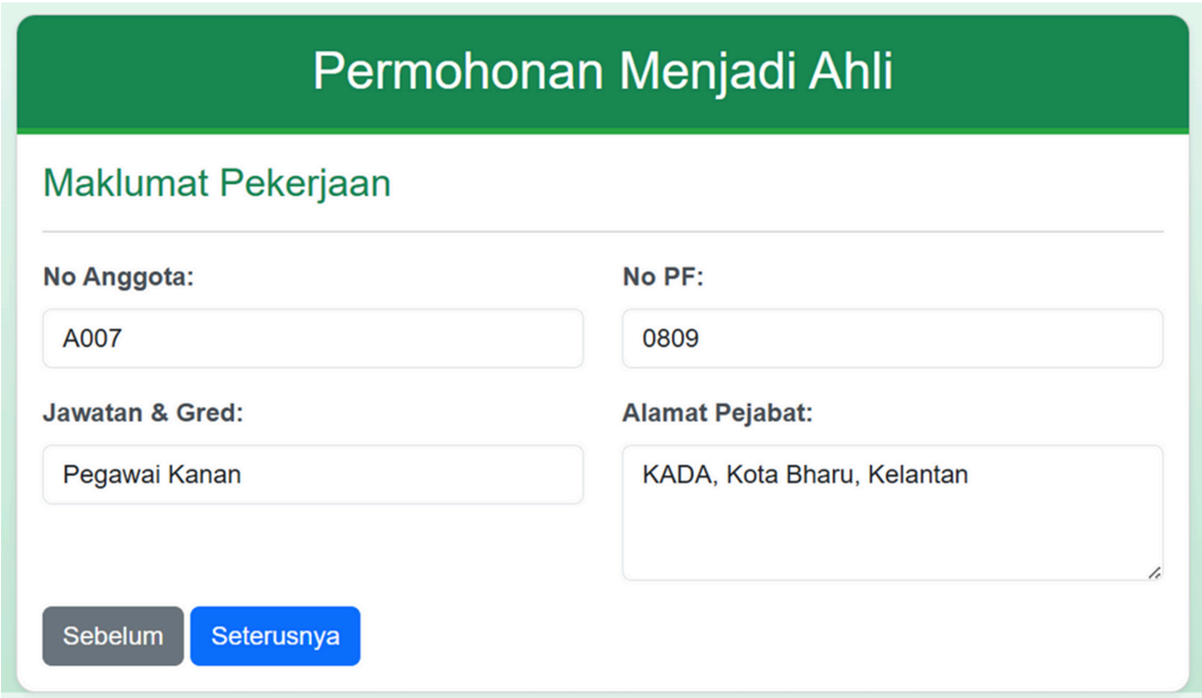


Figure 6.2.5 Interface for <Register>

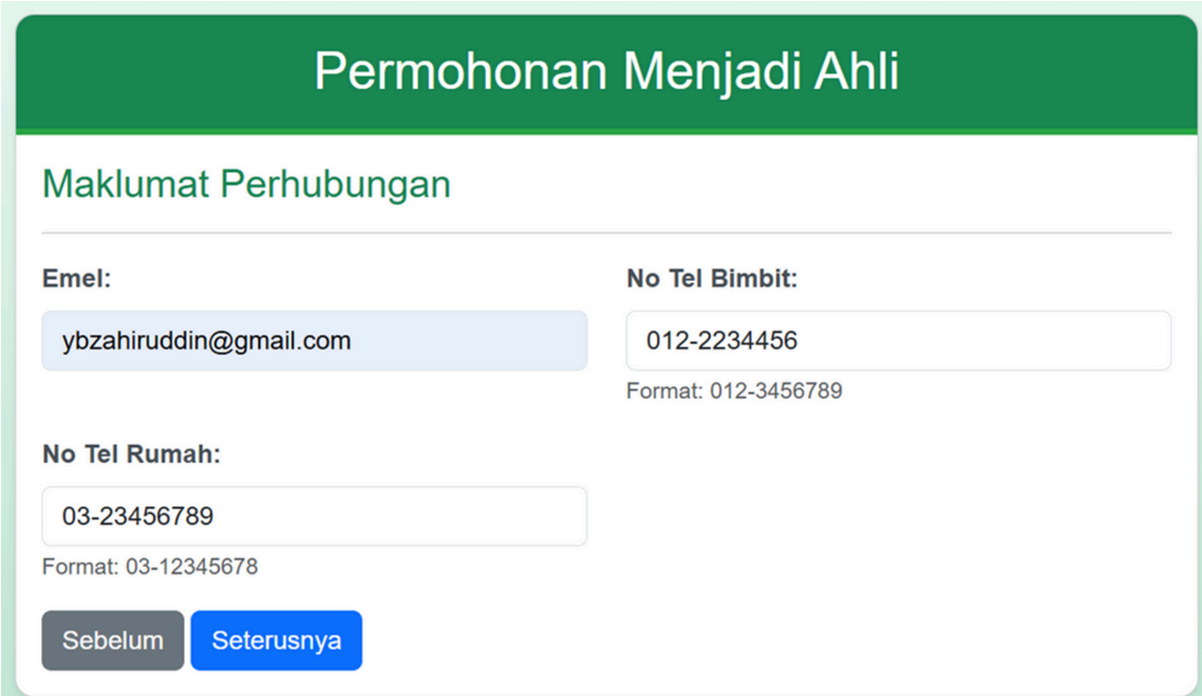


Figure 6.2.6 Interface for <Register>

Figure 6.2.7 Interface for <Register>

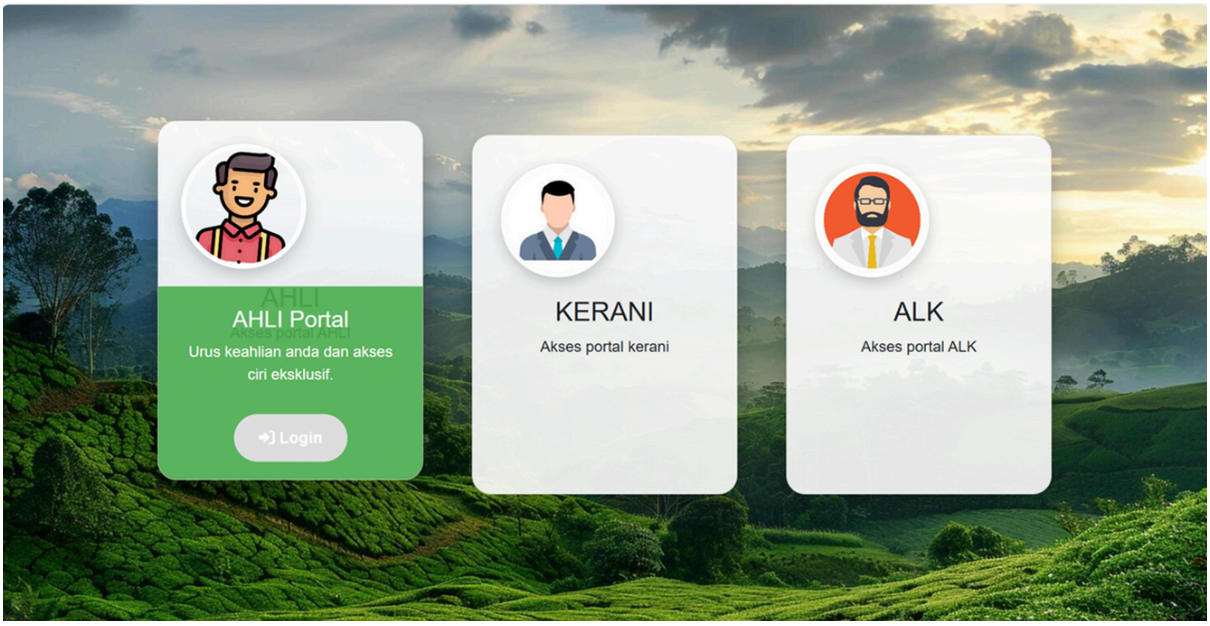


Figure 6.2.8 Interface for <Login>

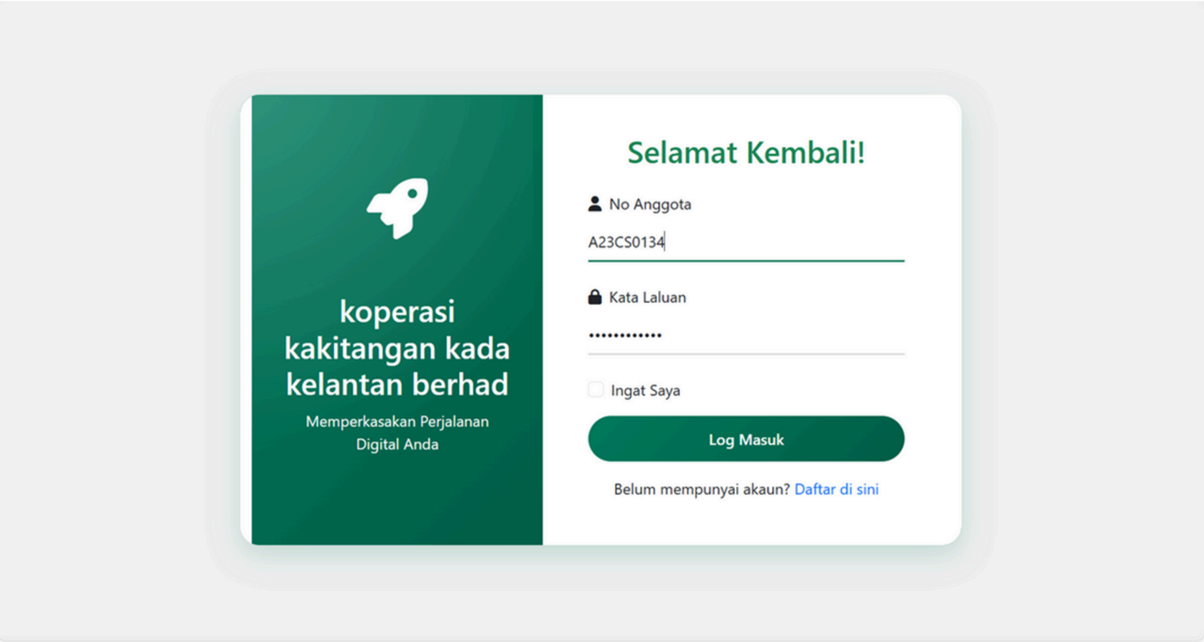


Figure 6.2.9 Interface for <Login>

Figure 6.2.10 Interface for <Manage Profile>

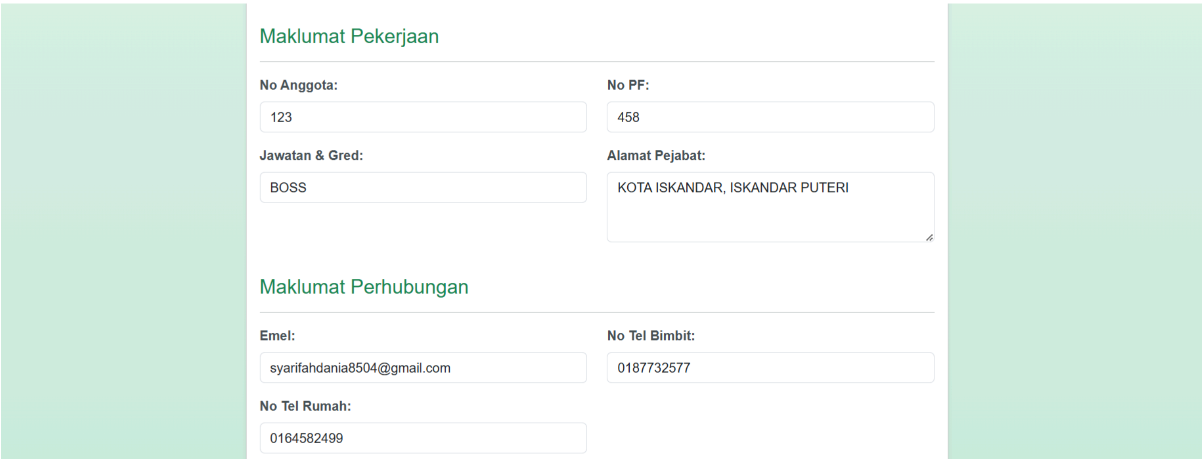


Figure 6.2.11 Interface for <Manage Profile>

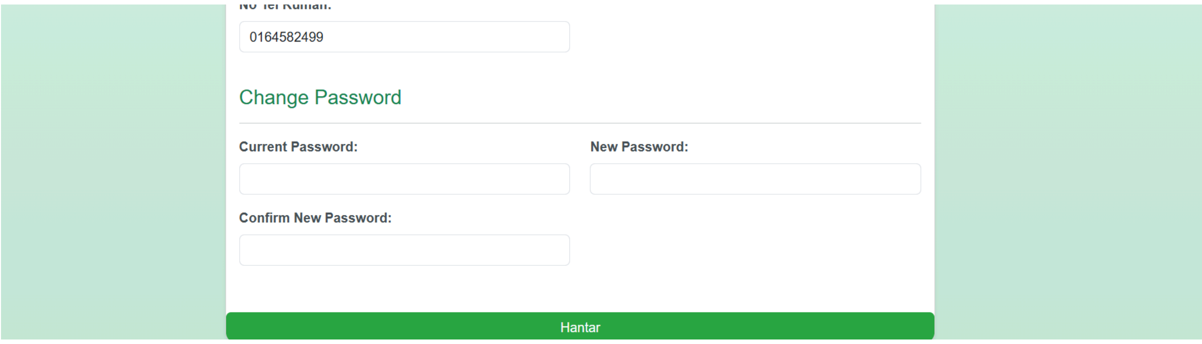


Figure 6.2.12 Interface for <Manage Profile>

Figure 6.2.13 <Logout>

Status Permohonan Pembiayaan

Nama	No. Anggota	Status	Tindakan
YASMIN BATRISYIA BINTI ZAHIRUDDIN	A007	pending	Maklumat lanjut...

Figure 6.2.14 Interface for <Loan Application>

Detail Pemohonan Pembiayaan

Name	YASMIN BATRISYIA BINTI ZAHIRUDDIN
Email	ybzahiruddin@gmail.com
Kad Pengenalan	040124140466
Tarikh Lahir	2004-01-24
Jantina	Perempuan
Agama	Islam
Bangsa	Melayu
Alamat Rumah	22, Elmina, Shah Alam, Selangor
Poskod	40140
Negeri	Selangor
No Anggota	A007
No PF	0809
Jawatan/Gred	Pegawai Kanan
Alamat Pejabat	KADA, Kota Bharu, Kelantan
Poskod Pejabat	5100
Nama Bank Akaun	RHB
No Bank Akaun	8190324412335678
Jenis Pembiayaan	Al_Bai
Amaun Dimohon	1000.00
Tempoh Bayaran	2
Bayaran Bulanan	45.17
Nama Penjamin 1	Naim
No KP Penjamin 1	001223145675
No Anggota Penjamin 1	A012
No PF Penjamin 1	12345
Nama Penjamin 2	Mukhritz
No KP Penjamin 2	020503145677
No Anggota Penjamin 2	A013
No PF Penjamin 2	45678
Gaji Pokok	10000.00
Gaji Bersih	7000.00
Borang Pengakuan Pemohon	View Document
Borang Pengesahan Majikan	View Document

Figure 6.2.15 Interface for <Loan Application>

Figure 6.2.16 Interface for <Loan Interest Calculator>

Figure 6.2.17 Interface for <Application to Resign as a Member>

Figure 6.2.18 Interface for <Financial status>

6.2.2 Module Clerk

Figure 6.2.1: Interface for <clerk interface>

Figure 6.2.2: Interface for <list of Membership Application>

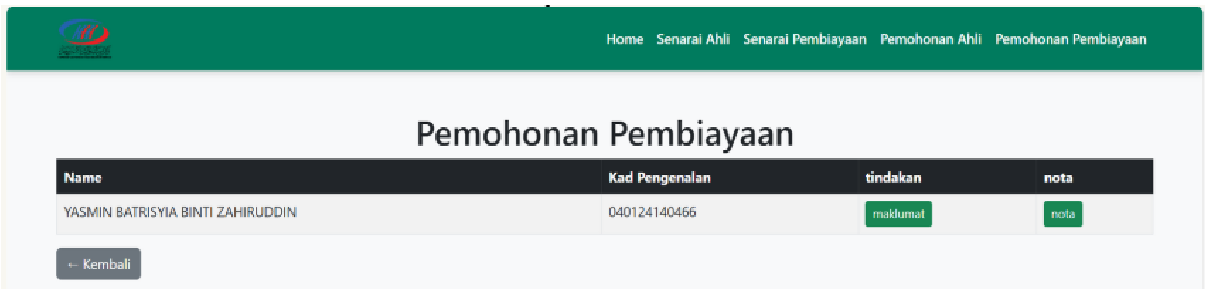
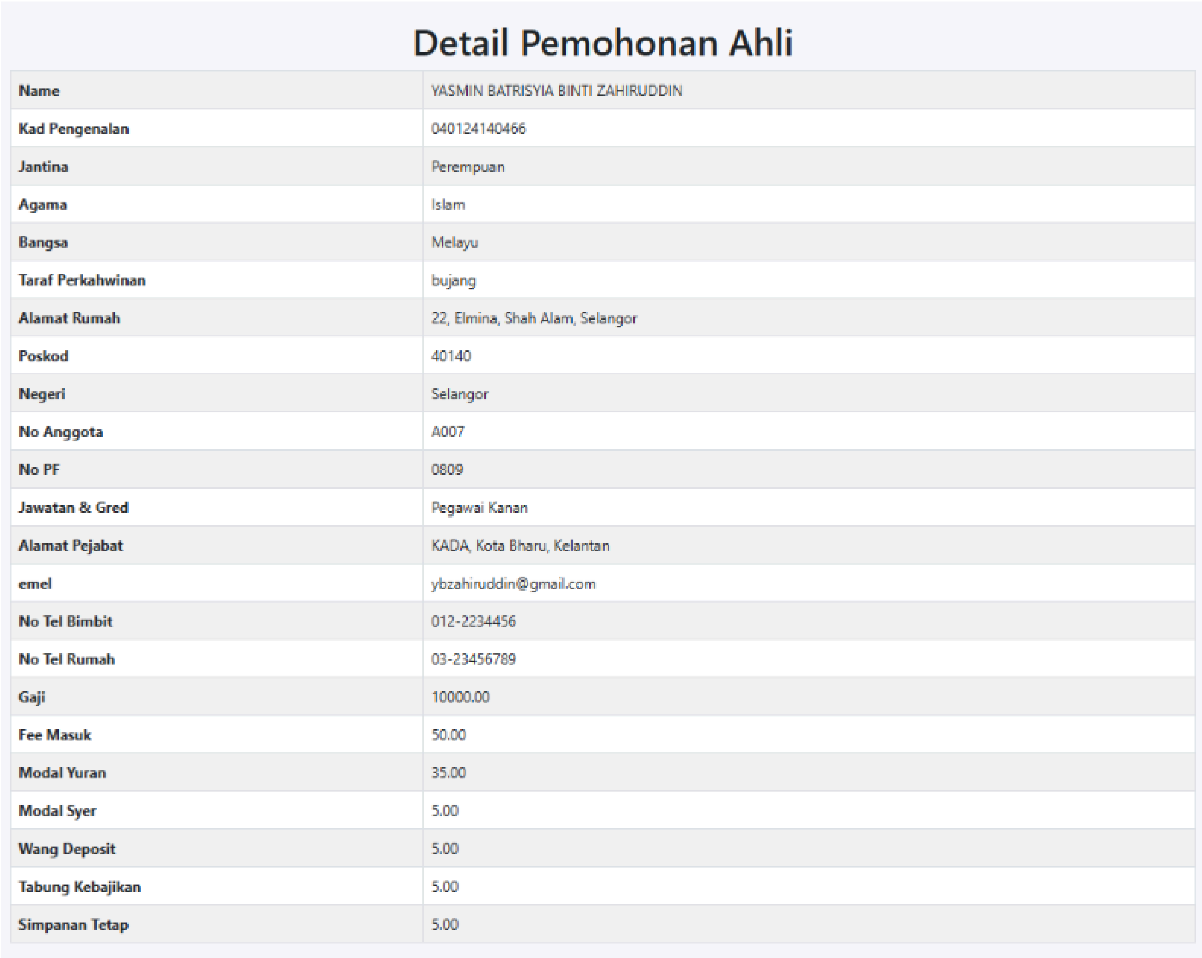


Figure 6.2.3: Interface for <list of requested Loan Application >

No_anggota	tindakan
A009	More
A007	More

Figure 6.2.4: Interface for <list of Member cooperative KADA>



Detail Pemohonan Ahli

Name	YASMIN BATRISYIA BINTI ZAHIRUDDIN
Kad Pengenalan	040124140466
Jantina	Perempuan
Agama	Islam
Bangsa	Melayu
Taraf Perkahwinan	bujang
Alamat Rumah	22, Elmina, Shah Alam, Selangor
Poskod	40140
Negeri	Selangor
No Anggota	A007
No PF	0809
Jawatan & Gred	Pegawai Kanan
Alamat Pejabat	KADA, Kota Bharu, Kelantan
emel	ybzahiruddin@gmail.com
No Tel Bimbit	012-2234456
No Tel Rumah	03-23456789
Gaji	10000.00
Fee Masuk	50.00
Modal Yuran	35.00
Modal Syer	5.00
Wang Deposit	5.00
Tabung Kebajikan	5.00
Simpanan Tetap	5.00

Figure 6.2.5: Interface for <Details the Member>

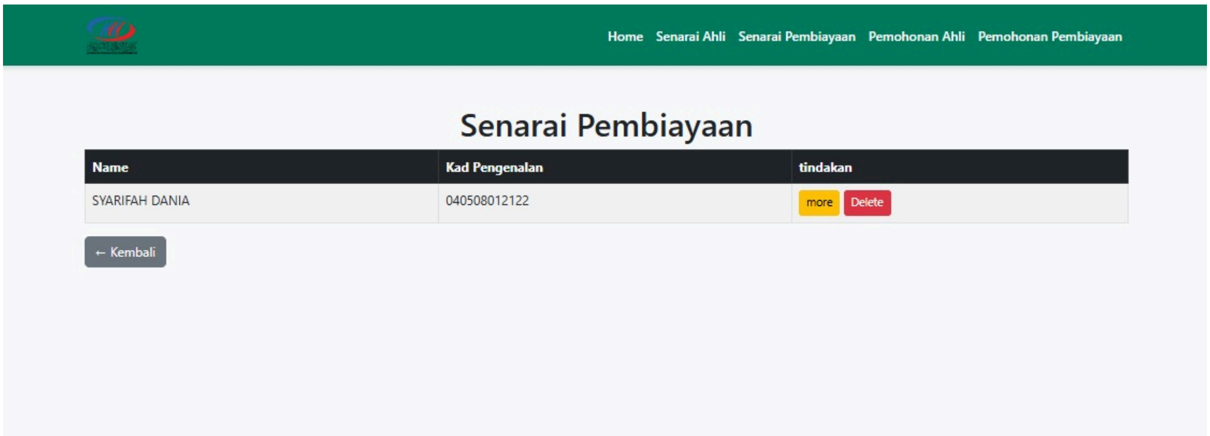
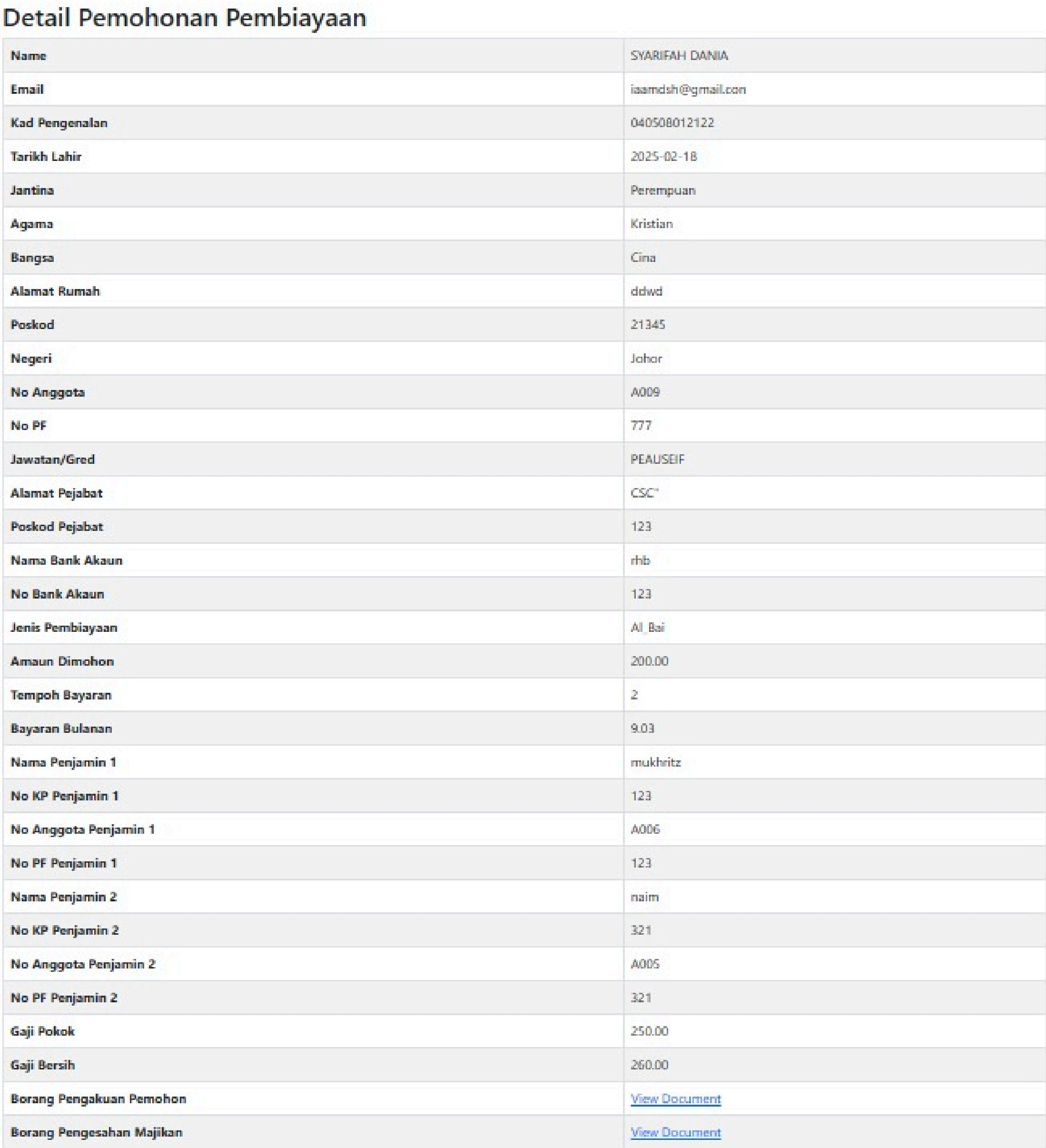


Figure 6.2.6: Interface for <list of Loan Applicant>



Detail Pemohonan Pembiayaan

Name	SYARIFAH DANIA
Email	iaamdsh@gmail.con
Kad Pengenalan	040508012122
Tarikh Lahir	2025-02-18
Jantina	Perempuan
Agama	Kristian
Bangsa	Cina
Alamat Rumah	ddwd
Poskod	21345
Negeri	Johor
No Anggota	A009
No PF	777
Jawatan/Gred	PEAUSEIF
Alamat Pejabat	CSC'
Poskod Pejabat	123
Nama Bank Akaun	rhb
No Bank Akaun	123
Jenis Pembiayaan	Al Bai
Amaun Dimohon	200.00
Tempoh Bayaran	2
Bayaran Bulanan	9.03
Nama Penjamin 1	mukhritz
No KP Penjamin 1	123
No Anggota Penjamin 1	A006
No PF Penjamin 1	123
Nama Penjamin 2	naim
No KP Penjamin 2	321
No Anggota Penjamin 2	A005
No PF Penjamin 2	321
Gaji Pokok	250.00
Gaji Bersih	260.00
Borang Pengakuan Pemohon	View Document
Borang Pengesahan Majikan	View Document

Figure 6.2.7: Interface for <details of Loan Applicant>

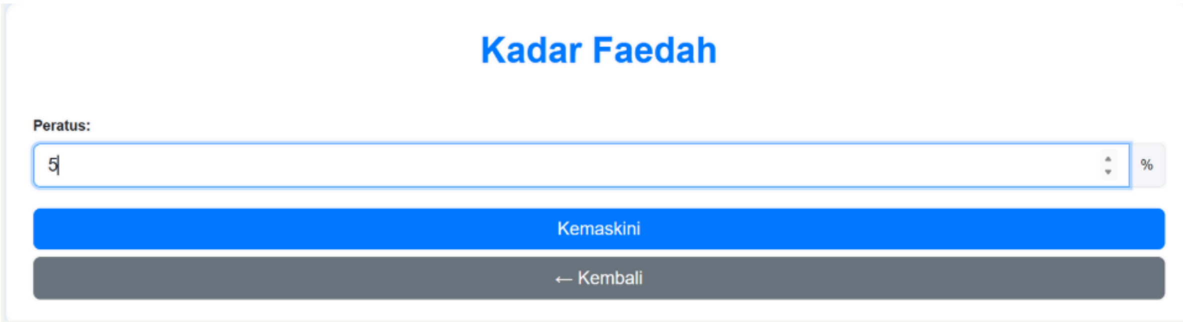


Figure 6.2.8: Interface for <interest Rate>

Figure 6.2.9: Interface for <Saving Payment>

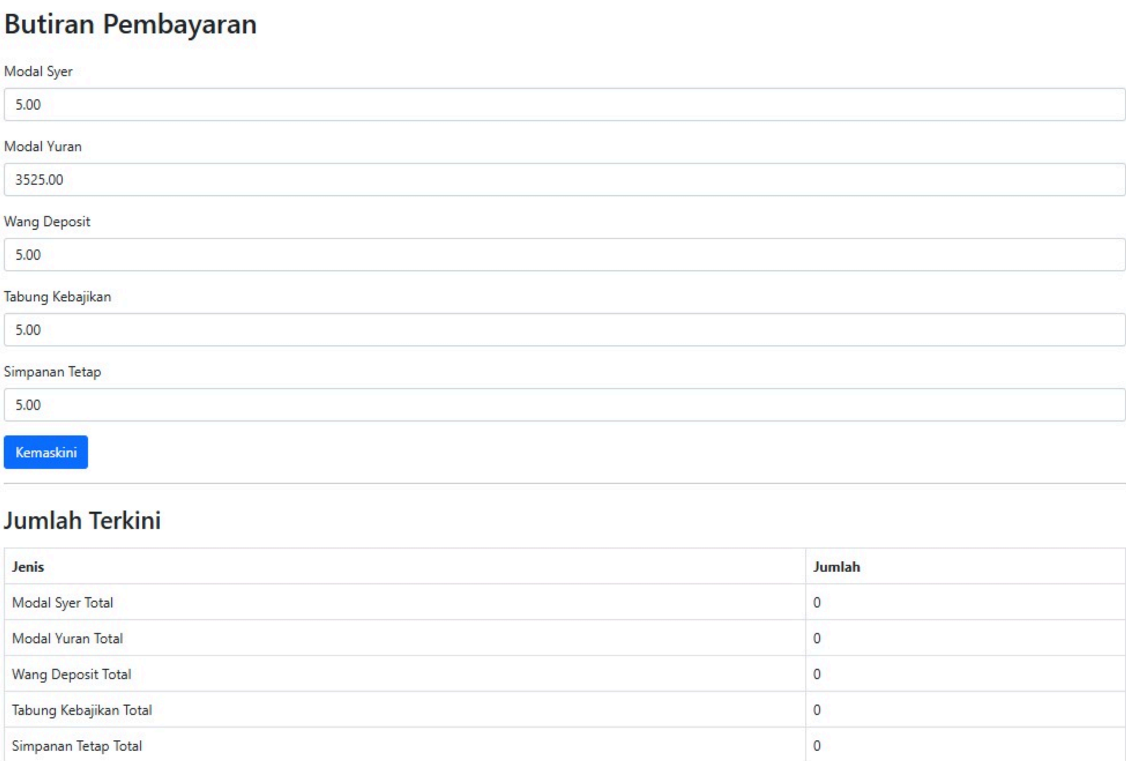


Figure 6.2.10: Interface for <Update Saving Payment>

Figure 6.2.11: Interface for <Loan Payment>

Figure 6.2.12: Interface for <update Loan Payment>

Figure 6.2.13: Interface for <List member that wanted to quit>

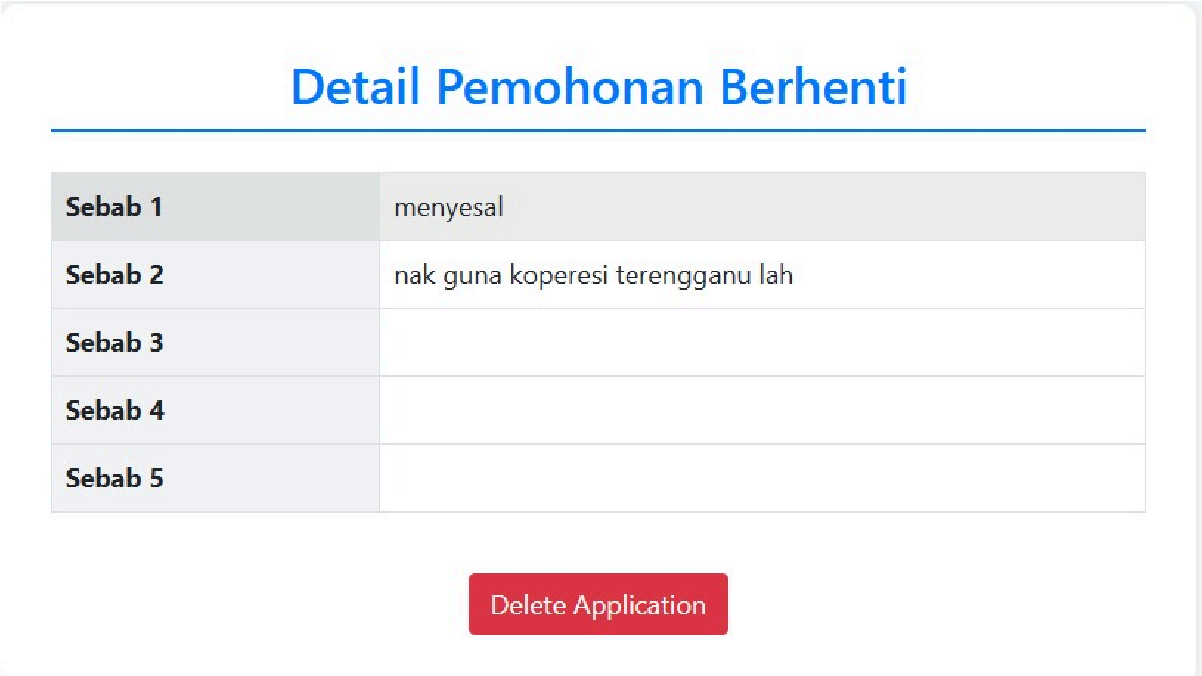


Figure 6.2.14: Interface for <Reason for Quitting >

6.3.1 Module Loan Application

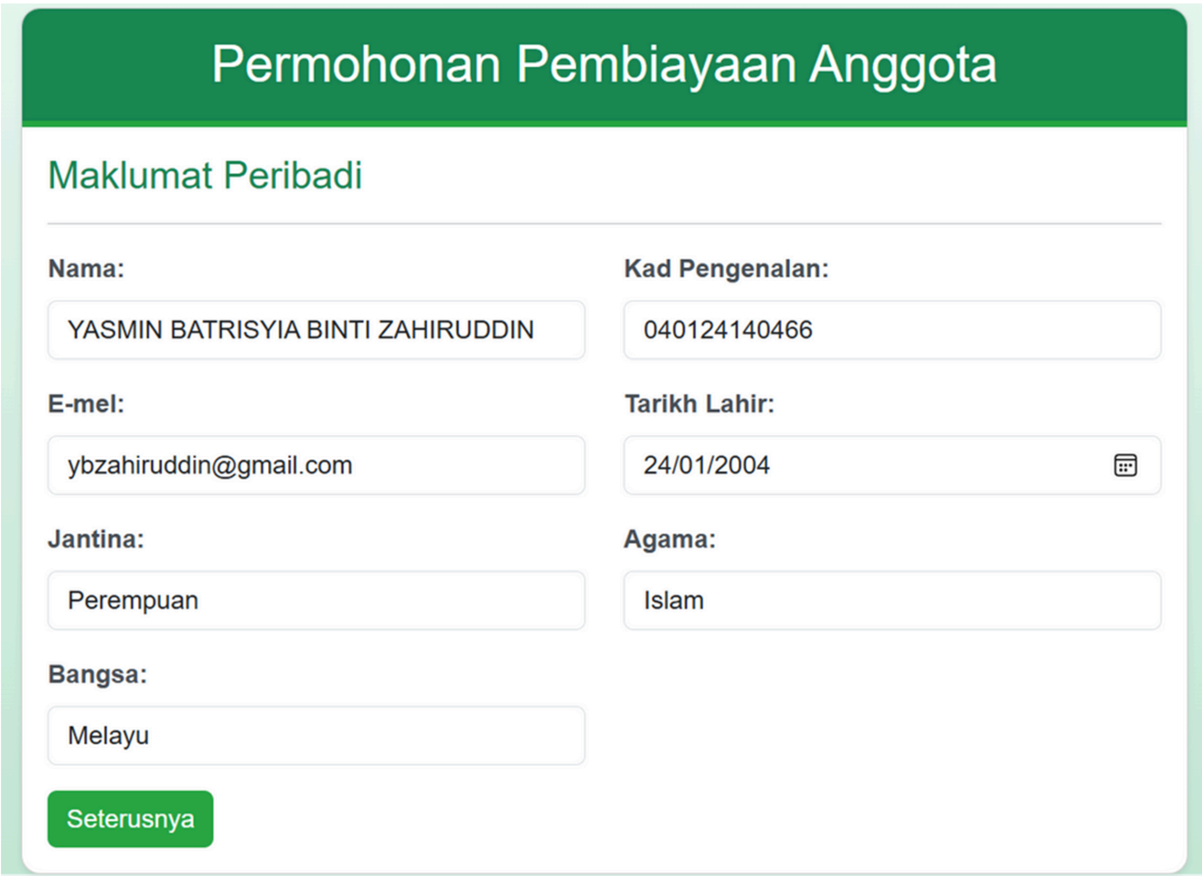


Figure 6.3.1: Interface for <Loan Application>

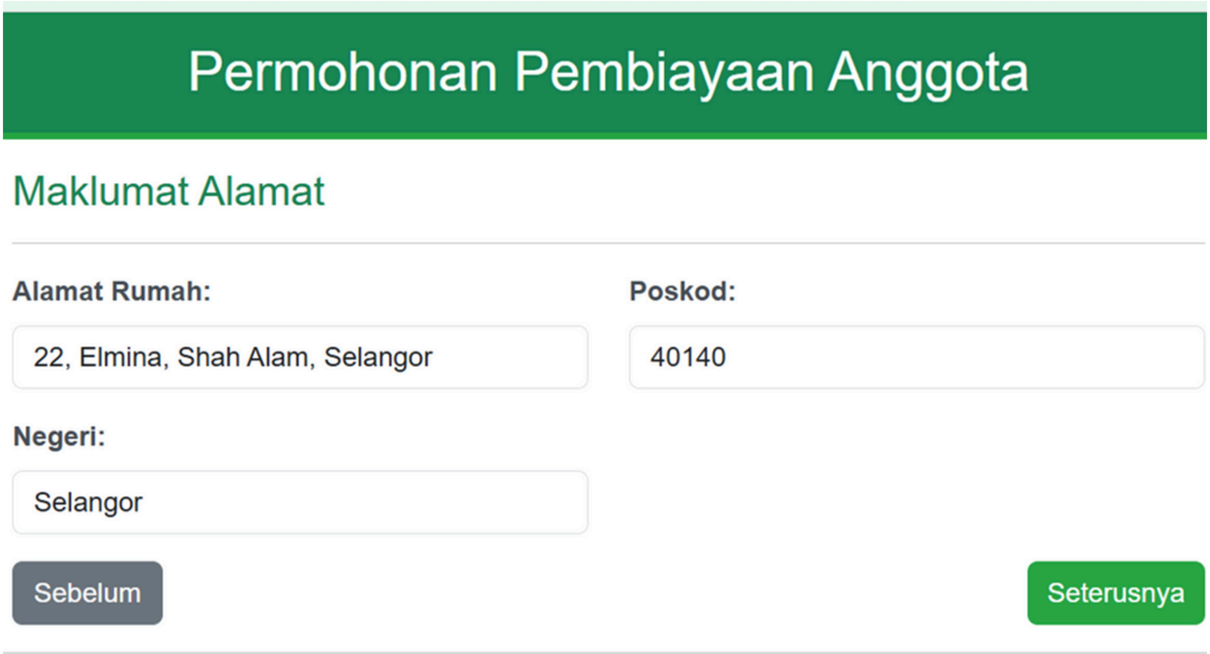


Figure 6.3.2 Interface for <Loan Application>

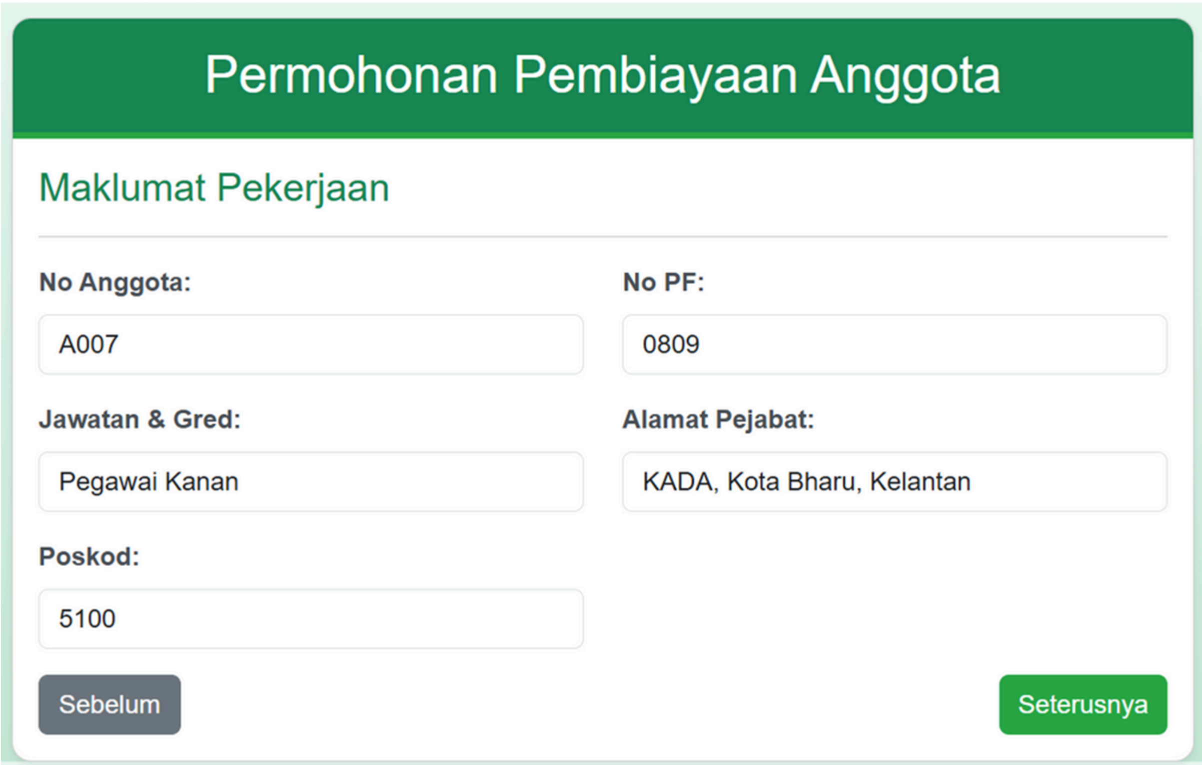


Figure 6.3.3 Interface for <Loan Application>

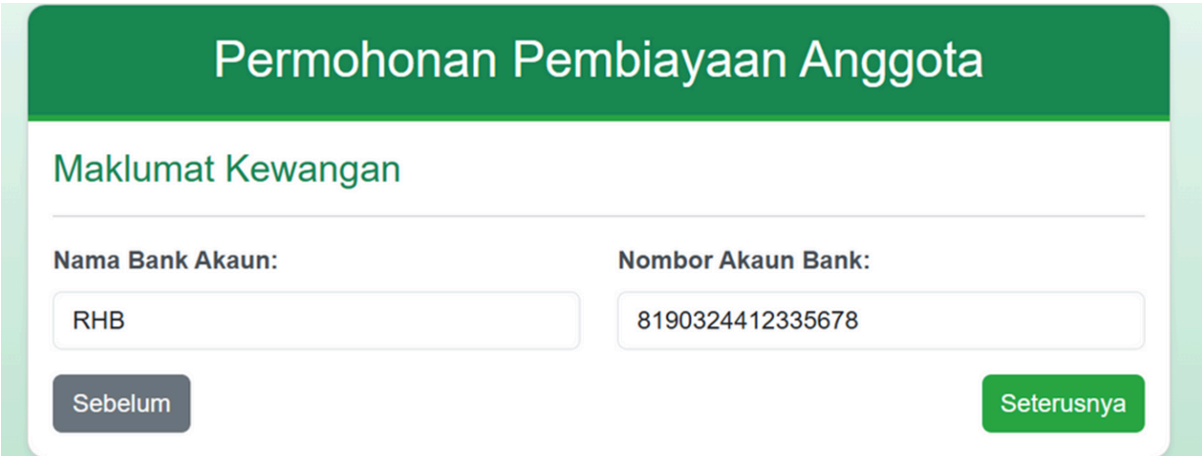


Figure 6.3.4 Interface for <Loan Application>

Figure 6.3.5 Interface for <Loan Application>

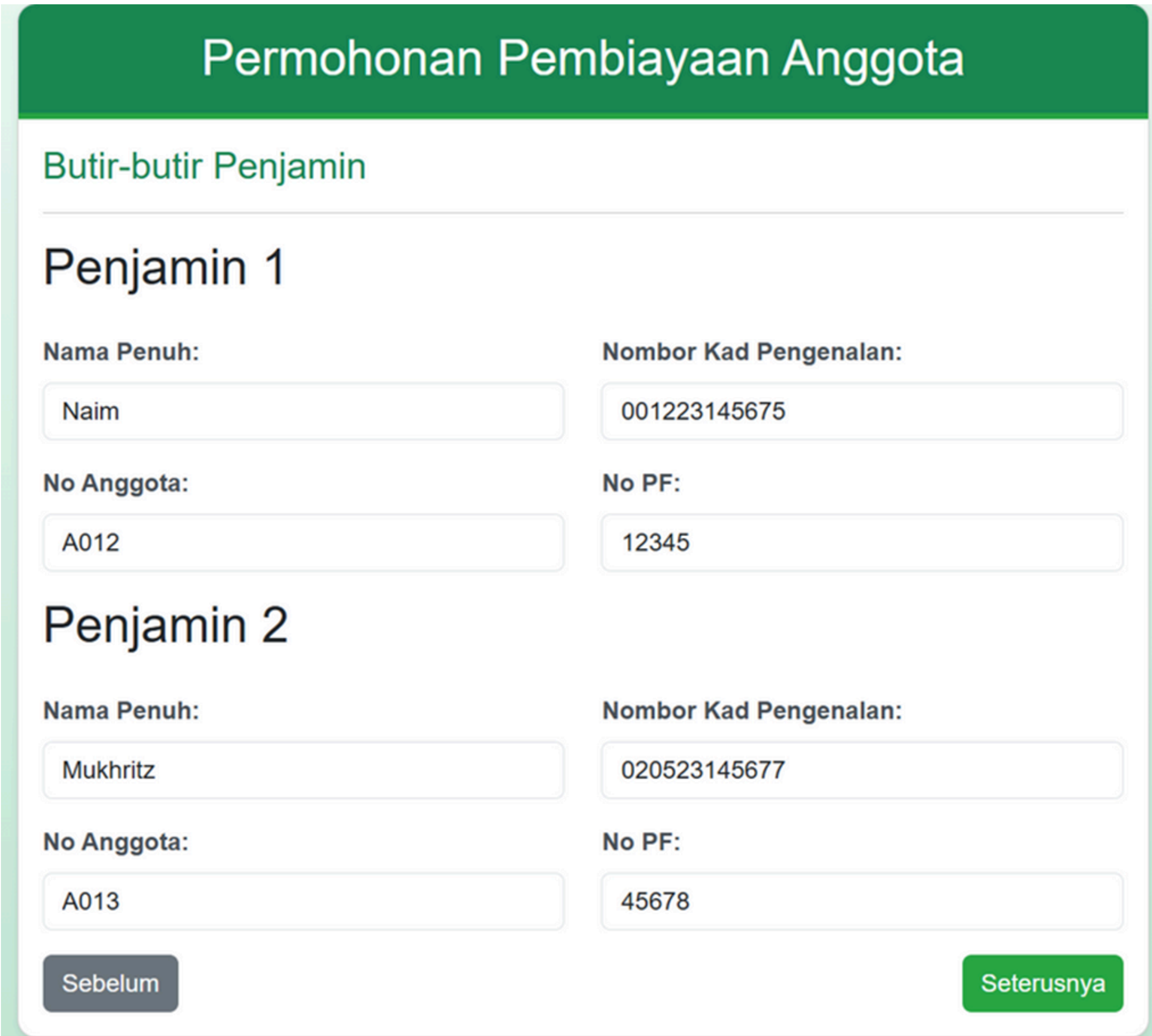


Figure 6.3.6 Interface for <Loan Application>

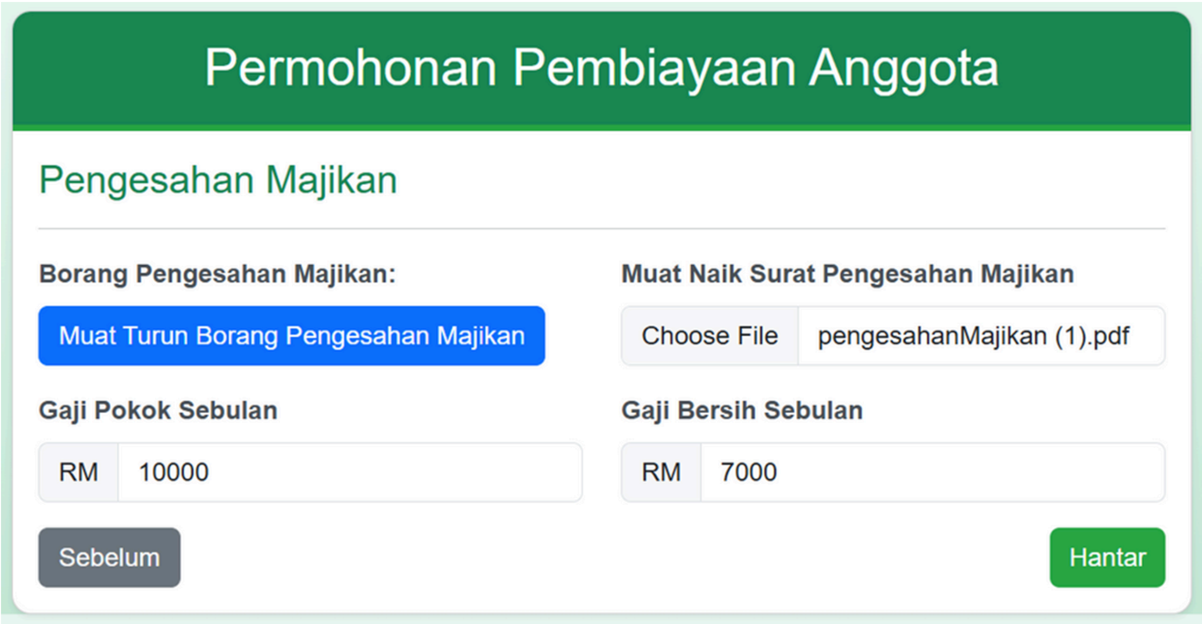


Figure 6.3.7 Interface for <Loan Application>

6.4.1 Module ALK

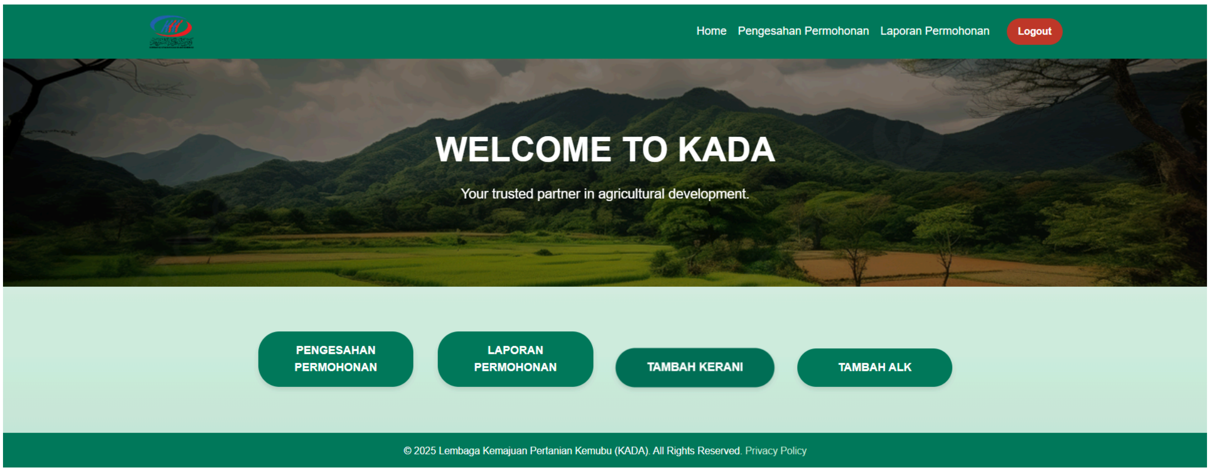


Figure 6.4.1: Interface for <ALK interface>

Figure 6.4.2: Interface for <verify loan application interface>

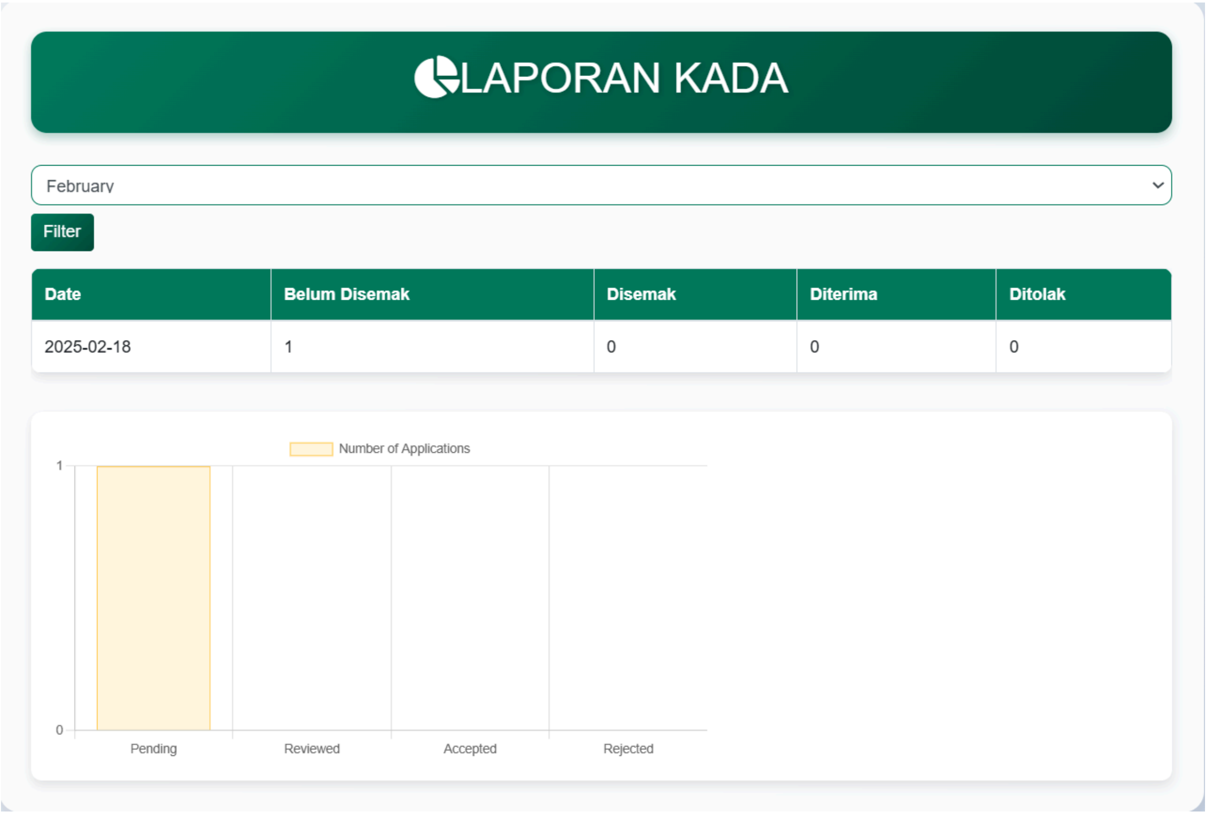


Figure 6.4.3: Interface for <Report interface>

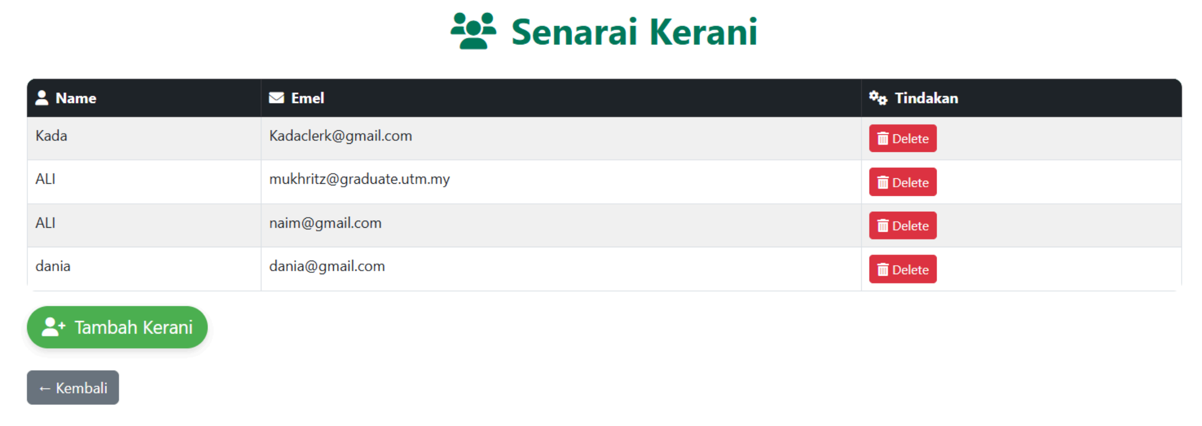


Figure 6.4.4: Interface for <add kerani interface>

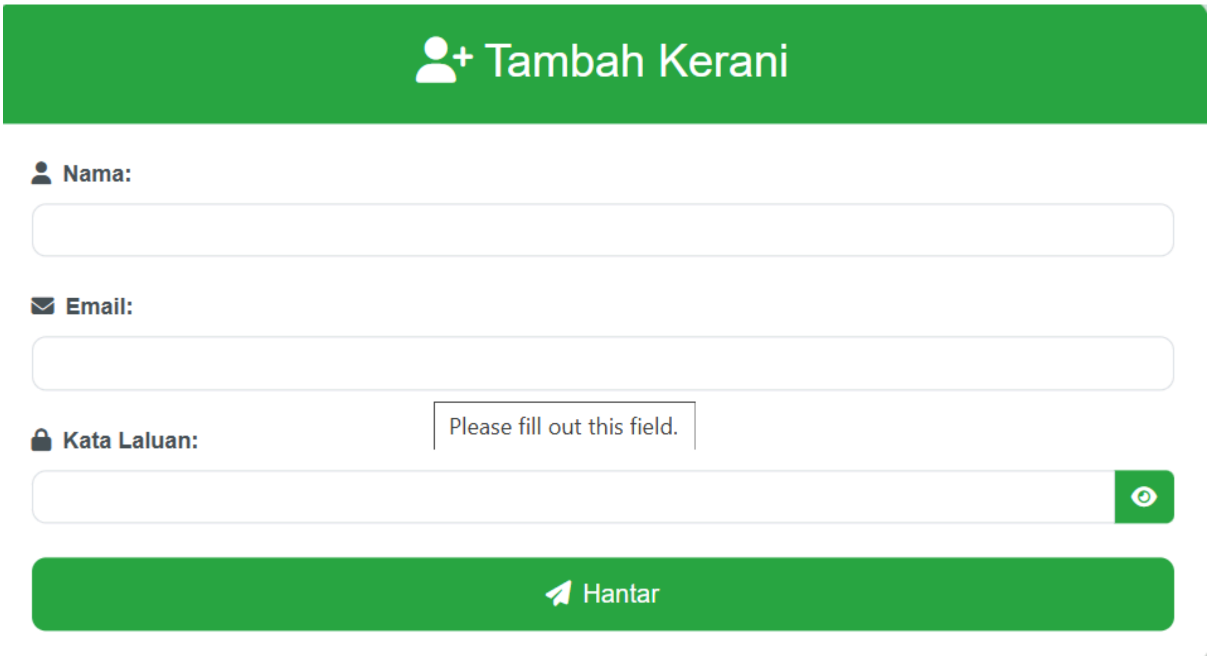


Figure 6.4.5: Interface for <add kerani details interface>

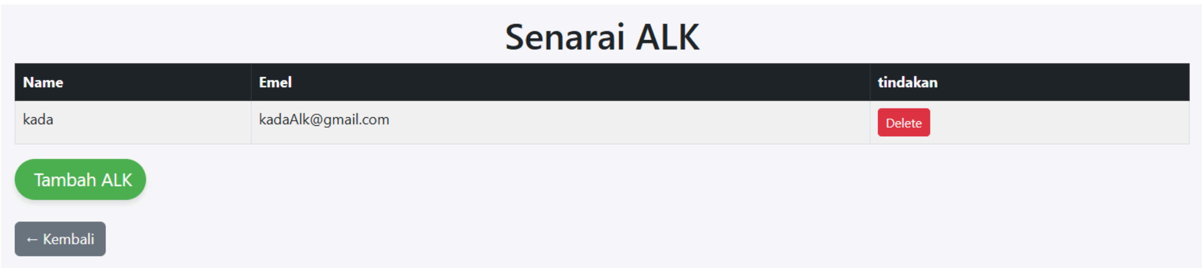


Figure 6.4.6: Interface for <add ALK interface>

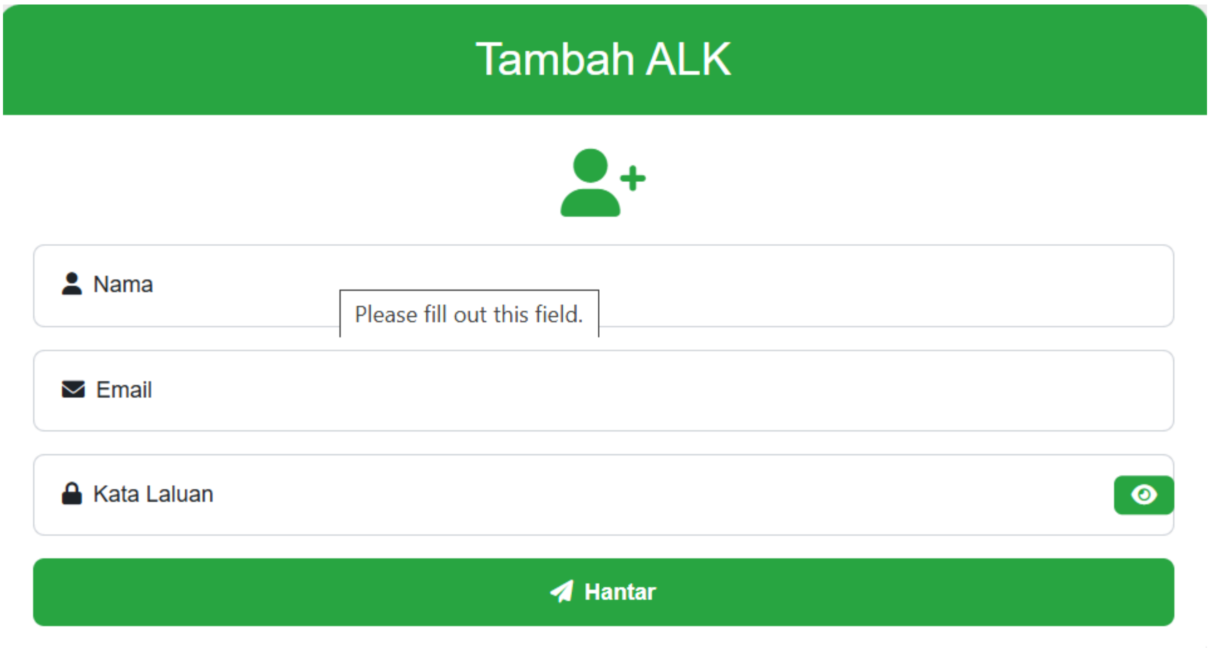


Figure 6.4.6: Interface for <add ALK interface>

7. Traceability

Table 7.1: Requirement Matrix

Package, UC, Sequence Diagram/Entity Class	Member	Clerk	ALK
P001,UC001,SD001	X		
P002,UC002,SD002	X		
P002,UC003,SD003	X		
P001,UC004,SD004	X	X	
P002,UC005,SD005	X	X	X
P002,UC006,SD006	X	X	X
P003,UC007,SD007			X
P002,UC008,SD008			X
P003,UC009,SD009		X	
P003,UC010,SD010		X	
P002,UC011,SD011		X	
P002,UC012,SD012	X		
P002,UC013,SD013		X	
P002,UC014,SD014		X	
P002,UC015,SD015			X

8. Test Cases

8.1 TC001: Test <P001> Subsystem: <Manage Profile(UC001)>

DESCRIPTION	VERSION	TEST DATE
Test user manage profile	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Yasmin	

Test Case ID **(TC001_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the manage profile at user interface		show the information of the applicant	As Expected	PASS	Yasmin	
2	change the user information		the information is changed	As Expected	PASS	Yasmin	
3	click "hantar"		the user new information is updated	As Expected	PASS	Yasmin	

8.2 TC002: Test <P002> Subsystem: <Applying a Loan (UC002)>

DESCRIPTION	VERSION	TEST DATE
Test to verify user loan application	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Muhammad Mukhritz	

Test Case ID **(TC002_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the loan applcation" in user interface by clicking "permohonan pembiayaan"		show the list of form to be fill up by the user to apply loan	As Expected	PASS	Yasmin	
2	fill up the form	loan information	the form is fill up	As Expected	PASS	Yasmin	
3	click "Hantar"		The information is sent to the database system	As Expected	PASS	Yasmin	

8.3 TC003: Test <P002> Subsystem: <Check Loan Status (UC003)>

DESCRIPTION	VERSION	TEST DATE
Test to check loan status	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Yasmin	

Test Case ID **(TC003_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the check loan status by clicking “status permohonan pembiayaan”		show the user loan information	As Expected	PASS	Yasmin	
2	click “Maklumat lanjut”		show the detail information of the user loan application	As Expected	PASS	Yasmin	

8.4 TC004: Test <P001> Subsystem: <Register (UC004)>

DESCRIPTION	VERSION	TEST DATE
Test member registration function	3.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
To ensure that all required fields are filled correctly, validations are in place, and the user can successfully register.	Syarifah Dania	

Test Case ID (TC004_01_01)	**Test Steps**	**Input Data**	**Expected Results**	**Actual Results**	**PASS/FAIL**	**Tested By**	**Tester Comments**
1	Navigate to the registration page.	-	The registration form page should load successfully.	As Expected	PASS	Syarifah Dania	-
2	Enter valid Kad Pengenalan (IC).	123456789012	The IC field accepts only a maximum of 12 digits without spaces or dashes.	As Expected	PASS	Syarifah Dania	-

3	Enter valid No Anggota.	ANG00123	The No Anggota field should accept a unique value and display no duplicates.	As Expected	PASS	Syarifah Dania	-
4	Enter a valid Email address.	user@example.com	The email field should accept only a valid, unique email address.	As Expected	PASS	Syarifah Dania	-
5	Enter valid Maklumat Kewangan (Financial Info).	5000	The financial information should accept positive values only, no negative numbers.	As Expected	PASS	Syarifah Dania	-
6	Click the "Register" button.	-	The user should be registered successfully, and a confirmation message should be displayed.	As Expected	PASS	Syarifah Dania	-
7	Verify if the user receives a confirmation email (if applicable).	-	A confirmation email should be sent to the user's provided email address.	As Expected	PASS	Syarifah Dania	-

8.5 TC005: Test <P002> Subsystem: <Login (UC005)>

DESCRIPTION	VERSION	TEST DATE
Test to verify user loan application	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Syarifah Dania	

Test Case ID **(TC005_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the login page of the system.		User manages to navigate to the login page	As Expected	PASS	Syarifah Dania	
2	Enter valid username and password and click login button	Username: accordingly Password: accordingly	The user should be logged in successfully and redirected to the dashboard.	As Expected	PASS	Syarifah Dania	Ensure that password is encrypted in the backend for security purposes.

8.6 TC006: Test <P002> Subsystem: <Logout (UC006)>

DESCRIPTION	VERSION	TEST DATE
Test Kad Pengenalan	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that Kad Pengenalan is entered correctly, with no duplicates and with a maximum of 12 digits, without dashes or spaces.	Syarifah Dania	

Test Case ID **(TC006_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Enter Kad Pengenalan in the input field.	040508011213	The Kad Pengenalan should accept only 12 digits with no dash or spaces.	As Expected	PASS	Syarifah Dania	
2	Verify no duplicate Kad Pengenalan in the system.	040508011312	The Kad Pengenalan should be unique in the system.	As Expected	PASS	Syarifah Dania	

8.7 TC007: Test <P003> Subsystem: <Verify Loan Application (UC007)>

DESCRIPTION	VERSION	TEST DATE
Test to verify user loan application	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Muhammad Mukhritz	

Test Case ID **(TC007_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the verify loan application in the ALK interface		show the list of user loan applicationr that are still in status pending	As Expected	PASS	Muhammad Mukhritz	
2	click Accept		accept the user loan application and send email to inform loan status	As Expected	PASS	Muhammad Mukhritz	

8.8 TC008: Test <P002> Subsystem: <View Annual Report (UC008)>

DESCRIPTION	VERSION	TEST DATE
Test to view the annual report of the membership application	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Muhammad Mukhritz	

Test Case ID **(TC008_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the report section in ALK interface by click button “laporan”		show the list of member application status with graph	As Expected	PASS	Muhammad Mukhritz	

8.9 TC009: Test <P003> Subsystem: <Create Account (UC009)>

a) TC009_01: Test <reject member application>

DESCRIPTION	VERSION	TEST DATE
Test to accept the membership application	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Muhammad Naim	

Test Case ID **(TC009_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the List of the member that still in status Pending by click button “Pemehonan Ahli”		show the list of member that are still in status pending	As Expected	PASS	Muhammad Naim	
2	click Reject		delete the membership application	As Expected	PASS	Muhammad Naim	

b) TC009_01: Test <accept membership application>

DESCRIPTION	VERSION	TEST DATE
Test to reject the membership application	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Muhammad Naim	

Test Case ID **(TC009_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the List of the member that still in status Pending by click button "Pemehonan Ahli"		show the list of member that are still in status pending	As Expected	PASS	Muhammad Naim	
2	click Accept		show the "created account" function	As Expected	PASS	Muhammad Naim	
3	Click Submit	password = 123	The account will be created, and an email will be sent.	As Expected	PASS	Muhammad Naim	

8.10 TC010: Test <P003> Subsystem: <Manage List Of Members (UC010)>

a) TC010_01: Test <View Details Member>

DESCRIPTION	VERSION	TEST DATE
Test the Login Functionality in Banking	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Muhammad Naim	

Test Case ID **(TC0010_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the List of the member by click button " senarai ahli"		show the list of the member that has been accepted	As Expected		naim	
2	click button "more"		show the details of the member	As Expected	PASS	naim	

b) TC010_02: Test <Edit Details Member>

DESCRIPTION	VERSION	TEST DATE
Test to edit the details of member	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Muhammad Naim	

Test Case ID **(TC0010_02 _01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the List of the member by click button " senarai ahli"		show the list of the member that has been accepted	As Expected	PASS	naim	
2	Click 'edit'		show the edit interface	As Expected	PASS	naim	
3	Click Submit	modal syer = 5 modal yuran = 35 Wang deposit = 5 simpanan tetap = 5 tabung kebajikan = 5	the value has been updated	As Expected	PASS	naim	

c) TC010_03: Test <Delete Account>

DESCRIPTION	VERSION	TEST DATE
Test to delete an member account	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Muhammad Naim	

Test Case ID **(TC0010_03 _01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the List of the member by click button " senarai ahli"		show the list of the member that has been accepted	As Expected	PASS	naim	
2	click "delete"		the account has been delete	As Expected	PASS	naim	

8.11 TC011: Test <P002> Subsystem: <Manage Loan application (UC011)>

a) TC011_01: Test <View Details Loan>

DESCRIPTION	VERSION	TEST DATE
View Details Loan of the applicant	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Muhammad Naim	

Test Case ID **(TC0011_01 _01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the list of the Loan by Click button "senarai Loan		show the list of the Loan that has been approve	As Expected	PASS	naim	
2	click button "more"		show the details of the loan applicant	As Expected	PASS	naim	

b) TC011_02: Test <Make A Remark>

DESCRIPTION	VERSION	TEST DATE
Make a Remark for the Loan Application	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Muhammad Naim	

Test Case ID **(TC0011_02_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the List of the loan Applicant that still in status Pending by click button “pemohonan pembiayaan”		Show the list of loan applicants who are still in pending status.	As Expected	PASS	naim	
2	click “nota”		Show the interface that allows writing a note.	As Expected	PASS	naim	
3	Click Submit	remark = this loan application is okay to approve	the remark has been updated	As Expected	PASS	naim	

c) TC011_03: Test <Delete Loan>

DESCRIPTION	VERSION	TEST DATE
Test to delete the loan application	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Muhammad Naim	

Test Case ID **(TC0011_03_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the list of the Loan by Click button “senarai Loan		show the list of the Loan that has been approve	As Expected	PASS	naim	
2	click button “delete”		The loan application has been deleted from the list of loans.	As Expected	PASS	naim	

8.12 TC012: Test <P002> Subsystem: <Calculate Loan Payment (UC012)>

DESCRIPTION	VERSION	TEST DATE
Test to update the Saving Payment	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Yasmin Batrisyia	

Test Case ID **(TC0012_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the "Calculate Loan Payment" section in the system.			As Expected	PASS	Yasmin Batrisyia	
2	Enter loan details including: Loan amount, interest rate and duration. Click “Kira” button.	Loan Amount: RM 10,000 Interest Rate: 5% Loan Term: 12 months (the values are according to the users)	The system should calculate the monthly payment based on the loan amount, interest rate, and loan term. Display the monthly payment amount correctly.	As Expected	PASS	Yasmin Batrisyia	

8.13 TC013: Test <P002> Subsystem: <Payment (UC013)>

a) TC013_01: Test <Record Saving Payment>

DESCRIPTION	VERSION	TEST DATE
Test to update the Saving Payment	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Muhammad Naim	

Test Case ID **(TC0013_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the Saving Payment interface by click button “pembayaran simpanann”		show the interface of the Saving Payment	As Expected	PASS	naim	
2	update Saving Payment	modal syer = 5 modal yuran = 35 Wang deposit = 5 simpanan tetap = 5 tabung kebajikan = 5	the saving paymentt interface will show updated value	As Expected	PASS	naim	

b) TC013_02: Test <Record Loan Payment>

DESCRIPTION	VERSION	TEST DATE
Test to update the Loan Payment	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Muhammad Naim	

Test Case ID **(TC0013_02 _01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to the Loan Payment interface by click button "pembayaran pembiayaan"	click button "pembayaran pembiayaan"	show the interface of the Loan Payment	As Expected	PASS	naim	
2	update Loan Payment	**bilangan bayaran = 1 jumlah bayaran = 100**	the loan payment interface will show updated value	As Expected	PASS	naim	

8.14 TC014: Test <P002> Subsystem: <Interest (UC014)>

a) TC014_01: Test <Update Interest>

DESCRIPTION	VERSION	TEST DATE
Test updating Interest Rate	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Muhammad Naim	

Test Case ID **(TC0014_01 _01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	navigate to the interest page by Click button "faedah"	Click button "faedah"	show the interface of the interest page	As Expected	PASS	naim	
2	Click Submit	interest rate = 1	the interest rate has been update	As Expected	PASS	naim	

8.15 TC015: Test <P002> Subsystem: <Add Staff(UC015)>

TC015_01: Test <add staff>

DESCRIPTION	VERSION	TEST DATE
Test add staff in the system	1.0	1-Jan-2024
TEST OBJECTIVE	AUTHOR	REVIEWER
	Muhammad Mukhritz	

Test Case ID **(TC0015_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	navigate to the senarai kerani in ALK interface and click "senarai kerani"	Click button "senarai kerani"	show the interface of the interest page	As Expected	PASS	Mukhritz	
2	Click tambah kerani		add clerk form to fill with data	As Expected	PASS	Mukhritz	
	fill the form and click submit	name, email and password	clerk is added	As expected	PASS	Mukhritz	

9. Traceability Matrix

Table 9.1: Example of Traceability Matrix for <system name>

User Story ID	User Story Description	Sequence Diagram ID	Sequence Diagram Description	Test Case ID
US_001	Manage Profile	SD001	Shows the process for updating a user's profile.	TC_001
US_002	Applying a loan	SD002	show the process how to apply loan	TC_002
US_003	Check loan Status	SD003	show the process how to check loan status	TC_003
US_004	Register	SD004	Illustrates the steps for user registration.	TC_004
US_005	Login	SD005	Depicts the user login process.	TC_005
US_006	Logout	SD006	show the process how to logout	TC_006
US_007	verify loan application	SD007	show the process how to verify loan application	TC_007
US_008	view annual report	SD008	show the process how to view annual report	TC_008
US_009	Create account	SD009	Shows the process of accept and reject a membership application	TC_009

US_010	manage list of member	SD010	Shows the process of edit, view and delete a member details	TC_010
US_011	manage loan application	SD011	Shows the process of view and delete a loan application details	TC_011
US_012	calculate loan repayment	SD012	show the process how to calculate the loan repayment	TC_012
US_013	payment	SD013	Shows the process of updating Saving payment and loan payment	TC_013
US_014	interest	SD014	show the process of updating interest rate	TC_014
US_015	Add Staff	SD015	show the process of adding ALK and clerk account	TC_015